

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286609

Kind Code

A1

Publication Date

September 11, 2025

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Access Node Farm for End-to-End Beamforming

Abstract

Methods and systems are described for providing end-to-end beamforming. For example, end-to-end beamforming systems include end-to-end relays and ground networks to provide communications to user terminals located in user beam coverage areas. The ground segment can include geographically distributed access nodes and a central processing system. Return uplink signals, transmitted from the user terminals, have multipath induced by a plurality of receive/transmit signal paths in the end to end relay and are relayed to the ground network. The ground network, using beamformers, recovers user data streams transmitted by the user terminals from return downlink signals. The ground network, using beamformers generates forward uplink signals from appropriately weighted combinations of user data streams that, after relay by the end-end-end relay, produce forward downlink signals that combine to form user beams.

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Applicant: ViaSat, Inc. (Carlsbad, CA)

Family ID: 60941388

Appl. No.: 19/056868

Filed: February 19, 2025

Related U.S. Application Data

parent US continuation 18616395 20240326 parent-grant-document US 12261679 child US 19056868

parent US continuation 17375485 20210714 parent-grant-document US 11973572 child US 18616395

parent US continuation 16131831 20180914 parent-grant-document US 11101877 child US 17375485

parent US continuation 15719249 20170928 parent-grant-document US 10084532 child US

16131831

parent US continuation PCT/US2017/013518 20170113 PENDING child US 15719249

parent US continuation-in-part PCT/US2016/026815 20160408 PENDING child US
PCT/US2017/013518

parent US continuation-in-part PCT/US2016/026815 20160408 PENDING child US 15719249

parent US continuation-in-part PCT/US2016/026813 20160408 PENDING child US 15719249

us-provisional-application US 62145810 20150410

us-provisional-application US 62145804 20150410

us-provisional-application US 62145810 20150410

us-provisional-application US 62145804 20150410

us-provisional-application US 62164456 20150520

us-provisional-application US 62164456 20150520

us-provisional-application US 62278368 20160113

us-provisional-application US 62278368 20160113

us-provisional-application US 62278368 20160113

us-provisional-application US 62298911 20160223

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us-provisional-application US 62312342 20160323

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us-provisional-application US 62314921 20160329

us-provisional-application US 62314921 20160329

us-provisional-application US 62314921 20160329

us-provisional-application US 62431416 20161207

Publication Classification

Int. Cl.: **H04B7/185** (20060101); **H01Q1/06** (20060101); **H01Q1/24** (20060101); **H01Q1/28** (20060101); **H01Q3/24** (20060101); **H01Q3/40** (20060101); **H01Q5/50** (20150101); **H01Q19/13** (20060101); **H01Q21/00** (20060101); **H01Q21/24** (20060101); **H01Q25/00** (20060101); **H04B7/024** (20170101); **H04B7/0413** (20170101); **H04B7/06** (20060101); **H04B7/204** (20060101); **H04B10/118** (20130101); **H04W4/00** (20180101); **H04W16/26** (20090101); **H04W16/28** (20090101); **H04W40/20** (20090101); **H04W76/10** (20180101); **H04W84/06** (20090101)

U.S. Cl.:

CPC **H04B7/18513** (20130101); **H01Q1/06** (20130101); **H01Q1/288** (20130101); **H01Q3/247** (20130101); **H01Q3/40** (20130101); **H01Q5/50** (20150115); **H01Q19/132** (20130101); **H01Q21/0025** (20130101); **H01Q21/24** (20130101); **H01Q25/001** (20130101); **H04B7/024** (20130101); **H04B7/0413** (20130101); **H04B7/0617** (20130101); **H04B7/18508** (20130101); **H04B7/18515** (20130101); **H04B7/18517** (20130101); **H04B7/18534** (20130101); **H04B7/1858** (20130101); **H04B7/18586** (20130101); **H04B7/18589** (20130101); **H04B7/204** (20130101); **H04B7/2041** (20130101); **H04B10/118** (20130101); **H04W4/00** (20130101); **H04W16/28** (20130101); **H04W40/20** (20130101); **H04W76/10** (20180201); **H04W84/06** (20130101); **H01Q1/247** (20130101); **H04W16/26** (20130101)

Background/Summary

TECHNICAL FIELD

[0001] The disclosed systems, methods, and apparatuses relate to end-to-end beamforming in a system using an end-to-end relay.

BACKGROUND

[0002] Wireless communication systems, such as satellite communication systems, provide a means by which data, including audio, video, and various other sorts of data, may be communicated from one location to another. Information originates at a first station, such as a first ground-based station, and is transmitted to a wireless relay, such as a communication satellite. Information received by the wireless relay is retransmitted to a second station, such as a second ground-based station. In some wireless relay communication systems, either the first or second station (or both) are mounted on a craft, such as an aircraft, watercraft, or landcraft. Information may be transmitted in just one direction (e.g., from a first ground-based station to a second ground-based station only) or may be transmitted in both directions (e.g., also from the second ground-based station to the first ground-based station).

[0003] In a wireless relay communication system in which the wireless relay is a satellite, the satellite may be a geostationary satellite, in which case the satellite's orbit is synchronized to the rotation of the Earth, keeping the coverage area of the satellite essentially stationary with respect to the Earth. In other cases, the satellite is in an orbit about the Earth that causes the coverage area of the satellite to move over the surface of the Earth as the satellite traverses its orbital path.

[0004] The signals that are directed to or from a first station may be directed by using an antenna that is shaped to focus the signal into a narrow beam. Such antennas typically have a paraboloid shaped reflector to focus the beam.

[0005] In some cases, a beam may be formed electronically by adjusting the gain and phase (or time delay) of signals that are transmitted, received, or both from several elements of a phased array antenna. By properly selecting the relative phase and gain transmitted and/or received by each element of a phased array antenna, the beam may be directed. In most cases, all of the energy being transmitted from a ground-based station is intended to be received by one wireless relay. Similarly, information received by the second station is typically received from one wireless relay at a time. Therefore, it is typical that a transmit beam that is formed to transmit information to the wireless relay (whether by use of electronic beamforming or by use of an antenna with a shaped reflector) is relatively narrow to allow as much of the transmitted energy as possible to be directed to the wireless relay. Likewise, a receive beam that is formed to receive information from the wireless relay is typically narrow to gather energy from the direction of the wireless relay with minimal interference from other sources.

[0006] In many cases of interest, the signals that are transmitted from the wireless relay to the first and second stations are not directed to a single station. Rather, the wireless relay is able to transmit signals over a relatively large geographic area. For example, in one satellite communication system, a satellite may service the entire continental United States. In such a case, the satellite is said to have a satellite coverage area that includes the entire continental United States. Nonetheless, in order to increase the amount of data that may be transmitted through a satellite, the energy transmitted by the satellite is focused into beams. The beams may be directed to geographic areas on the Earth.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0007] The drawings are provided for purposes of illustration only and merely depict examples. These drawings are provided to facilitate the reader's understanding of the disclosed method and apparatus. They do not limit the breadth, scope, or applicability of the claimed invention. For clarity and ease of illustration, these drawings are not necessarily made to scale.

[0008] FIG. **1** is an illustration of an example of a satellite communication system.

[0009] FIG. **2** is a diagram showing an example pattern of beams that covers the continental United States.

[0010] FIG. **3** is an illustration of an example of the forward link of a satellite communication system in which the satellite has a phased array multi-feed per beam on-board beamforming capability.

[0011] FIG. **4** is an illustration of an example of the forward link of a satellite communication system having ground-based beamforming.

[0012] FIG. **5** is an illustration of an example end-to-end beamforming system.

[0013] FIG. **6** is an illustration of example signal paths for signals in the return direction.

[0014] FIG. **7** is an illustration of example signal paths in the return direction from a user terminal.

[0015] FIG. **8** is a simplified illustration of an example end-to-end return channel matrix model.

[0016] FIG. **9** is an illustration of example signal paths in the forward direction.

[0017] FIG. **10** is an illustration of example signal paths in the forward direction to a user terminal located within a user beam coverage area.

[0018] FIG. **11** is a simplified illustration of an example end-to-end forward channel matrix model.

[0019] FIG. **12** is an illustration of an example end-to-end relay satellite supporting forward and return data.

[0020] FIG. **13** is an illustration of an example of an uplink frequency range being divided into two portions.

[0021] FIG. **14** is an illustration of an example end-to-end relay being time multiplexed between forward data and return data.

[0022] FIG. **15** is a block diagram of components of an example end-to-end relay implemented as a satellite.

[0023] FIG. **16** is a block diagram of an example transponder including a phase shifter.

[0024] FIG. **17** is a graph of example signal strength patterns of several antenna elements.

[0025] FIG. **18** is an illustration of example 3 dB signal strength contours for several antenna elements.

[0026] FIG. **19** is an illustration of example overlapping signal strength patterns of several antenna elements.

[0027] FIG. **20A-20E** is an illustration of example overlapping 3 dB signal strength contours for several antenna elements.

[0028] FIG. **21** is an illustration of an example enumeration of 16 antenna elements and their overlapping 3 dB signal strength contours.

[0029] FIG. **22** is a table showing example mappings of receive antenna elements to transmit antenna elements through 16 transponders.

[0030] FIG. **23** is an illustration of a cross-section of a paraboloid antenna reflector and an array of elements centered at the focal point of the parabola.

[0031] FIG. **24** is an illustration of a cross-section of a paraboloid antenna reflector and an array of elements placed away from the focal point of the parabola.

[0032] FIG. **25** is an illustration of an example relay coverage area (shown with single cross-hatching) and the area (shown with double cross-hatching) defined by the points within the relay coverage area that are also contained within six antenna element coverage areas.

[0033] FIG. **26** is an illustration of an example relay antenna pattern in which all of the points

within a relay coverage area are also contained within at least four antenna element coverage areas.
[0034] FIG. **27** is an illustration of an example distribution of access nodes (ANs) and user beam coverage areas.

[0035] FIG. **28** is an example graph of normalized forward and return link capacity as a function of the number of ANs deployed.

[0036] FIG. **29** is a block diagram of an example ground segment **502** for an end-to-end beamforming system.

[0037] FIG. **30** is a block diagram of an example forward/return beamformer.

[0038] FIG. **31** is a block diagram of an example forward beamformer comprising multiple return time-slice beamformers with time-domain de-multiplexing and multiplexing.

[0039] FIG. **32** is an illustration of a simplified example ground segment showing the operation of a forward time-slice beamformer.

[0040] FIG. **33** is a block diagram of an example return beamformer comprising multiple return time-slice beamformers with time-domain de-multiplexing and multiplexing.

[0041] FIG. **34** is an illustration of a simplified example ground segment showing the operation of a return beamformer employing time-domain multiplexing.

[0042] FIG. **35** is a block diagram of an example multi-band forward/return beamformer that employs sub-band de-multiplexing and multiplexing.

[0043] FIG. **36** and FIG. **37** is an illustration of example timing alignment for the forward link.

[0044] FIG. **38** is a block diagram of an example AN.

[0045] FIG. **39** is a block diagram of part of an example of an AN.

[0046] FIG. **40** is a block diagram of an example AN **515** in which multiple frequency sub-bands are processed separately.

[0047] FIG. **41** is an illustration of an example end-to-end beamforming system for enabling distinct user-link and feeder-link coverage areas.

[0048] FIG. **42** is an illustration of an example model of signal paths for signals carrying return data on the end-to-end return link.

[0049] FIG. **43** is an illustration of an example model of signal paths for signals carrying forward data on the end-to-end forward link.

[0050] FIGS. **44A** and **44B** are an illustration of an example forward signal path and return signal path, respectively.

[0051] FIGS. **45A**, **45B**, **45C**, **45D**, **45E**, **45F**, and **45G** are illustrations of examples of an end-to-end relay visible coverage areas.

[0052] FIGS. **46A** and **46B** are an illustration of an example of an end-to-end relay Earth coverage area and North American coverage area, respectively.

[0053] FIGS. **47A** and **47B** are block diagrams of an example forward signal path and return signal path, respectively, each having selective activation of multiple user-link antenna subsystems.

[0054] FIGS. **48A** and **48B** are an illustration of an example of an end-to-end relay coverage area that includes multiple, selectively activated user coverage areas.

[0055] FIGS. **49A** and **49B** are block diagrams of example forward and return signal paths, respectively, each having selective activation of multiple user-link antenna subsystems and multiple feeder-link antenna subsystem.

[0056] FIGS. **50A**, **50B**, and **50C** illustrate examples of one or more user coverage areas with multiple access node areas.

[0057] FIGS. **51A** and **51B** show example forward and return signal paths, respectively, each having selective activation of multiple user-link antenna element arrays and multiple feeder-link antenna element arrays.

[0058] FIGS. **52A** and **52B** show example forward and return receive/transmit signal paths for concurrent use of multiple AN clusters, respectively.

[0059] FIGS. **53A** and **53B** illustrate example transponders allowing selective coupling between

multiple feeder-link constituent elements and a single user-link constituent element.

[0060] FIGS. **54A** and **54B** illustrate forward and return link transponders, respectively.

[0061] FIGS. **55A**, **55B**, and **55C** illustrate example loopback transponders.

[0062] FIG. **56A** illustrates an end-to-end relay that includes one or more reflectors.

[0063] FIG. **56B** illustrates an antenna subsystem with multiple feed clusters.

[0064] FIG. **57** illustrates an antenna subsystem that includes a compound reflector.

[0065] FIG. **58** shows an end-to-end relay system with portions disposed on one or more offshore (e.g., fixed or floating) platforms.

[0066] FIGS. **59A** and **59B** are illustrations of examples of end-to-end relay visible coverage areas supporting distinct frequency ranges.

[0067] FIGS. **60A** and **60B** show example forward/return receive/transmit signal paths supporting multiple frequency bands.

[0068] FIGS. **61A** and **61B** show example forward/return receive/transmit signal paths supporting multiple frequency bands.

[0069] FIG. **62** shows an example antenna element array with spatially interleaved subsets of constituent antenna elements.

[0070] FIGS. **63A** and **63B** are illustrations of example frequency allocations.

[0071] FIGS. **64A** and **64B** are illustrations of example frequency allocations.

[0072] FIGS. **65A** and **65B** are illustrations of example frequency allocations.

[0073] FIGS. **66A** and **66B** show example forward/return receive/transmit signal paths.

[0074] Reference designators (e.g., **100**) are used herein to refer to aspects of the drawings. Similar or like aspects are typically shown using like numbers. A group of similar or like elements may be referred to collectively by a single reference designator (e.g., **200**), while individual elements of the group may be referred to by the reference designator with an appended letter (e.g., **200a**, **200b**).

[0075] The figures are not intended to be exhaustive or to limit the claimed invention to the precise form disclosed. The disclosed method and apparatus may be practiced with modification and alteration, and that the invention is limited only by the claims and the equivalents thereof.

DETAILED DESCRIPTION

[0076] This detailed description is organized as follows. First, an introduction to wireless relay communication systems using satellite communication and beamforming are described. Second, end-to-end beamforming is described generally and at the system level using satellite end-to-end beamforming as an example, although application of end-to-end beamforming is not limited to satellite communications. Third, operation of forward and return data is described in context of end-to-end beamforming. Fourth, end-to-end relays and their antennas are described using a communication satellite as an example. Next, ground networks to form the end-to-end beams are described, including related aspects, such as delay equalization, feeder-link impairment removal, and beam weight computation. Finally, end-to-end beamforming with distinct user-link and feeder-link coverage areas is described, as well as systems with multiple coverage areas.

Satellite Communication

[0077] FIG. **1** is an illustration of an example of a hub and spoke satellite communication system **100**. The satellite serves as an example of a wireless relay. Though many examples are described throughout this disclosure in context of a satellite or satellite communication system, such examples are not intended to be limited to satellite; any other suitable wireless relay may be used and operate in a similar fashion. The system **100** comprises a ground-based Earth station **101**, a communication satellite **103**, and an Earth transmission source, such as a user terminal **105**. A satellite coverage area may be broadly defined as that area from which, and/or to which, either an Earth transmission source, or an Earth receiver, such as a ground-based Earth station or a user terminal, can communicate through the satellite. In some systems, the coverage area for each link (e.g., forward uplink coverage area, forward downlink coverage area, return uplink coverage area, and return downlink coverage area) can be different. The forward uplink coverage area and return

uplink coverage area are collectively referred to as the uplink satellite coverage area. Similarly, the forward downlink coverage area and the return downlink coverage area are collectively referred to as the downlink satellite coverage area. While the satellite coverage area is only active for a satellite that is in service (e.g., in a service orbit), the satellite can be considered as having (e.g., can be designed to have) a satellite antenna pattern that is independent of the relative location of the satellite with respect to the Earth. That is, the satellite antenna pattern is a pattern of distribution of energy transmitted from an antenna of a satellite (either transmitted from or received by the antenna of the satellite). The satellite antenna pattern illuminates (transmits to, or receives from) a particular satellite coverage area when the satellite is in a service orbit. The satellite coverage area is defined by the satellite antenna pattern, an orbital position and attitude for which the satellite is designed, and a given antenna gain threshold. In general, the intersection of an antenna pattern (at a particular effective antenna gain, e.g. 3 dB, 4 dB, 6 dB 10 dB from peak gain) with a particular physical region of interest (e.g., an area on or near the earth surface) defines the coverage area for the antenna. Antennas can be designed to provide a particular antenna pattern (and/or coverage area) and such antenna patterns can be determined computationally (e.g., by analysis or simulation) and/or measured experimentally (e.g., on an antenna test range or in actual use).

[0078] While only one user terminal **105** is shown in the figure for the sake of simplicity, there are typically many user terminals **105** in the system. The satellite communication system **100** operates as a point to multi-point system. That is, the Earth station **101** within the satellite coverage area can send information to, and receive information from, any of the user terminals **105** within the satellite coverage area. However, the user terminals **105** only communicate with the Earth station **101**. The Earth station **101** receives forward data from a communication network **107**, modulates the data using a feeder link modem **109** and transmits the data to the satellite **103** on a forward feeder uplink **111**. The satellite **103** relays this forward data to user terminals **105** on the forward user downlink (sometimes called a forward service downlink) **113**. In some cases, the forward direction communication from the Earth station **101** is intended for several of the user terminals **105** (e.g., information is multicast to the user terminals **105**). In some cases, the forward communication from the Earth station **101** is intended for only one user terminal **105** (e.g., unicast to a particular user terminal **105**). The user terminals **105** transmit return data to the satellite **103** on a return user uplink (sometimes called a return service uplink) **115**. The satellite **103** relays the return data to the Earth station **101** on a return feeder downlink **117**. A feeder-link modem **109** demodulates the return data, which is forwarded to the communication network **107**. This return-link capability is generally shared by a number of user terminals **105**.

[0079] FIG. 2 is a diagram showing an example of one configuration of beam coverage areas of a satellite to service the continental United States. Seventy beams are shown in the example configuration. A first beam **201** covers approximately two thirds of the state of Washington. A second beam **203** adjacent to the first beam **201** covers an area immediately to the east of the first beam **201**. A third beam **205** approximately covers Oregon to the south of the first beam **201**. A fourth beam **207** covers an area roughly southeast of the first beam **201**. Typically, there is some overlap between adjacent beams. In some cases, a multi-color (e.g., two, three or four-color re-use pattern) is used. In an example of a four-color pattern, the beams **201**, **203**, **205**, **207** are individually allocated a unique combination of frequency (e.g., a frequency range or ranges or one or more channels) and/or antenna polarization (e.g., in some cases an antenna may be configured to transmit signals with a right-hand circular polarization (RHCP) or a left-hand circular polarization (LHCP); other polarization techniques are available). Accordingly, there may be relatively little mutual interference between signals transmitted on different beams **201**, **203**, **205**, **207**. These combinations of frequency and antenna polarization may then be re-used in the repeating non-overlapping “four-color” re-use pattern. In some situations, a desired communication capacity may be achieved by using a single color. In some cases, time sharing among beams and/or other interference mitigation techniques can be used.

[0080] Within some limits, focusing beams into smaller areas and thus increasing the number of beams, increases the data capacity of the satellite by allowing greater opportunity for frequency re-use. However, increasing the number of beams can increase the complexity of the system, and in many cases, the complexity of the satellite.

[0081] Complexity in the design of a satellite typically results in larger size, more weight, and greater power consumption. Satellites are expensive to launch into orbit. The cost of launching a satellite is determined in part by the weight and size of the satellite. In addition, there are absolute limits on the weight and size of a satellite if the satellite is to be launched using presently available rocket technology. This leads to tradeoffs between features that may be designed into a satellite. Furthermore, the amount of power that may be provided to components of a satellite is limited. Therefore, weight, size, and power consumption are parameters to be considered in the design of a satellite.

[0082] Throughout this disclosure, the term receive antenna element refers to a physical transducer that converts an electro-magnetic signal to an electrical signal, and the term transmit antenna element refers to a physical transducer that launches an electro-magnetic signal when excited by an electrical signal. The antenna element can include a horn, septum polarized horn (e.g., which may function as two combined elements with different polarizations), multi-port multi-band horn (e.g., dual-band 20 GHz/30 GHz with dual polarization LHCP/RHCP), cavity-backed slot, inverted-F, slotted waveguide, Vivaldi, Helical, loop, patch, or any other configuration of antenna element or combination of interconnected sub-elements. An antenna element has a corresponding antenna pattern, which describes how the antenna gain varies as a function of direction (or angle). An antenna element also has a coverage area which corresponds to an area (e.g., a portion of the Earth surface) or volume (e.g., a portion of the Earth surface plus airspace above the surface) over which the antenna element provides a desired level of gain (e.g., within 3 dB, 6 dB, 10 dB, or other value relative to a peak gain of the antenna element). The coverage area of the antenna element may be modified by various structures such as a reflector, frequency selective surface, lens, radome, and the like. Some satellites, including those described herein, can have several transponders, each able to independently receive and transmit signals. Each transponder is coupled to antenna elements (e.g., a receive element and a transmit element) to form a receive/transmit signal path that has a different radiation pattern (antenna pattern) from the other receive/transmit signal paths to create unique beams that may be allocated to different beam coverage areas. It is common for a single receive/transmit signal path to be shared across multiple beams using input and/or output multiplexers. In both cases, the number of simultaneous beams that may be formed is generally limited by the number of receive/transmit signal paths that are deployed on the satellite.

Beamforming

[0083] Beamforming for a communication link may be performed by adjusting the signal phase (or time delay), and sometimes signal amplitude, of signals transmitted and/or received by multiple elements of one or more antenna arrays with overlapping coverage areas. In some cases, some or all antenna elements are arranged as an array of constituent receive and/or transmit elements that cooperate to enable end-to-end beamforming, as described below. For transmissions (from transmit elements of the one or more antenna arrays), the relative phases, and sometimes amplitudes, of the transmitted signals are adjusted, so that the energy transmitted by transmit antenna elements will constructively superpose at a desired location. This phase/amplitude adjustment is commonly referred to as “applying beam weights” to the transmitted signals. For reception (by receive elements of the one or more antenna arrays), the relative phases, and sometimes amplitudes, of the received signals are adjusted (i.e., the same or different beam weights are applied) so that the energy received from a desired location by receive antenna elements will constructively superpose at those receive antenna elements. In some cases, the beamformer computes the desired antenna element beam weights. The term beamforming may refer in some cases to the application of the beam weights. Adaptive beamformers include the function of dynamically computing the beam

weights. Computing the beam weights may require direct or indirect discovery of the communication channel characteristics. The processes of beam weight computation and beam weight application may be performed in the same or different system elements.

[0084] The antenna beams may be steered, selectively formed, and/or otherwise reconfigured by applying different beam weights. For example, the number of active beams, coverage area of beams, size of beams, relative gain of beams, and other parameters may be varied over time. Such versatility is desirable in certain situations. Beamforming antennas can generally form relatively narrow beams. Narrow beams may allow the signals transmitted on one beam to be distinguished from signals transmitted on the other beams (e.g., to avoid interference). Accordingly, narrow beams can allow frequency and polarization to be re-used to a greater extent than when larger beams are formed. For example, beams that are narrowly formed can service two discontinuous coverage areas that are non-overlapping. Each beam can use both a right hand polarization and a left hand polarization. Greater reuse can increase the amount of data transmitted and/or received.

[0085] Some satellites use on-board beamforming (OBBF) to electronically steer an array of antenna elements. FIG. 3 is an illustration of a satellite system 300 in which the satellite 302 has phased array multi-feed per beam (MFPB) on-board beamforming capability. In this example, the beam weights are computed at a ground based computation center and then transmitted to the satellite or pre-stored in the satellite for application (not shown). The forward link is shown in FIG. 3, although this architecture may be used for forward links, return links, or both forward and return links. Beamforming may be employed on the user link, the feeder link, or both. The illustrated forward link is the signal path from one of a plurality of gateways (GWs) 304 to one or more of a plurality of user terminals within one or more spot beam coverage areas 306. The satellite 302 has a receive antenna array 307, a transmit antenna array 309, a down-converter (D/C) and gain module 311, a receive beamformer 313, and a transmit beamformer 315. The satellite 302 can form beams on both the feeder link 308 and the user link 310. Each of the L elements of the receive array 307 receives K signals from the K GWs 304. For each of the K feeder link beams that are to be created (e.g., one beam per GW 304), a different beam weight is applied (e.g., a phase/amplitude adjustment is made) by the receive beamformer 313 to each signal received by each of the L receive antenna array elements (of receive antenna array 307). Accordingly, for K beams to be formed using a receive antenna array 307 having L receive antenna elements, K different beam weight vectors of length L are applied to the L signals received by the L receive antenna array elements. The receive beamformer 313 within the satellite 302 adjusts the phase/amplitude of the signals received by the L receive antenna array elements to create K receive beam signals. Each of the K receive beams are focused to receive a signal from one GW 304. Accordingly, the receive beamformer 313 outputs K receive beam signals to the D/C and gain module 311. One such receive beam signal is formed for the signal received from each transmitting GW 304.

[0086] The D/C and gain module 311 down-converts each of the K receive beam signals and adjusts the gain appropriately. K signals are output from the D/C and gain module 311 and coupled to the transmit beamformer 315. The transmit beamformer 315 applies a vector of L weights to each of the K signals for a total of $L \times K$ transmit beam weights to form K beams on the user downlink 310.

[0087] In some cases, significant processing capability may be needed within the satellite to control the phase and gain of each antenna element that is used to form the beams. Such processing power increases the complexity of the satellite. In some cases, satellites may operate with ground-based beamforming (GBBF) to reduce the complexity of the satellite while still providing the advantage of electronically forming narrow beams.

[0088] FIG. 4 is an illustration of one example of a satellite communication system 400 having forward GBBF. GBBF is performed on the forward user link 317 via an L element array similar to that described above. The phases/amplitudes of the signals transmitted on the user link 317 are weighted such that beams are formed. The feeder link 319 uses a Single Feed per Beam (SFPB)

scheme in which each receive and transmit antenna element of an antenna **324** is dedicated to one feeder link beam.

[0089] Prior to transmission from a GW or GWs **304**, for each of the K forward feeder link beams, a transmit beamformer **321** applies a respective one of K beam weight vectors, each of length L, to each of K signals to be transmitted. Determining the K vectors of L weights and applying them to the signals enables K forward beams to be formed on the ground for the forward user downlink **317**. On the feeder uplink **319**, each of the L different signals is multiplexed into a frequency division multiplexed (FDM) signal by a multiplexer **323** (or the like). Each FDM signal is transmitted by the GWs **304** to one of the receive antenna elements in the antenna **324** on the feeder link **319**. An FDM receiver **325** on the satellite **327** receives the signals from the antenna **324**. An analog to digital converter (A/D) **326** converts the received analog signals to digital signals. A digital channel processor **328** demultiplexes the FDM signals, each of which was appropriately weighted by the beamformer **321** for transmission through one of the L elements of an array of transmit antenna elements of a transmit antenna **329**. The digital channel processor **328** outputs the signals to a digital to analog converter (D/A) **331** to be converted back to analog form. The analog outputs of the D/A **331** are up-converted and amplified by an up-converter (U/C) and gain stage **330** and transmitted by the associated element of the transmit antenna **329**. A complimentary process occurs in reverse for the return beams. Note that in this type of system the FDM feeder link requires L times as much bandwidth as the user beams making it impractical for systems with wide data bandwidths or systems that have a large number of elements L.

End-to-End Beamforming Systems

[0090] The end-to-end beamforming systems described herein form end-to-end beams through an end-to-end relay. An end-to-end beamforming system can connect user terminals with data sources/sinks. In contrast to the beamforming systems discussed above, in an end-to-end beamforming system, beam weights are computed at a central processing system (CPS) and end-to-end beam weights are applied within the ground network (rather than at a satellite). The signals within the end-to-end beams are transmitted and received at an array of access nodes (ANs), which may be satellite access node (SANs). As described above, any suitable type of end-to-end relays can be used in an end-to-end beamforming system, and different types of ANs may be used to communicate with different types of end-to-end relays. The term “central” refers to the fact that the CPS is accessible to the ANs that are involved in signal transmission and/or reception, and does not refer to a particular geographic location at which the CPS resides. A beamformer within a CPS computes one set of end-to-end beam weights that accounts for: (1) the wireless signal uplink paths up to the end-to-end relay; (2) the receive/transmit signal paths through the end-to-end relay; and (3) the wireless signal downlink paths down from the end-to-end relay. The beam weights can be represented mathematically as a matrix. As discussed above, OBBF and GBBF satellite systems have beam weight vector dimensions set by the number of antenna elements on the satellite. In contrast, end-to-end beam weight vectors have dimensions set by the number of ANs, not the number of elements on the end-to-end relay. In general, the number of ANs is not the same as the number of antenna elements on the end-to-end relay. Further, the formed end-to-end beams are not terminated at either transmit or receive antenna elements of the end-to-end relay. Rather, the formed end-to-end beams are effectively relayed, since the end-to-end beams have uplink signal paths, relay signal paths (via a satellite or other suitable end-to-end relay), and downlink signal paths.

[0091] Because the end-to-end beamforming takes into account both the user link and the feeder link (as well as the end-to-end relay) only a single set of beam weights is needed to form the desired end-to-end user beams in a particular direction (e.g., forward user beams or return user beams). Thus, one set of end-to-end forward beam weights (hereafter referred to simply as forward beam weights) results in the signals transmitted from the ANs, through the forward uplink, through the end-to-end relay, and through the forward downlink to combine to form the end-to-end forward

user beams (hereafter referred to as forward user beams). Conversely, signals transmitted from return users through the return uplink, through the end-to-end relay, and the return downlink have end-to-end return beam weights (hereafter referred to as return beam weights) applied to form the end-to-end return user beams (hereafter referred to as return user beams). Under some conditions, it may be very difficult or impossible to distinguish between the characteristics of the uplink and the downlink. Accordingly, formed feeder link beams, formed user beam directivity, and individual uplink and downlink carrier to interference ratio (C/I) may no longer have their traditional role in the system design, while concepts of uplink and downlink signal-to-noise ratio (E_s/N_0) and end-to-end C/I may still be relevant.

[0092] FIG. 5 is an illustration of an example end-to-end beamforming system 500. The system 500 includes: a ground segment 502; an end-to-end relay 503; and a plurality of user terminals 517. The ground segment 502 comprises M ANs 515, spread geographically over an AN area. The ANs 515 cooperate in transmitting forward uplink signals 521 to form user beams 519 and return downlink signals 527 are collectively processed to recover return uplink transmissions 525. A set of ANs 515 that are within a distinct (e.g., geographically separated or otherwise orthogonally configured) AN area and cooperate to perform end-to-end beamforming for forward and/or return user beams is referred to herein as an “AN cluster.” In some examples, multiple AN clusters in different AN areas may also cooperate. AN clusters may also be referred to as “AN farms” or “SAN farms.” ANs 515 and user terminals 517 can be collectively referred to as Earth receivers, Earth transmitters, or Earth transceivers, depending upon the particular functionality at issue, since they are located on, or near, the Earth and both transmit and receive signals. In some cases, user terminals 517 and/or ANs 515 can be located in aircraft, watercraft or mounted on landcraft, etc. In some cases, the user terminals 517 can be geographically distributed. The ANs 515 can be geographically distributed. The ANs 515 exchange signals with a CPS 505 within the ground segment 502 via a distribution network 518. The CPS 505 is connected to a data source (not shown), such as, for example, the internet, a video headend or other such entity.

[0093] User terminals 517 may be grouped with other nearby user terminals 517 (e.g., as illustrated by user terminals 517a and 517b). In some cases, such groups of user terminals 517 are serviced by the same user beam and so reside within the same geographic forward and/or return user beam coverage area 519. A user terminal 517 is within a user beam if the user terminal 517 is within the coverage area serviced by that user beam. While only one such user beam coverage area 519 is shown in FIG. 5 to have more than one user terminal 517, in some cases, a user beam coverage area 519 can have any suitable number of user terminals 517. Furthermore, the depiction in FIG. 5 is not intended to indicate the relative size of different user beam coverage areas 519. That is, the user beam coverage areas 519 may all be approximately the same size. Alternatively, the user beam coverage areas 519 may be of varying sizes, with some user beam coverage areas 519 much larger than others. In some cases, the number of ANs 515 is not equal to the number of user beam coverage areas 519.

[0094] The end-to-end relay 503 relays signals wirelessly between the user terminals 517 and a number of network access nodes, such as the ANs 515 shown in FIG. 5. The end-to-end relay 503 has a plurality of signal paths. For example, each signal path can include at least one receive antenna element, at least one transmit antenna element, and at least one transponder (as is discussed in detail below). In some cases, the plurality of receive antenna elements are arranged to receive signals reflected by a receive reflector to form a receive antenna array. In some cases, the plurality of transmit antenna elements is arranged to transmit signals and thus to form a transmit antenna array.

[0095] In some cases, the end-to-end relay 503 is provided on a satellite. In other cases, the end-to-end relay 503 is provided on an aircraft, blimp, tower, underwater structure or any other suitable structure or vehicle in which an end-to-end relay 503 can reside. In some cases, the system uses different frequency ranges (in the same or different frequency bands) for the uplinks and

downlinks. In some cases, the feeder links and user links are in different frequency ranges. In some cases, the end-to-end relay **503** acts as a passive or active reflector.

[0096] As described herein, various features of the end-to-end relay **503** enable end-to-end beamforming. One feature is that the end-to-end relay **503** includes multiple transponders that, in the context of end-to-end beamforming systems, induce multipath between the ANs **515** and the user terminals **517**. Another feature is that the antennas (e.g., one or more antenna subsystems) of the end-to-end relay **503** contribute to end-to-end beamforming, so that forward and/or return user beams are formed when properly beam-weighted signals are communicated through the multipath induced by the end-to-end relay **503**. For example, during forward communications, each of multiple transponders receives a respective superposed composite of (beam weighted) forward uplink signals **521** from multiple (e.g., all) of the ANs **515** (referred to herein as composite input forward signals), and the transponders output corresponding composite signals (referred to herein as forward downlink signals). Each of the forward downlink signals can be a unique composite of the beam-weighted forward uplink signals **521**, which, when transmitted by the transmit antenna elements of the end-to-end relay **503**, superpose to form the user beams **519** in desired locations (e.g., recovery locations within forward user beams, in this case). Return end-to-end beamforming is similarly enabled. Thus, the end-to-end relay **503** can cause multiple superpositions to occur, thereby enabling end-to-end beamforming over induced multipath channels.

Return Data

[0097] FIG. **6** is an illustration of an example model of signal paths for signals carrying return data on the end-to-end return link. Return data is the data that flows from user terminals **517** to the ANs **515**. Signals in FIG. **6** flow from right to left. The signals originate with user terminals **517**. The user terminals **517** transmit return uplink signals **525** (which have return user data streams) up to the end-to-end relay **503**. Return uplink signals **525** from user terminals **517** in K user beam coverage areas **519** are received by an array of L receive/transmit signal paths **1702**. In some cases, an uplink coverage area for the end-to-end relay **503** is defined by that set of points from which all of the L receive antenna elements **406** can receive signals. In other cases, the relay coverage area is defined by that set of points from which a subset (e.g., a desired number more than 1, but less than all) of the L receive antenna elements **406** can receive signals. Similarly, in some cases, the downlink coverage area is defined by the set of points to which all of the L transmit antenna elements **409** can reliably send signals. In other cases, the downlink coverage area for the end-to-end relay **503** is defined as that set of points to which a subset of the transmit antenna elements **409** can reliably send signals. In some cases, the size of the subset of either receive antenna elements **406** or transmit antenna elements **409** is at least four. In other cases, the size of the subset is 6, 10, 20, 100, or any other number that provides the desired system performance.

[0098] For the sake of simplicity, some examples are described and/or illustrated as all L receive antenna elements **406** receiving signals from all points in the uplink coverage area and/or all L transmit antenna elements **409** transmitting to all points in the downlink coverage area. Such descriptions are not intended to require that all L elements receive and/or transmit signals at a significant signal level. For example, in some cases, a subset of the L receive antenna elements **406** receives an uplink signal (e.g., a return uplink signal **525** from a user terminal **517**, or a forward uplink signal **521** from an AN **515**), such that the subset of receive antenna elements **406** receives the uplink signal at a signal level that is close to a peak received signal level of the uplink signal (e.g., not substantially less than the signal level corresponding to the uplink signal having the highest signal level); others of the L receive antenna elements **406** that are not in the subset receive the uplink signal at an appreciably lower level (e.g., far below the peak received signal level of the uplink signal). In some cases, the uplink signal received by each receive antenna element of a subset is at a signal level within 10 dB of a maximum signal level received by any of the receive antenna elements **406**. In some cases, the subset includes at least 10% of the receive antenna elements **406**. In some cases, the subset includes at least 10 receive antenna elements **406**.

[0099] Similarly, on the transmit side, a subset of the L transmit antenna elements **409** transmits a downlink signal to an Earth receiver (e.g., a return downlink signal **527** to an AN **515**, or a forward downlink signal **522** to a user terminal **517**), such that the subset of transmit antenna elements **409** transmits the downlink signal to the receiver with a received signal level that is close to a peak transmitted signal level of the downlink signal (e.g., not substantially less than the signal level corresponding to the downlink signal having the highest received signal level); others of the L transmit antenna elements **409** that are not in the subset transmit the downlink signal such that it is received at an appreciably lower level (e.g., far below the peak transmitted signal level of the downlink signal). In some cases, the signal level is within 3 dB of a signal level corresponding to a peak gain of the transmit antenna element **409**. In other cases, the signal level is within 6 dB of the signal level corresponding to a peak gain of the transmit antenna element **409**. In yet other cases, the signal level is within 10 dB of the signal level corresponding to a peak gain of the transmit antenna element **409**.

[0100] In some cases, the signal received by each receive antenna element **406** originates at the same source (e.g., one of the user terminals **517**) due to overlap in the receive antenna pattern of each receive antenna element. However, in some cases, there may be points within the end-to-end relay coverage area at which a user terminal is located and from which not all of the receive antenna elements can receive the signal. In some such cases, there may be a significant number of receive antenna elements that do not (or cannot) receive the signal from user terminals that are within the end-to-end relay coverage area. However, as described herein, inducing multipath by the end-to-end relay **503** can rely on receiving the signal by at least two receive elements.

[0101] As shown in FIG. 6 and discussed in greater detail below, in some cases, a receive/transmit signal path **1702** comprises a receive antenna element **406**, a transponder **410**, and a transmit antenna element **409**. In such cases, the return uplink signals **525** are received by each of a plurality of transponders **410** via a respective receive antenna element **406**. The output of each receive/transmit signal path **1702** is a return downlink signal **527** corresponding to a respective composite of received return uplink signals. The return downlink signal is created by the receive/transmit signal path **1702**. The return downlink signal **527** is transmitted to the array of M ANs **515**. In some cases, the ANs **515** are placed at geographically distributed locations (e.g., reception or recovery locations) throughout the end-to-end relay coverage area. In some cases, each transponder **410** couples a respective one of the receive antenna elements **406** with a respective one of the transmit antenna elements **409**. Accordingly, there are L different ways for a signal to get from a user terminal **517** located in a user beam coverage area **519** to a particular AN **515**. This creates L paths between a user terminal **517** and an AN **515**. The L paths between one user terminal **517** and one AN **515** are referred to collectively as an end-to-end return multipath channel **1908** (see FIG. 8). Accordingly receiving the return uplink signal **525** from a transmission location within a user beam coverage area **519**, through the L transponders **410**, creates L return downlink signals **527**, each transmitted from one of the transponders **410** (i.e., through L collocated communication paths). Each end-to-end return multipath channel **1908** is associated with a vector in the uplink radiation matrix $A_{sub,r}$, the payload matrix E, and a vector in downlink radiation matrix C_t . Note that due to antenna element coverage patterns, in some cases, some of the L paths may have relatively little energy (e.g., 6 dB, 10 dB, 20 dB, 30 dB, or any other suitable power ratio less than other paths). A superposition **1706** of return downlink **527** signal is received at each of the ANs **515** (e.g., at M geographically distributed reception or recovery locations). Each return downlink signal **527** comprises a superposition of a plurality of the transmitted return downlink signals **527**, resulting in a respective composite return signal. The respective composite return signals are coupled to the return beamformer **531** (see FIGS. 5 and 29).

[0102] FIG. 7 illustrates an example end-to-end return link **523** from one user terminal **517** located within a user beam coverage area **519** to the ANs **515**. The return uplink signal **525** transmitted from the user terminal **517** is received by the array of L receive antenna elements **406** on the end-

to-end relay **503** (e.g., or received by a subset of the L receive antenna elements **406**).

[0103] A_r is the $L \times K$ return uplink radiation matrix. The values of the return uplink radiation matrix model the signal path from a reference location in the user beam coverage area **519** to the end-to-end relay receive antenna elements **406**. For example, $A_r.\text{sub}.L,1$ is the value of one element of the return uplink radiation matrix (i.e. the amplitude and phase of the path) from a reference location in the $1.\text{sup}.st$ user beam coverage area **519** to the $L.\text{sup}.th$ receive antenna element. In some cases, all of the values in the return uplink radiation matrix A_r may be non-zero (e.g., there is a significant signal path from the reference location to each of the receive antenna elements of the receive antenna array).

[0104] E (dimension $L \times L$) is the payload matrix and provides the model (amplitude and phase) of the paths from the receive antenna elements **406** to the transmit antenna elements **409**. A “payload” of an end-to-end relay **503**, as used herein, generally includes the set of components of the end-to-end relay **503** that affect, and/or are affected by, signal communications as they are received by, relayed through, and transmitted from the end-to-end relay **503**. For example, an end-to-end relay payload can include antenna elements, reflectors, transponders, etc.; but the end-to-end relay can further include batteries, solar cells, sensors, and/or other components not considered herein as part of the payload (since they do not affect signals when operating normally). Consideration of the set of components as a payload can enable mathematically modeling the overall impact of the end-to-end relay as a single payload matrix E). The predominant path from each receive antenna element **406** to each corresponding transmit antenna element **409** is modeled by the value that lies on the diagonal of the payload matrix E . Assuming there is no crosstalk between receive/transmit signal paths, the off-diagonal values of the payload matrix are zero. In some cases, the crosstalk may not be zero. Isolating the signal paths from each other will minimize crosstalk. In some cases, since the crosstalk is negligible, the payload matrix E can be estimated by a diagonal matrix. In some cases, the off-diagonal values (or any other suitable values) of the payload matrix can be treated as zero, even where there is some signal impact corresponding to those values, to reduce mathematical complexity and/or for other reasons.

[0105] C_t is the $M \times L$ return downlink radiation matrix. The values of the return downlink radiation matrix model the signal paths from the transmit antenna elements **409** to the ANs **515**. For example, $C_t.\text{sub}.3,2$ is the value of the return downlink radiation matrix (e.g., the gain and phase of the path) from the second transmit antenna element $409.\text{sub}.b$ to the third AN $515.\text{sub}.c$. In some cases, all of the values of the downlink radiation matrix C_t may be non-zero. In some cases, some of the values of the downlink radiation matrix C_t are essentially zero (e.g., the antenna pattern established by a corresponding transmit antenna elements **409** of the transmit antenna array is such that the transmit antenna element **409** does not transmit useful signals to some of the ANs **515**).

[0106] As can be seen in FIG. 7, the end-to-end return multipath channel from a user terminal **517** in a particular user beam coverage area **519** to a particular AN **515** is the sum of the L different paths. The end-to-end return multipath channel has multipath induced by the L unique paths through the transponders **410** in the end-to-end relay. As with many multipath channels, the paths' amplitudes and phases can add up favorably (constructively) to produce a large end-to-end channel gain or unfavorably (destructively) to produce a low end-to-end channel gain. When the number of different paths, L , between a user terminal and an AN is large, the end-to-end channel gain can have a Rayleigh distribution of the amplitude. With such a distribution, it is not uncommon to see some end-to-end channel gains from a particular user terminal **517** to a particular AN **515** that are 20 dB or more below the average level of the channel gain from a user terminal **517** to an AN **515**. This end-to-end beamforming system intentionally induces a multipath environment for the end-to-end path from any user terminal to any AN.

[0107] FIG. 8 is a simplified illustration of an example model of all the end-to-end return multipath channels from user beam coverage areas **519** to ANs **515**. There are $M \times K$ such end-to-end return multipath channels in the end-to-end return link (i.e., M from each of the K user beam coverage

areas **519**). Channels **1908** connect user terminals in one user beam coverage area **519** to one AN **515** over L different receive/transmit signal paths **1702**, each path going through a different one of the L receive/transmit signal paths (and associated transponders) of the relay. While this effect is referred to as “multipath” herein, this multipath differs from conventional multipath (e.g., in a mobile radio or multiple-input multiple-output (MIMO) system), as the multiple paths herein are intentionally induced (and, as described herein, affected) by the L receive/transmit signal paths. Each of the $M \times K$ end-to-end return multipath channels that originate from a user terminal **517** within a particular user beam coverage area **519** can be modeled by an end-to-end return multipath channel. Each such end-to-end return multipath channel is from a reference (or recovery) location within the user beam coverage area **519** to one of the ANs **515**.

[0108] Each of the $M \times K$ end-to-end return multipath channels **1908** may be individually modeled to compute a corresponding element of an $M \times K$ return channel matrix H_{ret} . The return channel matrix H_{ret} has K vectors, each having dimensionality equal to M , such that each vector models the end-to-end return channel gains for multipath communications between a reference location in one of a respective K user beam coverage areas and the M ANs **515**. Each end-to-end return multipath channel couples one of the M ANs **515** with a reference location within one of K return user beams via L transponders **410** (see FIG. 7). In some cases, only a subset of the L transponders **410** on the end-to-end relay **503** is used to create the end-to-end return multipath channel (e.g., only a subset is considered to be in the signal path by contributing significant energy to the end-to-end return multipath channel). In some cases, the number of user beams K is greater than the number of transponders L that is in the signal path of the end-to-end return multipath channel. Furthermore, in some cases, the number of ANs M is greater than the number of transponders L that is in the signal path of the end-to-end return multipath channel **1908**. In an example, the element $H_{ret.sub.4,2}$ of the return channel matrix H_{ret} is associated with the channel from a reference location in the second user beam coverage area **1903** to the fourth AN **1901**. The matrix H_{ret} models the end-to-end channel as the product of the matrices $C_t \times E \times A_r$ (see FIG. 6). Each element in H_{ret} models the end-to-end gain of one end-to-end return multipath channel **1908**. Due to the multipath nature of the channel, the channel can be subject to a deep fade. Return user beams may be formed by the CPS **505**. The CPS **505** computes return beam weights based on the model of these $M \times K$ signal paths and forms the return user beams by applying the return beam weights to the plurality of composite return signals, each weight being computed for each end-to-end return multipath channel that couples the user terminals **517** in one user beam coverage area with one of the plurality of ANs **515**. In some cases, the return beam weights are computed before receiving the composite return signal. There is one end-to-end return link from each of the K user beam coverage areas **519** to the M ANs **515**. The weighting (i.e., the complex relative phase/amplitude) of each of the signals received by the M ANs **515** allows those signals to be combined to form a return user beam using the beamforming capability of the CPS **505** within the ground segment **502**. The computation of the beam weight matrix is used to determine how to weight each end-to-end return multipath channel **1908**, to form the plurality of return user beams, as described in more detail below. User beams are not formed by directly adjusting the relative phase and amplitude of the signals transmitted by one end-to-end relay antenna element with respect to the phase and amplitude of the signals transmitted by the other end-to-end relay antenna elements. Rather, user beams are formed by applying the weights associated with the $M \times K$ channel matrix to the M AN signals. It is the plurality of ANs that provide the receive path diversity, single transmitter (user terminal) to multiple receivers (ANs), to enable the successful transmission of information from any user terminal in the presence of the intentionally induced multipath channel.

Forward Data

[0109] FIG. 9 is an illustration of an example model of signal paths for signals carrying forward data on the end-to-end forward link **501**. Forward data is the data that flows from ANs **515** to user terminals **517**. Signals in this figure flow from right to left. The signals originate with M ANs **515**,

which are located in the footprint of the end-to-end relay **503**. There are K user beam coverage areas **519**. Signals from each AN **515** are relayed by L receive/transmit signal paths **2001**.

[0110] The receive/transmit signal paths **2001** transmit a relayed signal to user terminals **517** in user beam coverage areas **519**. Accordingly, there may be L different ways for a signal to get from a particular AN **515** to a user terminal **517** located in a user beam coverage area **519**. This creates L paths between each AN **515** and each user terminal **517**. Note that due to antenna element coverage patterns, some of the L paths may have less energy than other paths.

[0111] FIG. **10** illustrates an example end-to-end forward link **501** that couples a plurality of access nodes at geographically distributed locations with a user terminal **517** in a user beam (e.g., located at a recovery location within a user beam coverage area **519**) via an end-to-end relay **503**. In some cases, the forward data signal is received at a beamformer prior to generating forward uplink signals. A plurality of forward uplink signals is generated at the beamformer and communicated to the plurality of ANs **515**. For example, each AN **515** receives a unique (beam weighted) forward uplink signal generated according to beam weights corresponding to that AN **515**. Each AN **515** has an output that transmits a forward uplink signal via one of M uplinks. Each forward uplink signal comprises a forward data signal associated with the forward user beam. The forward data signal is “associated with” the forward user beam, since it is intended to be received by user terminals **517** serviced by the user beam. In some cases, the forward data signal comprises two or more user data streams. The user data streams can be multiplexed together by time-division or frequency-division multiplexing, etc. In some cases, each user data stream is for transmission to one or more of a plurality of user terminals within the same forward user beam.

[0112] As is discussed in greater detail below, each forward uplink signal is transmitted in a time-synchronized manner by its respective transmitting AN **515**. The forward uplink signals **521** transmitted from the ANs **515** are received by a plurality of transponders **410** on the end-to-end relay **503** via receive antenna elements **406** on the end-to-end relay **503**. The superposition **550** of the forward uplink signals **521** received from geographically distributed locations creates a composite input forward signal **545**. Each transponder **410** concurrently receives a composite input forward signal **545**. However, each transponder **410** will receive the signals with slightly different timing due to the differences in the location of the receive antenna element **406** associated with each transponder **401**.

[0113] Cr is the $L \times M$ forward uplink radiation matrix. The values of the forward uplink radiation matrix model the signal path (amplitude and phase) from the ANs **515** to the receive antenna elements **406**. E is the $L \times L$ payload matrix and provides the model of the transponder signal paths from the receive antenna elements **406** to the transmit antenna elements **409**. The direct path gain from each receive antenna element **406** through a corresponding one of a plurality of transponders to each corresponding transmit antenna element **409** is modeled by the diagonal values of the payload matrix. As noted above with respect to the return link, assuming there is no cross-talk between antenna elements, the off-diagonal elements of the payload matrix are zero. In some cases, the crosstalk may not be zero. Isolating the signal paths from each other will minimize crosstalk. In this example, each of the transponders **410** couples a respective one of the receive antenna elements **406** with a respective one of the transmit antenna elements **409**. Accordingly, a forward downlink signal **522** output from each of the transponders **410** is transmitted by each of the plurality of transponders **410** (see FIG. **9**) via the transmit antenna elements **409**, such that the forward downlink signals **522** form a forward user beam (by constructively and destructively superposing in desired geographic recovery locations to form the beam). In some cases, a plurality of user beams is formed, each corresponding to a geographic user beam coverage area **519** that services a respective set of user terminals **517** within the user beam coverage area **519**. The path from the first transmit antenna element **409a** (see FIG. **10**) to a reference (or recovery) location in the first user beam coverage area **519** is given in the At.sub.11 value of the forward downlink radiation matrix. As noted with regard to the return link, this end-to-end beamforming system intentionally induces a

multipath environment for the end-to-end path from any AN 515 to any user terminal 517. In some cases, a subset of the transmit antenna elements 409 transmits forward downlink signals 522 with significant energy to a user terminal 517. The user terminal 517 (or, more generally, a reference or recovery location in the user beam coverage area 519 for receiving and/or recovery) receives the plurality of forward downlink signals 522 and recovers at least a portion of the forward data signal from the received plurality of forward downlink signals 522. The transmitted forward downlink signals 522 may be received by the user terminal 517 at a signal level that is within 10 dB of a maximum signal level from any of the other signals transmitted by the transmit antenna elements 409 within the subset. In some cases, the subset of transmit antenna elements includes at least 10% of the plurality of transmit antenna elements present in the end-to-end relay 503. In some cases, the subset of transmit antenna elements include at least 10 transmit antenna elements, regardless of how many transmit antenna elements 409 are present in the end-to-end relay 503. In one case, receiving the plurality of forward downlink signals comprises receiving a superposition 551 of the plurality of forward downlink signals.

[0114] FIG. 11 is a simplified illustration of a model of all the end-to-end forward multipath channels 2208 from the M ANs 515 to the K user beam coverage areas 519. As shown in FIG. 11, there is an end-to-end forward multipath channel 2208 that couples each AN 515 to each user beam coverage area 519. Each channel 2208 from one AN 515 to one user beam coverage area 519 has multipath induced as a result of L unique paths from the AN 515 through the plurality of transponders to the user beam coverage area 519. As such, the $K \times M$ multipath channels 2208 may be individually modeled and the model of each serves as an element of a $K \times M$ forward channel matrix H_{fwd} . The forward channel matrix H_{fwd} has M vectors, each having dimensionality equal to K, such that each vector models the end-to-end forward gains for multipath communications between a respective one of the M ANs 515 and reference (or recovery) locations in K forward user beam coverage areas. Each end-to-end forward multipath channel couples one of the M ANs 515 with user terminals 517 serviced by one of K forward user beams via L transponders 410 (see FIG. 10). In some cases, only a subset of the L transponders 410 on the end-to-end relay 503 are used to create the end-to-end forward multipath channel (i.e., are in the signal path of the end-to-end forward multipath channel). In some cases, the number of user beams K is greater than the number of transponders L that are in the signal path of the end-to-end forward multipath channel. Furthermore, in some cases, the number of ANs M is greater than the number of transponders L that are in the signal path of the end-to-end forward multipath channel.

[0115] H_{fwd} may represent the end-to-end forward link as the product of matrices $A_t \times E \times C_r$. Each element in H_{fwd} is the end-to-end forward gain due to the multipath nature of the path and can be subject to a deep fade. An appropriate beam weight may be computed for each of the plurality of end-to-end forward multipath channels 2208 by the CPS 505 within the ground segment 502 to form forward user beams from the set of M ANs 515 to each user beam coverage area 519. The plurality of ANs 515 provide transmit path diversity, by using multiple transmitters (ANs) to a single receiver (user terminal), to enable the successful transmission of information to any user terminal 517 in the presence of the intentionally induced multipath channel.

Combined Forward and Return Data

[0116] FIG. 12 illustrates an example end-to-end relay supporting both forward and return communications. In some cases, the same end-to-end relay signal paths (e.g., set of receive antenna elements, transponders, and transmit antenna elements) may be used for both the end-to-end forward link 501 and the end-to-end return link 523. Some other cases include forward link transponders and return link transponders, which may or may not share receive and transmit antenna elements. In some cases, the system 1200 has a plurality of ANs and user terminals that are located in the same general geographic region 1208 (which may be, for example, a particular state, an entire country, a region, an entire visible area, or any other suitable geographic region 1208). A single end-to-end relay 1202 (disposed on a satellite or any other suitable end-to-end relay)

receives forward uplink signals **521** from ANs and transmits forward downlink signals **522** to user terminals. At alternate times, or on alternate frequencies, the end-to-end relay **1202** also receives return uplink signals **525** from the user terminals and transmits return downlink signals **527** to the ANs. In some cases, the end-to-end relay **1202** is shared between forward and return data using techniques such as time domain duplexing, frequency domain duplexing, and the like. In some cases, time domain duplexing between forward and return data uses the same frequency range: forward data is transmitted during different (non-overlapping) time intervals than those used for transmitting return data. In some cases, with frequency domain duplexing, different frequencies are used for forward data and return data, thereby permitting concurrent, non-interfering transmission of forward and return data.

[0117] FIG. **13** is an illustration of an uplink frequency range being divided into two portions. The lower-frequency (left) portion of the range is allocated to the forward uplink and the upper-frequency (right) portion of the range is allocated to the return uplink. The uplink range may be divided into multiple portions of either forward or return data.

[0118] FIG. **14** is an illustration of the forward data and return data being time division multiplexed. A data frame period is shown in which forward data is transported during the first time interval of the frame, while return data is transported during the last time interval of the frame. The end-to-end relay receives from one or more access nodes during a first (forward) receive time interval and from one or more user terminals during a second (return) receive time interval that doesn't overlap the first receive time interval. The end-to-end relay transmits to one or more user terminals during a first (forward) transmit time interval and to one or more access nodes during a second (return) transmit time interval that doesn't overlap the first receive time interval. The data frame may be repeated or may change dynamically. The frame may be divided into multiple (e.g., non-contiguous) portions for forward and return data.

End-to-End Beamforming Satellites

[0119] In some cases, the end-to-end relay **503** is implemented on a satellite, so that the satellite is used to relay the signals from the ANs (which can be referred to as satellite access nodes (SANs) in such cases) to the user terminals and vice versa. In some cases, the satellite is in geostationary orbit. An example satellite operating as an end-to-end relay has an array of receive antenna elements, an array of transmit antenna elements, and a number of transponders that connect the receive antenna elements to the transmit antenna elements. The arrays have a large number of antenna elements with overlapping antenna element coverage areas, similar to traditional single link phased array antennas. It is the overlapping antenna element coverage areas on both the transmit antenna elements and receive antenna elements that create the multipath environment previously described. In some cases, the antenna patterns established by the corresponding antenna elements, and those that result in the overlapping antenna element coverage areas (e.g., overlapping component beam antenna patterns), are identical. For the purposes of this disclosure, the term “identical” means that they follow essentially the same distribution of power over a given set of points in space, taking the antenna element as the point of reference for locating the points in space. It is very difficult to be perfectly identical. Therefore, patterns that have relatively small deviations from one pattern to another are within the scope of “identical” patterns. In other cases, receive component beam antenna patterns may not be identical, and in fact may be significantly different. Such antenna patterns may yet result in overlapping antenna element coverage areas, however, those resulting coverage areas will not be identical.

[0120] Antenna types include, but are not limited to, array fed reflectors, confocal arrays, direct radiating arrays and other forms of antenna arrays. Each antenna can be a system including additional optical components to aid in the receipt and/or transmission of signals, such as one or more reflectors. In some cases, a satellite includes components that assist in system timing alignment and beamforming calibration.

[0121] FIG. **15** is a diagram of an example satellite **1502** that can be used as an end-to-end relay

503. In some cases, the satellite **1502** has an array fed reflector transmit antenna **401** and an array fed reflector receive antenna **402**. The receive antenna **402** comprises a receive reflector (not shown) and an array of receive antenna elements **406**. The receive antenna elements **406** are illuminated by the receive reflector. The transmit antenna **401** comprises a transmit reflector (not shown) and an array of transmit antenna elements **409**. The transmit antenna elements **409** are arranged to illuminate the transmit reflector. In some cases, the same reflector is used for both receive and transmit. In some cases, one port of the antenna element is used for receiving and another port for transmission. Some antennas have the ability to distinguish between signals of different polarizations. For example, an antenna element can include four waveguide ports for right-hand circular polarization (RHCP) receive, left-hand circular polarization (LHCP) receive, RHCP transmit, and LHCP transmit, respectively. In some cases, dual polarizations may be used to increase capacity of the system; in other cases, single polarization may be used to reduce interference (e.g., with other systems using a different polarization).

[0122] The example satellite **1502** also comprises a plurality of transponders **410**. A transponder **410** connects the output from one receive antenna element **406** to the input of a transmit antenna element **409**. In some cases, the transponder **410** amplifies the received signal. Each receive antenna element outputs a unique received signal. In some cases, a subset of receive antenna elements **406** receive a signal from an Earth transmitter, such as either a user terminal **517** in the case of a return link signal or an AN **515** in the case of a forward link signal. In some of these cases, the gain of each receive antenna element in the subset for the received signal is within a relatively small range. In some cases, the range is 3 dB. In other cases, the range is 6 dB. In yet other cases, the range is 10 dB. Accordingly, the satellite will receive a signal at each of a plurality of receive antenna elements **406** of the satellite, the communication signal originating from an Earth transmitter, such that a subset of the receive antenna elements **406** receives the communication signal at a signal level that is not substantially less than a signal level corresponding to a peak gain of the receive antenna element **406**.

[0123] In some cases, at least 10 transponders **410** are provided within the satellite **1502**. In another case, at least 100 transponders **410** are provided in the satellite **1502**. In yet another case, the number of transponders per polarity may be in the range of 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024 or numbers in-between or greater. In some cases, the transponder **410** includes a low noise amplifier (LNA) **412**, a frequency converter and associated filters **414** and a power amplifier (PA) **420**. In some cases in which the uplink frequency and downlink frequency are the same, the transponder does not include a frequency converter. In other cases, the plurality of receive antenna elements operate at a first frequency. Each receive antenna element **406** is associated with one transponder **410**. The receive antenna element **406** is coupled to the input of the LNA **412**. Accordingly, the LNA independently amplifies the unique received signal provided by the receive antenna element associated with the transponder **410**. In some cases, the output of the LNA **412** is coupled to the frequency converter **414**. The frequency converter **414** converts the amplified signal to a second frequency.

[0124] The output of the transponder is coupled to an associated one of the transmit antenna elements. In these examples, there is a one to one relationship between a transponder **410**, an associated receive antenna element **406**, and an associated transmit antenna element **409**, such that the output of each receive antenna element **406** is connected to the input of one and only one transponder and the output of that transponder is connected to the input of one and only one transmit antenna element.

[0125] FIG. **16** is an illustration of an example transponder **410**. The transponder **410** can be an example of a transponder of an end-to-end relay **503**, as described above (e.g., the satellite **1502** of FIG. **15**). In this example, the transponder includes a phase shifter **418** in addition to the LNA **412**, frequency converter and associated filters **414**, and power amplifier (PA) of transponder **410**. As illustrated in FIG. **16**, the example transponder **410** can also be coupled with a phase shift

controller **427**. For example, the phase shift controller **427** can be coupled (directly or indirectly) with each of some or all of the transponders of an end-to-end relay **503**, so that the phase shift controller **427** can individually set the phases for each transponder. The phase shifters may be helpful for calibration, for example, as discussed below.

Antennas

[0126] To create the multipath environment, antenna element coverage areas can overlap with antenna element coverage areas of at least one other antenna element of the same polarity, frequency, and type (transmit or receive, respectively). In some cases, a plurality of receive component beam antenna patterns, operable at the same receive polarization and receive frequency (e.g., having at least a portion of the receive frequency in common), overlap with one another. For example, in some cases, at least 25% of the receive component beam antenna patterns, operable at the same receive polarization and receive frequency (e.g., having at least a portion of the receive frequency in common), overlap with at least five other receive component beam antenna patterns of the receive antenna elements. Similarly, in some cases, at least 25% of the transmit component beam antenna patterns, operable at the same transmit polarization and transmit frequency (e.g., having at least a portion of the transmit frequency in common), overlap with at least five other transmit component beam antenna patterns. The amount of overlap will vary from system to system. In some cases, at least one of the receive antenna elements **406** has component beam antenna patterns that overlap with the antenna patterns of other receive antenna elements **406** operable at the same receive frequency (e.g., having at least a portion of the receive frequency in common) and same receive polarization. Therefore, at least some of the plurality of receive antenna elements are capable of receiving the same signals from the same source. Similarly, at least one of the transmit antenna elements **409** has a component beam antenna pattern that overlaps with the antenna patterns of other transmit antenna elements **409** operable at the same transmit frequency (e.g., having at least a portion of the transmit frequency in common) and transmit polarization. Therefore, at least some of the plurality of transmit antenna elements are capable of transmitting signals having the same frequency at the same polarization to the same receiver. In some cases, overlapping component beam antenna patterns may have gains that differ by less than 3 dB (or any other suitable value) over a common geographic area. The antenna elements, whether receive or transmit, may have a broad component beam antenna pattern, and thus a relatively broad antenna element coverage area. In some cases, signals transmitted by an Earth transmitter, such as a user terminal **517** or access node **515**, are received by all of the receive antenna elements **406** of the end-to-end relay (e.g., satellite). In some cases, a subset of the elements **406** receives the signals from an Earth transmitter. In some cases, the subset includes at least 50% of the receive antenna elements. In other cases, the subset includes at least 75% of the receive antenna elements. In still other cases, the subset includes at least 90% (e.g., up to and including all) of the receive antenna elements. Different subsets of the receive antenna elements **406** may receive signals from different Earth transmitters. Similarly, in some cases, a subset of the elements **409** transmits signals that may be received by a user terminal **517**. In some cases, the subset includes at least 50% of the transmit antenna elements. In other cases, the subset includes at least 75% of the transmit antenna elements. In still other cases, the subset includes at least 90% (e.g., up to and including all) of the transmit antenna elements. Different subsets of the elements **409** may transmit signals that are received by different user terminals. Furthermore, user terminals may be within several formed user beam coverage areas **519**. For the purpose of this disclosure, an antenna pattern is a pattern of distribution of energy transmitted to, or received from, an antenna. In some cases, the energy may be directly radiated from/to the antenna element. In other cases, the energy from one or more transmit antenna elements may be reflected by one or more reflectors that shape the antenna element pattern. Similarly, a receive element may receive energy directly, or after the energy has reflected off one or more reflectors. In some cases, antennas can be made up of several elements, each having a component beam antenna pattern that establishes a corresponding antenna element coverage area.

Similarly, all or a subset of receive and transmit antenna elements that receive and transmit signals to ANs **515** may overlap, such that a plurality of receive antenna elements receives signals from the same AN **515** and/or a plurality of transmit antenna elements transmits signals to the same AN **515**. [0127] FIG. **17** is an illustration of component beam antenna patterns produced by several antenna elements (either receive antenna elements **406**, or transmit antenna elements **409**) that intersect at the 3 dB points. The component beam antenna pattern **1301** of a first antenna element has peak component beam antenna gain along the boresight **1303**. The component beam antenna pattern **1301** is shown to attenuate about 3 dB before it intersects with the component beam antenna pattern **1305**. Since each pair of two adjacent component beam antenna patterns overlap about the 3 dB line **1307** for only a relatively small portion of the component beam antenna pattern, the antenna elements that produce these component beam antenna patterns are considered not to be overlapping.

[0128] FIG. **18** shows idealized 3 dB antenna contours **3901**, **3902**, **3903** of several elements **406**, **409** with the peak gain designated with the letter 'x'. The contours **3901**, **3902**, **3903** are referred to herein as "idealized" because the contours are shown as circular for the sake of simplicity.

However, the contours **3901**, **3902**, **3903** need not be circular. Each contour indicates the place at which the transmitted or received signal is 3 dB below the peak level. Outside the contour, the signal is more than 3 dB below the peak. Inside the contour, the signal is less than 3 dB below the peak (i.e., within 3 dB of the peak). In a system in which the coverage area of a receive component beam antenna pattern is all points for which the receive component beam antenna gain is within 3 dB of peak receive component beam antenna gain, the area inside the contour is referred to as the antenna element coverage area. The 3 dB antenna contour for each element **406**, **409** is not overlapping. That is, only a relatively small portion of the area inside the 3 dB antenna contour **3901** overlaps with the area that is inside the adjacent 3 dB antenna patterns **3902**, **3903**.

[0129] FIG. **19** is an illustration of the antenna patterns **1411**, **1413**, **1415** of several antenna elements (either receive antenna elements **406** or transmit antenna elements **409**). In contrast to the component beam antenna patterns of FIG. **17**, the component beam antenna patterns shown in FIG. **19** intersect **1417** above the 3 dB line **1307**.

[0130] FIG. **20A** through FIG. **20E** illustrate 3 dB antenna contours for several antenna elements **406**, **409** with the beam center point (peak gain) designated with the letter 'x'. FIG. **20A** shows the particular antenna contour **1411** of a first antenna element **406**. FIG. **20B** shows the 3 dB antenna contours **1411**, **1413** for two particular elements **406**. FIG. **20C** shows the 3 dB antenna contours for three elements **406**. FIG. **20D** shows the 3 dB antenna contours for four antenna elements **406**. FIG. **20E** shows the 3 dB antenna contours for an array of 16 antenna elements **406**. The 3 dB antenna contours are shown to overlap **1418** (e.g., 16 such 3 dB antenna contours are shown). The antenna elements in either the receive or transmit antenna may be arranged in any of several different configurations. For example, if elements have a generally circular feed horn, the elements may be arranged in a honeycomb configuration to tightly pack the elements in a small amount of space. In some cases, the antenna elements are aligned in horizontal rows and vertical columns.

[0131] FIG. **21** is an example illustration of relative positions of receive antenna 3 dB antenna contours associated with receive antenna elements **406**. The element **406** beam centers are numbered 1-16, with element **406.sub.4** identified by the number '4' to the upper left of the beam center indicator 'x'. In some cases, there may be many more than 16 receive antenna elements **406**. However, for the sake of simplicity, only 16 are shown in FIG. **21**. A corresponding array of transmit antenna elements **409** and their associated 3 dB antenna contours will look similar to FIG. **21**. Therefore, for the sake of simplicity, only the array of receive antenna elements **406** are shown. The area **2101** in the center is where all of the antenna element coverage areas overlap.

[0132] In some cases, at least one point within the relay coverage area (e.g., satellite coverage area) falls within the 3 dB antenna contour of the component beams of several antenna elements **406**. In one such case, at least one point is within the 3 dB antenna contour of at least 100 different antenna

elements **406**. In another case, at least 10% of the relay coverage area lies within the 3 dB antenna contours of at least 30 different antenna elements. In another case, at least 20% of the relay coverage area lies within the 3 dB antenna contours of at least 20 different antenna elements. In another case, at least 30% of the relay coverage area lies within the 3 dB antenna contours of at least 10 different antenna elements. In another case, at least 40% of the relay coverage area lies within the 3 dB antenna contours of at least eight different antenna elements. In another case, at least 50% of the relay coverage area lies within the 3 dB antenna contours of at least four different antenna elements. However, in some cases, more than one of these relationships may be true.

[0133] In some cases, the end-to-end relay has a relay coverage area (e.g., satellite coverage area) in which at least 25% of the points in the uplink relay coverage area are within (e.g., span) overlapping coverage areas of at least six receive antenna elements **406**. In some cases, 25% of the points within the uplink relay coverage area are within (e.g., span) overlapping coverage areas of at least four receive antenna elements **406**. In some cases, the end-to-end relay has a coverage area in which at least 25% of the points in the downlink relay coverage area are within (e.g., span) overlapping coverage areas of at least six transmit antenna elements **409**. In some cases, 25% of the points within the downlink relay coverage area are within (e.g., span) overlapping coverage areas of at least four transmit antenna elements **409**.

[0134] In some cases, the receive antenna **402** may be pointed roughly at the same coverage area as the transmit antenna **401**, so that some receive antenna element coverage areas may naturally correspond to particular transmit antenna element coverage areas. In these cases, the receive antenna elements **406** may be mapped to their corresponding transmit antenna elements **409** via the transponders **410**, yielding similar transmit and receive antenna element coverage areas for each receive/transmit signal path. In some cases, however, it may be advantageous to map receive antenna elements **406** to transmit antenna elements **409** that do not correspond to the same component beam coverage area. Accordingly, the mapping of the elements **406** of the receive antenna **402** to the elements **409** of the transmit antenna **401** may be randomly (or otherwise) permuted. Such permutation includes the case that results in the receive antenna elements **406** not being mapped to the transmit antenna elements **409** in the same relative location within the array or that have the same coverage area. For example, each receive antenna element **406** within the receive antenna element array may be associated with the same transponder **410** as the transmit antenna element **409** located in the mirror location of the transmit antenna element array. Any other permutation can be used to map the receive antenna elements **406** to the transmit antenna elements **409** according to a permutation (e.g., pair each receive antenna element **406** with the same transponder to which an associated transmit antenna element **409** is coupled in accordance with a particular permutation of the receive antenna element **406** and the transmit antenna element **409**).

[0135] FIG. 22 is a table **4200** showing example mappings of receive antenna elements **406** to transmit antenna elements **409** through 16 transponders **410**. Each transponder **410** has an input that is exclusively coupled to an associated receive antenna element **406** and an output that is exclusively coupled to an associated transmit antenna element **409** (e.g., there is a one to one relationship between each receive antenna element **406**, one transponder **410** and one transmit antenna element **409**). In some cases, other receive antenna elements, transponders and transmit antenna elements may be present on the end-to-end relay (e.g., satellite) that are not configured in a one to one relationship (and do not operate as a part of the end-to-end beamforming system).

[0136] The first column **4202** of the table **4200** identifies a transponder **410**. The second column **4204** identifies a receive antenna element **406** to which the transponder **410** of the first column is coupled. The third column **4206** of the table **4200** identifies an associated transmit antenna element **409** to which the output of the transponder **410** is coupled. Each receive antenna element **406** is coupled to the input of the transponder **410** identified in the same row of the table **4200**. Similarly, each transmit antenna element **409** is coupled to the output of the transponder **410** identified in the same row of the table **4200**. The third column of the table **4200** shows an example of direct

mapping in which each receive antenna element **406** of the receive antenna array is coupled to the same transponder **410** as a transmit antenna element **409** in the same relative location within the transmit antenna array. The fourth column **4208** of table **4200** shows an example of interleaved mapping in which the first receive antenna element **406** is coupled to the first transponder **410** and to the tenth transmit antenna element **409**. The second receive antenna element **406** is coupled to the second transponder **410** and to the ninth transmit antenna element **409**, and so on. Some cases have other permutations, including a random mapping in which the particular pairing of the receive antenna element **406** and the transmit element **409** with a transponder **410** are randomly selected. [0137] The direct mapping, which attempts to keep the transmit and receive antenna element coverage areas as similar as possible for each receive/transmit signal path, generally yields the highest total capacity of the system. Random and interleaved permutations generally produce slightly less capacity but provide a more robust system in the face of AN outages, fiber outages in the terrestrial network, or loss of receive/transmit signal paths due to electronic failure on the end-to-end relay (e.g., in one or more transponders). Random and interleaved permutations allow lower cost non-redundant ANs to be used. Random and interleaved permutations also provide less variation between the capacity in the best performing beam and the capacity in the worst performing beam. Random and interleaved permutations may also be more useful to initially operate the system with just a fraction of the ANs resulting in only a fraction of the total capacity being available but no loss in coverage area. An example of this is an incremental rollout of ANs, where the system was initially operated with only 50% of the ANs deployed. This may provide less than the full capacity, while still allowing operation over the entire coverage area. As the demand increases, more ANs can be deployed to increase the capacity until the full capacity is achieved with all the ANs active. In some cases, a change in the composition of the ANs results in a recalculation of the beam weights. A change in composition may include changing the number or characteristics of one or more ANs. This may require a re-estimation of the end-to-end forward and/or return gains.

[0138] In some cases, the antenna is an array-fed reflector antenna with a paraboloid reflector. In other cases, the reflector does not have a paraboloid shape. An array of receive antenna elements **406** may be arranged to receive signals reflected by the reflector. Similarly, an array of transmit antenna elements **409** may be arranged to form an array for illuminating the reflector. One way to provide elements with overlapping component beam antenna patterns is to have the elements **406**, **409** defocused (unfocused) as a consequence of the focal plane of the reflector being behind (or in front of) the array of elements **406**, **409** (i.e., the receive antenna array being located outside the focal plane of the receive reflector).

[0139] FIG. **23** is an illustration of a cross-section of a center-fed paraboloid reflector **1521**. A focal point **1523** lies on a focal plane **1525** that is normal to the central axis **1527** of the reflector **1521**. Received signals that strike the reflector **1521** parallel to the central axis **1527** are focused onto the focal point **1523**. Likewise, signals that are transmitted from an antenna element located at the focal point and that strike the reflector **1521** will be reflected in a focused beam from the reflector **1521** parallel to the central axis **1527**. Such an arrangement is often used in Single Feed per Beam systems to maximize the directivity of each beam and minimize overlap with beams formed by adjacent feeds.

[0140] FIG. **24** is an illustration of another paraboloid reflector **1621**. By locating antenna elements **1629** (either receive antenna elements or transmit antenna elements **406**, **409**, **3416**, **3419**, **3426**, **3429**) outside the focal plane (e.g., in front of the focal plane **1625** of the reflector **1621**), the path of transmitted signals **1631** that strike the reflector **1621** will not be parallel to one another as they reflect off the reflector **1621**, resulting in a wider beam width than in the focused case. In some cases, reflectors that have shapes other than paraboloids are used. Such reflectors may also result in defocusing the antenna. The end-to-end beamforming system may use this type of defocused antenna to create overlap in the coverage area of adjacent antenna elements and thus provide a

large number of useful receive/transmit paths for given beam locations in the relay coverage area. [0141] In one case, a relay coverage area is established, in which 25% of the points within the relay coverage area are within the antenna element coverage areas of at least six component beam antenna patterns when the end-to-end relay is deployed (e.g., an end-to-end satellite relay is in a service orbit). Alternatively, 25% of the points within the relay coverage area are within the antenna element coverage areas of at least four receive antenna elements. FIG. 25 is an illustration of an example relay coverage area (for an end-to-end satellite relay, also referred to as satellite coverage area) **3201** (shown with single cross-hatching) and the area **3203** (shown with double cross-hatching) defined by the points within the relay coverage area **3201** that are also contained within six antenna element coverage areas **3205, 3207, 3209, 3211, 3213, 3215**. The coverage area **3201** and the antenna element coverage areas **3205, 3207, 3209, 3211, 3213, 3215** may be either receive antenna element coverage areas or transmit antenna element coverage areas and may be associated with only the forward link or only the return link. The size of the antenna element coverage areas **3205, 3207, 3209, 3211, 3213, 3215** is determined by the desired performance to be provided by the system. A system that is more tolerant of errors may have antenna element coverage areas that are larger than a system that is less tolerant. In some cases, each antenna element coverage area **3205, 3207, 3209, 3211, 3213, 3215** is all points for which the component beam antenna gain is within 10 dB of the peak component beam antenna gain for the antenna element establishing the component beam antenna pattern. In other cases, each antenna element coverage area **3205, 3207, 3209, 3211, 3213, 3215** is all points for which the component beam antenna gain is within 6 dB of peak component beam antenna gain. In still other cases, each antenna element coverage area **3205, 3207, 3209, 3211, 3213, 3215** is all points for which the component beam antenna gain is within 3 dB of peak component beam antenna gain. Even when an end-to-end relay has not yet been deployed (e.g., an end-to-end satellite relay is not in a service orbit, the end-to-end relay still has component beam antenna patterns that conform to the above definition. That is, antenna element coverage areas corresponding to an end-to-end relay in orbit can be calculated from the component beam antenna patterns even when the end-to-end relay is not in a service orbit. The end-to-end relay may include additional antenna elements that do not contribute to beamforming and thus may not have the above-recited characteristics.

[0142] FIG. 26 is an illustration of an end-to-end relay (e.g., satellite) antenna pattern **3300** in which all of the points within a relay coverage area **3301** (e.g. satellite coverage area) are also contained within at least four antenna element coverage areas **3303, 3305, 3307, 3309**. Other antenna elements may exist on the end-to-end relay and can have antenna element coverage areas **3311** that contain less than all of the points within the relay coverage area **3301**.

[0143] The system may operate in any suitable spectrum. For example, an end-to-end beamforming system may operate in the C, L, S, X, V, Ka, Ku, or other suitable band or bands. In some such systems, the receive means operates in the C, L, S, X, V, Ka, Ku, or other suitable band or bands. In some cases, the forward uplink and the return uplink may operate in the same frequency range (e.g., in vicinity of 30 GHz); and the return downlink and the forward downlink may operate in a non-overlapping frequency range (e.g., in the vicinity of 20 GHz). The end-to-end system may use any suitable bandwidth (e.g., 500 MHz, 1 GHz, 2 GHz, 3.5 GHz, etc.). In some cases, the forward and return links use the same transponders.

[0144] To assist in system timing alignment, path lengths among the L transponders are set to match signal path time delays in some cases, for example through appropriate cable length selection. The end-to-end relay (e.g., satellite) in some cases has a relay beacon generator **426** (e.g. satellite beacon) within a calibration support module **424** (see FIG. 15). The beacon generator **426** generates a relay beacon signal. The end-to-end relay broadcasts the relay beacon signal to further aid in system timing alignment as well as support feeder link calibration. In some cases, the relay beacon signal is a pseudo-random (known as PN) sequence, such as a PN direct sequence spread spectrum signal that runs at a high chip rate (e.g., 100, 200, 400, or 800 million chips per second

(Mcps), or any other suitable value). In some cases, a linearly polarized relay (e.g., satellite) beacon, receivable by both RHCP and LHCP antennas, is broadcast over a wide coverage area by an antenna, such as an antenna horn (not shown) or coupled into one or more of the transponders **410** for transmission through the associated transmit antenna element **409**. In an example system, beams are formed in multiple 500 MHz bandwidth channels over the Ka band, and a 400 Mcps PN code is filtered or pulse-shaped to fit within a 500 MHz bandwidth channel. When multiple channels are used, the same PN code may be transmitted in each of the channels. The system may employ one beacon for each channel, or one beacon for two or more channels.

[0145] Since there may be a large number of receive/transmit signal paths in an end-to-end relay, redundancy of individual receive/transmit signal paths may not be required. Upon failure of a receive/transmit signal path, the system may still perform very close to its previous performance level, although modification of beamforming coefficients may be used to account for the loss.

Ground Networks

[0146] The ground network of an example end-to-end beamforming system contains a number of geographically distributed Access Node (AN) Earth stations pointed at a common end-to-end relay. Looking first at the forward link, a Central Processing System (CPS) computes beam weights for transmission of user data and interfaces to the ANs through a distribution network. The CPS also interfaces to the sources of data being provided to the user terminals. The distribution network may be implemented in various ways, for example using a fiber optic cable infrastructure. Timing between the CPS and SANs may be deterministic (e.g., using circuit-switched channels) or non-deterministic (e.g., using a packet-switched network). In some cases, the CPS is implemented at a single site, for example using custom application specific integrated circuits (ASICs) to handle signal processing. In some cases, the CPS is implemented in a distributed manner, for example using cloud computing techniques.

[0147] Returning to the example of FIG. 5, the CPS **505** may include a plurality of feeder link modems **507**. For the forward link, the feeder link modems **507** each receive forward user data streams **509** from various data sources, such as the internet, a video headend (not shown), etc. The received forward user data streams **509** are modulated by the modems **507** into K forward beam signals **511**. In some cases, K may be in the range of 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024 or numbers in-between or greater. Each of the K forward beam signals carries forward user data streams to be transmitted on one of K forward user beams. Accordingly, if K=400, then there are 400 forward beam signals **511**, each to be transmitted over an associated one of 400 forward user beams to a forward user beam coverage area **519**. The K forward beam signals **511** are coupled to a forward beamformer.

[0148] If M ANs **515** are present in the ground segment **502**, then the output of the forward beamformer is M access node-specific forward signals **516**, each comprising weighted forward beam signals corresponding to some or all of the K forward beam signals **511**. The forward beamformer may generate the M access node-specific forward signals **516** based on a matrix product of the $K \times M$ forward beam weight matrix with the K forward data signals. A distribution network **518** distributes each of the M access node-specific forward signals to a corresponding one of the M ANs **515**. Each AN **515** transmits a forward uplink signal **521** comprising a respective access node-specific forward signal **516**. Each AN **515** transmits its respective forward uplink signal **521** for relay to one or more (e.g., up to and including all) of the forward user beam coverage areas via one or more (e.g., up to and including all) of the forward receive/transmit signal paths of the end-to-end relay. Transponders **410**, **411** within the end-to-end relay **503** receive a composite input forward signal comprising a superposition **550** of forward uplink signals **521** transmitted by a plurality (e.g., up to and including all) of the ANs **515**. Each transponder (e.g., each receive/transmit signal path through the relay) relays the composite input forward signal as a respective forward downlink signal to the user terminals **517** over the forward downlink.

[0149] FIG. 27 is an illustration of an example distribution of ANs **515**. Each of the smaller

numbered circles represents the location of an AN **515**. Each of the larger circles indicates a user beam coverage area **519**. In some cases, the ANs **515** are spaced approximately evenly over the coverage area of the end-to-end relay **503**. In other cases, the ANs **515** may be distributed unevenly over the entire coverage area. In yet other cases, the ANs **515** may be distributed evenly or unevenly over one or more sub-regions of the relay coverage area. Typically, system performance is best when the ANs **515** are uniformly distributed over the entire coverage area. However, considerations may dictate compromises in the AN placement. For example, an AN **515** may be placed based on the amount of interference, rain, or other environmental conditions, cost of real estate, access to the distribution network, etc. For example, for a satellite-based end-to-end relay system that is sensitive to rain, more of the ANs **515** may be placed in areas that are less likely to experience rain-induced fading (e.g., the western United States). As another example, ANs **515** may be placed more densely in high rain regions (e.g., the southeastern United States) to provide some diversity gain to counteract the effects of rain fading. ANs **515** may be located along fiber routes to reduce distribution costs associated with the ANs **515**.

[0150] The number of ANs **515**, M , is a selectable parameter that can be selected based upon several criteria. Fewer ANs can result in a simpler, lower cost ground segment, and lower operational costs for the distribution network. More ANs can result in larger system capacity. FIG. **28** shows a simulation of the normalized forward and return link capacity as a function of the number of ANs deployed in an example system. Normalized capacity is the capacity with M ANs divided by the capacity obtained with the largest number of ANs in the simulation. The capacity increases as the number of ANs increases, but it does not increase without bound. Both forward link and return link capacities approach an asymptotic limit as the number of ANs is increased. This simulation was performed with $L=517$ transmit and receive antenna elements and with the ANs distributed uniformly over the coverage area, but this asymptotic behavior of the capacity can be seen with other values for L and other AN spatial distributions. Curves like those shown in FIG. **28** can be helpful in selection of the number of ANs, M , to be deployed and in understanding how the system capacity can be phased in as ANs are incrementally deployed, as discussed previously.

[0151] FIG. **29** is a block diagram of an example ground segment **502** for an end-to-end beamforming system. FIG. **29** may illustrate, for example, ground segment **502** of FIG. **5**. The ground segment **502** comprises CPS **505**, distribution network **518**, and ANs **515**. CPS **505** comprises beam signal interface **524**, forward/return beamformer **513**, distribution interface **536**, and beam weight generator **910**.

[0152] For the forward link, beam signal interface **524** obtains forward beam signals (FBS) **511** associated with each of the forward user beams. Beam signal interface **524** may include forward beam data multiplexer **526** and forward beam data stream modulator **528**. Forward beam data multiplexer **526** may receive forward user data streams **509** comprising forward data for transmission to user terminals **517**. Forward user data streams **509** may comprise, for example, data packets (e.g., TCP packets, UDP packets, etc.) for transmission to the user terminals **517** via the end-to-end beamforming system **500** of FIG. **5**. Forward beam data multiplexer **526** groups (e.g., multiplexes) the forward user data streams **509** according to their respective forward user beam coverage areas to obtain forward beam data streams **532**. Forward beam data multiplexer **526** may use, for example, time-domain multiplexing, frequency-domain multiplexing, or a combination of multiplexing techniques to generate forward beam data streams **532**. Forward beam data stream modulator **528** may modulate the forward beam data streams **532** according to one or more modulation schemes (e.g., mapping data bits to modulation symbols) to create the forward beam signals **511**, which are passed to the forward/return beamformer **513**. In some cases, the modulator **528** may frequency multiplex multiple modulated signals to create a multi-carrier beam signal **511**. Beam signal interface **524** may, for example, implement the functionality of feeder link modems **507** discussed with reference to FIG. **5**.

[0153] Forward/return beamformer **513** may include forward beamformer **529** and return

beamformer **531**. Beam weight generator **910** generates an $M \times K$ forward beam weight matrix **918**. Techniques for generating the $M \times K$ forward beam weight matrix **918** are discussed in more detail below. Forward beamformer **529** may include a matrix multiplier that calculates M access-node specific forward signals **516**. For example, this calculation can be based on a matrix product of the $M \times K$ forward beam weight matrix **918** and a vector of the K forward beam signals **511**. In some examples, each of the K forward beam signals **511** may be associated with one of F forward frequency sub-bands. In this case, the forward beamformer **529** may generate samples for the M access-node specific forward signals **516** for each of the F forward frequency sub-bands (e.g., effectively implementing the matrix product operation for each of the F sub-bands for respective subsets of the K forward beam signals **511**). Distribution interface **536** distributes (e.g., via distribution network **518**) the M access node-specific forward signals **516** to the respective ANs **515**.

[0154] For the return link, the distribution interface **536** obtains composite return signals **907** from ANs **515** (e.g., via distribution network **518**). Each return data signal from user terminals **517** may be included in multiple (e.g., up to and including all) of the composite return signals **907**. Beam weight generator **910** generates a $K \times M$ return beam weight matrix **937**. Techniques for generating the $K \times M$ return beam weight matrix **937** are discussed in more detail below. Return beamformer **531** calculates K return beam signals **915** for the K return user beam coverage areas. For example, this calculation can be based on a matrix product of the return beam weight matrix **937** and a vector of the respective composite return signals **907**. Beam signal interface **524** may include return beam signal demodulator **552** and return beam data de-multiplexer **554**. Return beam signal demodulator **552** may demodulate each of the return beam signals to obtain K return beam data streams **534** associated with the K return user beam coverage areas. Return beam data de-multiplexer **554** may de-multiplex each of the K return beam data streams **534** into respective return user data streams **535** associated with the return data signals transmitted from user terminals **517**. In some examples, each of the return user beams may be associated with one of R return frequency sub-bands. In this case, the return beamformer **531** may generate respective subsets of the return beam signals **915** associated with each of the R return frequency sub-bands (e.g., effectively implementing the matrix product operation for each of the R return frequency sub-bands to generate respective subsets of the return beam signals **915**).

[0155] FIG. **30** is a block diagram of an example forward/return beamformer **513**. The forward/return beamformer **513** comprises a forward beamformer **529**, a forward timing module **945**, a return beamformer **531**, and a timing module **947**. The forward timing module **945** associates each of the M access node-specific forward signals **516** with a time stamp (e.g., multiplexes the time stamp with the access node-specific forward signal in a multiplexed access node-specific forward signal) that indicates when the signal is desired to arrive at the end-to-end relay. In this way, the data of the K forward beam signals **511** that is split in a splitting module **904** within the forward beamformer **529** may be transmitted at the appropriate time by each of the ANs **515**. The timing module **947** aligns the receive signals based on time stamps. Samples of the M AN composite return signals (CRS) **907** are associated with time stamps indicating when the particular samples were transmitted from the end-to-end relay. Timing considerations and generation of the time stamps are discussed in greater detail below.

[0156] The forward beamformer **529** has a data input **925**, a beam weights input **920** and an access node output **923**. The forward beamformer **529** applies the values of an $M \times K$ beam weight matrix to each of the K forward data signals **511** to generate M access node specific forward signals **521**, each having K weighted forward beam signals. The forward beamformer **529** may include a splitting module **904** and M forward weighting and summing modules **533**. The splitting module **904** splits (e.g., duplicates) each of the K forward beam signals **511** into M groups **906** of K forward beam signals, one group **906** for each of the M forward weighting and summing modules **533**. Accordingly, each forward weighting and summing module **533** receives all K forward data

signals **511**.

[0157] A forward beam weight generator **917** generates an $M \times K$ forward beam weight matrix **918**. In some cases, the forward beam weight matrix **918** is generated based on a channel matrix in which the elements are estimates of end-to-end forward gains for each of the $K \times M$ end-to-end forward multipath channels to form a forward channel matrix, as discussed further below. Estimates of the end-to-end forward gain are made in a channel estimator module **919**. In some cases, the channel estimator has a channel data store **921** that stores data related to various parameters of the end-to-end multipath channels, as is discussed in further detail below. The channel estimator **919** outputs an estimated end-to-end gain signal to allow the forward beam weight generator **917** to generate the forward beam weight matrix **918**. Each of the weighting and summing modules **533** are coupled to receive respective vectors of beamforming weights of the forward beam weight matrix **918** (only one such connection is shown in FIG. **30** for simplicity). The first weighting and summing module **533** applies a weight equal to the value of the 1,1 element of the $M \times K$ forward beam weight matrix **918** to the first of the K forward beam signals **511** (discussed in more detail below). A weight equal to the value of the 1,2 element of the $M \times K$ forward beam weight matrix **918** is applied to the second of the K forward beam signals **511**. The other weights of the matrix are applied in like fashion, on through the K .sup.th forward beam signal **511**, which is weighted with the value equal to the 1, K element of the $M \times K$ forward beam weight matrix **918**. Each of the K weighted forward beam signals **903** are then summed and output from the first weighting and summing module **533** as an access node-specific forward signal **516**. The access node-specific forward signal **516** output by the first weighting and summing module **533** is then coupled to the timing module **945**. The timing module **945** outputs the access node-specific forward signal **516** to the first AN **515** through a distribution network **518** (see FIG. **5**). Similarly, each of the other weighting and summing modules **533** receive the K forward beam signals **511**, and weight and sum the K forward beam signals **511**. The outputs from each of the M weighting and summing modules **533** are coupled through the distribution network **518** to the associated M ANs **515** so that the output from the m .sup.th weighting and summing module is coupled to the m .sup.th AN **515**. In some cases, jitter and uneven delay through the distribution network, as well as some other timing considerations, are handled by the timing module **945** by associating a time stamp with the data. Details of an example timing technique are provided below with regard to FIGS. **36** and **37**.

[0158] As a consequence of the beam weights applied by the forward beamformers **529** at the ground segment **502**, the signals that are transmitted from the ANs **515** through the end-to-end relay **503** form user beams. The size and location of the beams that are able to be formed may be a function of the number of ANs **515** that are deployed, the number and antenna patterns of relay antenna elements that the signal passes through, the location of the end-to-end relay **503**, and/or the geographic spacing of the ANs **515**.

[0159] Referring now to the end-to-end return link **523** shown in FIG. **5**, a user terminal **517** within one of the user beam coverage areas **519** transmits signals up to the end-to-end relay **503**. The signals are then relayed down to the ground segment **502**. The signals are received by ANs **515**.

[0160] Referring once again to FIG. **30**, M return downlink signals **527** are received by the M ANs **515** and are coupled, as composite return signals **907**, from the M ANs **515** through the distribution network **518** and received in an access node input **931** of the return beamformer **531**. Timing module **947** aligns the composite return signals from the M ANs **515** to each other and outputs the time-aligned signals to the return beamformer **531**. A return beam weight generator **935** generates the return beam weights as a $K \times M$ return beam weight matrix **937** based on information stored in a channel data store **941** within a channel estimator **943**. The return beamformer **531** has a beam weights input **939** through which the return beamformer **531** receives the return beam weight matrix **937**. Each of the M AN composite return signals **907** is coupled to an associated one of M splitter and weighting modules **539** within the return beamformer **531**. Each splitter and weighting

module **539** splits the time-aligned signal into K copies **909**. The splitter and weighting modules **539** weight each of the K copies **909** using the k, m element of the K×M return beam weight matrix **937**. Further details regarding the K×M return beam weight matrix are provided below. Each set of K weighted composite return signals **911** is then coupled to a combining module **913**. In some cases, the combining module **913** combines the k.sup.th weighted composite return signal **911** output from each splitter and weighting module **539**. The return beamformer **531** has a return data signal output **933** that outputs K return beam signals **915**, each having the samples associated with one of the K return user beams **519** (e.g., the samples received through each of the M ANs). Each of the K return beam signals **915** may have samples from one or more user terminals **517**. The K combined and aligned, beamformed return beam signals **915** are coupled to the feeder link modems **507** (see FIG. 5). Note that the return timing adjustment may be performed after the splitting and weighting. Similarly, for the forward link, the forward timing adjustment may be performed before the beamforming.

[0161] As discussed above, forward beamformer **529** may perform matrix product operations on input samples of K forward beam signals **511** to calculate M access node-specific forward signal **516** in real-time. As the beam bandwidth increases (e.g., to support shorter symbol duration) and/or K and M become large, the matrix product operation becomes computationally intensive and may exceed the capabilities of a single computing node (e.g., a single computing server, etc.). The operations of return beamformer **531** are similarly computationally intensive. Various approaches may be used to partition computing resources of multiple computing nodes in the forward/return beamformer **513**. In one example, the forward beamformer **529** of FIG. 30 may be partitioned into separate weighting and summing modules **533** for each of the M ANs **515**, which may be distributed into different computing nodes. Generally, the considerations for implementations include cost, power consumption, scalability relative to K, M, and bandwidth, system availability (e.g., due to node failure, etc.), upgradeability, and system latency. The example above is per row (or column). Vice versa is possible. Other manners of grouping the matrix operations may be considered (e.g., split into four with [1,1 to K/2, M/2], [. . .], computed individually and summed up).

[0162] In some cases, the forward/return beamformer **513** may include a time-domain multiplexing architecture for processing of beam weighting operations by time-slice beamformers. FIG. 31 is a block diagram of an example forward beamformer **529** comprising multiple forward time-slice beamformers with time-domain de-multiplexing and multiplexing. The forward beamformer **529** includes a forward beam signal de-multiplexer **3002**, N forward time-slice beamformers **3006**, and a forward access node signal multiplexer **3010**.

[0163] Forward beam signal de-multiplexer **3002** receives forward beam signals **511** and de-multiplexes the K forward beam signals **511** into forward time slice inputs **3004** for input to the N forward time-slice beamformers **3006**. For example, the forward beam signal de-multiplexer **3002** sends a first time-domain subset of samples for the K forward beam signals **511** to a first forward time-slice beamformer **3006**, which generates samples associated with the M access node-specific forward signals corresponding to the first time-domain subset of samples. The forward time-slice beamformer **3006** outputs the samples associated with the M access node-specific forward signals for the first time-domain subset of samples via its forward time slice output **3008** to the forward access node signal multiplexer **3010**. The forward time-slice beamformer **3006** may output the samples associated with each of the M access node-specific forward signals with synchronization timing information (e.g., the corresponding time-slice index, etc.) used by the access nodes to cause (e.g., by pre-correcting) the respective access node-specific forward signals to be synchronized when received by the end-to-end relay. The forward access node signal multiplexer **3010** multiplexes time-domain subsets of samples for the M access node-specific forward signals received via the N forward time slice outputs **3008** to generate the M access node-specific forward signals **516**. Each of the forward time-slice beamformers **3006** may include a data buffer, a beam

matrix buffer, and beam weight processor implementing the matrix product operation. That is, each of the forward time-slice beamformers **3006** may implement computations mathematically equivalent to the splitting module **904** and forward weighting and summing modules **533** shown for forward beamformer **529** of FIG. **30** during processing of the samples of one time slice-index. Updating of the beam weight matrix may be performed incrementally. For example, the beam weight matrix buffers for forward time-slice beamformers may be updated during idle time in a rotation of time-slice indices t through the N forward time-slice beamformers **3006**. Alternatively, each forward time-slice beamformer may have two buffers that can be used in a ping-pong configuration (e.g., one can be updated while the other is being used). In some cases, multiple buffers can be used to store beam weights corresponding to multiple user beam patterns (e.g., multiple user coverage areas). Beam weight buffers and data buffers for forward time-slice beamformers **3006** may be implemented as any type of memory or storage including dynamic or static random access memory (RAM). Beam weight processing may be implemented in an application specific integrated circuit (ASIC) and/or a field programmable gate array (FPGA), and may include one or more processing cores (e.g., in a cloud computing environment). Additionally or alternatively, the beam weight buffer, data buffer, and beam weight processor may be integrated within one component.

[0164] FIG. **32** illustrates a simplified example ground segment showing the operation of a forward time-slice beamformer **529**. In the example of FIG. **32**, forward beamformer **529** receives four forward beam signals (e.g., $K=4$), generates access node-specific forward signals for five ANs (e.g., $M=5$), and has three forward time-slice beamformers (e.g., $N=3$). The forward beam signals are denoted by $FB_k:t$, where k is the forward beam signal index and t is the time-slice index (e.g., corresponding to a time-domain subset of samples). The forward beam signal de-multiplexer **3002** receives four time-domain subsets of samples of the forward beam signals associated with four forward user beams and de-multiplexes each forward beam signal so that one forward time slice input **3004** includes, for a particular time-slice index t , the time-domain subsets of samples from each of the forward beam signals **511**. For example, time-domain subsets can be a single sample, a contiguous block of samples, or a discontinuous (e.g., interleaved) block of samples as described below. The forward time-slice beamformers **3006** generate (e.g., based on the forward beam signals **511** and forward beam weight matrix **918**) each of the M access-node specific forward signals for the time-slice index t , denoted by $AF_m:t$. For example, the time-domain subsets of samples $FB1:0$, $FB2:0$, $FB3:0$, and $FB4:0$ for time-slice index $t=0$ are input to the first forward time-slice beamformer $TSBF[1]3006$, which generates corresponding samples of access node-specific forward signals $AF1:0$, $AF2:0$, $AF3:0$, $AF4:0$, and $AF5:0$ at a forward time slice output **3008**. For subsequent time-slice index values $t=1, 2$, the time-domain subsets of samples of forward beam signals **511** are de-multiplexed by the forward beam signal de-multiplexer **3002** for input to second and third forward time-slice beamformers **3006**, which generate access node-specific forward signals associated with the corresponding time-slice indices t at forward time slice outputs **3008**. FIG. **32** also shows that at time-slice index value $t=3$, the first forward time-slice beamformer generates access node-specific forward signals associated with the corresponding time-slice index **3**. The matrix product operation performed by each forward time-slice beamformer **3006** for one time-slice index value t may take longer than the real time of the time-domain subset of samples (e.g., the number of samples S multiplied by the sample rate $t.sub.S$). However, each forward time-slice beamformer **3006** may only process one time-domain subset of samples every N time-slice indices t . Forward access node signal multiplexer **3010** receives forward time slice outputs **3030** from each of the forward time-slice beamformers **3006** and multiplexes the time-domain subsets of samples to generate the M access node-specific forward signals **516** for distribution to respective ANs.

[0165] FIG. **33** is a block diagram of an example return beamformer **531** comprising multiple return time-slice beamformers with time-domain de-multiplexing and multiplexing. The return

beamformer **531** includes a return composite signal de-multiplexer **3012**, N return time-slice beamformers **3016**, and a return beam signal multiplexer **3020**. Return composite signal de-multiplexer **3012** receives M composite return signals **907** (e.g., from M ANs) and de-multiplexes the M composite return signals **907** into return time slice inputs **3014** for input to the N return time-slice beamformers **3016**. Each of the return time-slice beamformers **3016** output the samples associated with the K return beam signals **915** for corresponding time-domain subsets of samples via respective return time slice outputs **3018** to the return beam signal multiplexer **3020**. The return beam signal multiplexer **3020** multiplexes the time-domain subsets of samples for the K return beam signals received via the N return time slice outputs **3018** to generate the K return beam signals **915**. Each of the return time-slice beamformers **3016** may include a data buffer, a beam matrix buffer, and beam weight processor implementing the matrix product operation. That is, each of the return time-slice beamformers **3016** may implement computations mathematically equivalent to the splitter and weighting modules **539** and combining module **913** shown for return beamformer **531** of FIG. **30** during processing of the samples of one time slice-index. As discussed above with the forward time-slice beamformers, updating of the beam weight matrix may be performed incrementally using a ping-pong beam weight buffer configuration (e.g., one can be updated while the other is being used). In some cases, multiple buffers can be used to store beam weights corresponding to multiple user beam patterns (e.g., multiple user coverage areas). Beam weight buffers and data buffers for return time-slice beamformers **3016** may be implemented as any type of memory or storage including dynamic or static random access memory (RAM). Beam weight processing may be implemented in an application specific integrated circuit (ASIC) and/or a field programmable gate array (FPGA), and may include one or more processing cores. Additionally or alternatively, the beam weight buffer, data buffer, and beam weight processor may be integrated within one component.

[0166] FIG. **34** illustrates a simplified example ground segment showing the operation of a return beamformer **531** employing time-domain multiplexing. In the example of FIG. **33**, return beamformer **531** receives five composite return signals (e.g., $M=5$), generates return beam signals for four return user beams (e.g., $K=5$), and has three time-slice beamformers (e.g., $N=3$). The composite return signals are denoted by $RC_m:t$, where m is the AN index and t is the time-slice index (e.g., corresponding to a time-domain subset of samples). The return composite signal de-multiplexer **3012** receives four time-domain subsets of samples of the composite return signals from five ANs and de-multiplexes each composite return signal so that one return time slice input **3014** includes, for a particular time-slice index t , the corresponding time-domain subsets of samples from each of the composite return signals **907**. For example, time-domain subsets can be a single sample, a contiguous block of samples, or a discontinuous (e.g., interleaved) block of samples as described below. The return time-slice beamformers **3016** generate (e.g., based on the composite return signals **907** and return beam weight matrix **937**) each of the K return beam signals for the time-slice index t , denoted by $RB_k:t$. For example, the time-domain subsets of samples $RC1:0$, $RC2:0$, $RC3:0$, $RC4:0$, and $RC5:0$ for time-slice index $t=0$ are input to a first return time-slice beam former **3016**, which generates corresponding samples of return beam signals $RB1:0$, $RB2:0$, $RB3:0$, and $RB4:0$ at a return time slice output **3018**. For subsequent time-slice index values $t=1, 2$, the time-domain subsets of samples of composite return signals **907** are de-multiplexed by the return composite signal de-multiplexer **3012** for input to a second and a third return time-slice beamformer **3016**, respectively, which generate samples for the return beam signals associated with the corresponding time-slice indices t at return time slice outputs **3018**. FIG. **34** also shows that at time-slice index value $t=3$, the first return time-slice beamformer generates samples of return beam signals associated with the corresponding time-slice index **3**. The matrix product operation performed by each return time-slice beamformer **3016** for one time-slice index value t may take longer than the real time of the time-domain subset of samples (e.g., the number of samples S multiplied by the sample rate $t_{sub}S$). However, each return time-slice

beamformer **3016** may only process one time-domain subset of samples every N time-slice indices t . Return beam signal multiplexer **3020** receives return time slice outputs **3018** from each of the return time-slice beamformers **3016** and multiplexes the time-domain subsets of samples to generate the K return beam signals **915**.

[0167] Although FIGS. **31-34** illustrate the same number N of forward time-slice beamformers **3006** as return time-slice beamformers **3016**, some implementations may have more or fewer forward time-slice beamformers **3006** than return time-slice beamformers **3016**. In some examples, forward beamformer **529** and/or return beamformer **531** may have spare capacity for robustness to node failure. For example, if each forward time-slice beamformer **3006** takes $t_{\text{sub.FTS}}$ to process one set of samples for a time-slice index t having a real-time time-slice duration $t_{\text{sub.D}}$, where $t_{\text{sub.FTS}} = N \cdot t_{\text{sub.D}}$, the forward beamformer **529** may have $N+E$ forward time-slice beamformers **3006**. In some examples, each of the $N+E$ forward time-slice beamformers **3006** are used in operation, with each forward time-slice beamformer **3006** having an effective extra capacity of E/N . If one forward time-slice beamformer **3006** fails, the operations may be shifted to another forward time-slice beamformer **3006** (e.g., by adjusting how time-domain samples (or groups of samples) are routed through the time-domain de-multiplexing and multiplexing). Thus, forward beamformer **529** may be tolerant of up to E forward time-slice beamformers **3006** failing before system performance is impacted. In addition, extra capacity allows for system maintenance and upgrading of time-slice beamformers while the system is operating. For example, upgrading of time-slice beamformers may be performed incrementally because the system is tolerant of different performance between time-slice beamformers. The data samples associated with a time-slice index t may be interleaved. For example, a first time-slice index t_0 may be associated with samples $0, P, 2P, \dots (S-1) \cdot P$, while a second time-slice index t_1 may be associated with samples $1, P+1, 2P+1, \dots (S-1) \cdot P+1$, etc., where S is the number of samples in each set of samples, and P is the interleaving duration. The interleaving may also make the system more robust to time-slice beamformer failures, because each time-slice beamformer block of samples are separated in time such that errors due to a missing block would be distributed in time, similarly to the advantage from interleaving in forward error correction. In fact, the distributed errors caused by time-slice beamformer failure may cause effects similar to noise and not result in any errors to user data, especially if forward error coding is employed. Although examples where $N=3$ have been illustrated, other values of N may be used, and N need not have any particular relationship to K or M .

[0168] As discussed above, forward beamformer **529** and return beamformer **531** illustrated in FIGS. **31** and **33**, respectively, may perform time-domain de-multiplexing and multiplexing for time-slice beamforming for one channel or frequency sub-band. Multiple sub-bands may be processed independently using an additional sub-band mux/demux switching layer. FIG. **35** is a block diagram of an example multi-band forward/return beamformer **513** that employs sub-band de-multiplexing and multiplexing. The multi-band forward/return beamformer **513** may support F forward sub-bands and R return sub-bands.

[0169] Multi-band forward/return beamformer **513** includes F forward sub-band beamformers **3026**, R return sub-band beamformers **3036**, and a sub-band multiplexer/de-multiplexer **3030**. For example, the forward beam signals **511** may be split up into F forward sub-bands. Each of the F forward sub-bands may be associated with a subset of the K forward user beam coverage areas. That is, the K forward user beam coverage areas may include multiple subsets of forward user beam coverage areas associated with different (e.g., different frequency and/or polarization, etc.) frequency sub-bands, where the forward user beam coverage areas within each of the subsets may be non-overlapping (e.g., at 3 dB signal contours, etc.). Thus, each of the forward sub-band beamformer inputs **3024** may include a subset $K_{\text{sub.1}}$ of the forward beam signals **511**. Each of the F forward beamformers **3026** may include the functionality of forward beamformer **529**, generating forward sub-band beamformer outputs **3028** that comprise the M access node-specific

forward signals associated with the subset of the forward beam signals **511** (e.g., a matrix product of the $K_{\text{sub}.1}$ forward beam signals with an $M \times K_{\text{sub}.1}$ forward beam weight matrix). Thus, each of the ANs **515** may receive multiple access node-specific forward signals associated with different frequency sub-bands (e.g., for each of the F forward sub-bands). The ANs may combine (e.g., sum) the signals in different sub-bands in the forward uplink signals, as discussed in more detail below. Similarly, ANs **515** may generate multiple composite return signals **907** for R different return sub-bands. Each of the R return sub-bands may be associated with a subset of the K return user beam coverage areas. That is, the K return user beam coverage areas may include multiple subsets of return user beam coverage areas associated with different frequency sub-bands, where the return user beam coverage areas within each of the subsets may be non-overlapping (e.g., at 3 dB signal contours, etc.). The sub-band multiplexer/de-multiplexer **3030** may split the composite return signals **907** into the R return sub-band beamformer inputs **3034**. Each of the return sub-band beamformers **3036** may then generate a return sub-band beamformer output **3038**, which may include the return beam signals **915** for a subset of the return user beams (e.g., to the feeder link modems **507** or return beam signal demodulator, etc.). In some examples, the multi-band forward/return beamformer **513** may support multiple polarizations (e.g., right-hand circular polarization (RHCP), left-hand circular polarization (LHCP), etc.), which in some cases may effectively double the number of sub-bands.

[0170] In some cases, time-slice multiplexing and de-multiplexing for forward beamformer **529** and return beamformer **531** (e.g., beam signal de-multiplexer **3002**, forward access node signal multiplexer **3010**, return composite signal de-multiplexer **3012**, return beam signal multiplexer **3020**) and sub-band multiplexing/de-multiplexing (sub-band multiplexer/de-multiplexer **3030**) may be performed by packet switching (e.g., Ethernet switching, etc.). In some cases, the time-slice and sub-band switching may be performed in the same switching nodes, or in a different order. For example, a fabric switching architecture may be used where each switch fabric node may be coupled with a subset of the ANs **515**, forward time-slice beamformers **3006**, return time-slice beamformers **3016**, or feeder link modems **507**. A fabric switching architecture may allow, for example, any AN to connect (e.g., via switches and/or a switch fabric interconnect) to any forward time-slice beamformer or return time-slice beamformer in a low-latency, hierarchically flat architecture. In one example, a system supporting $K \leq 600$, $M \leq 600$, and a 500 MHz bandwidth (e.g., per sub-band) with fourteen sub-bands for the forward or return links may be implemented by a commercially available interconnect switch platform with 2048 10GigE ports.

Delay Equalization

[0171] In some cases, differences in the propagation delays on each of the paths between the end-to-end relay **503** and the CPS **505** are insignificant. For example, on the return link, when the same signal (e.g., data to or from a particular user) is received by multiple ANs **515**, each instance of the signal may arrive at the CPS essentially aligned with each other instance of the signal. Likewise, when the same signal is transmitted to a user terminal **517** through several ANs **515**, each instance of the signal may arrive at the user terminal **517** essentially aligned with each other instance of the signal. In other words, signals may be phase and time aligned with sufficient precision that signals will coherently combine, such that the path delays and beamforming effects are small relative to the transmitted symbol rate. As an illustrative example, if the difference in path delays is 10 microseconds, the beamforming bandwidth can be on the order of tens of kHz and one can use a narrow bandwidth signal, say ≈ 10 ksps with a small possible degradation in performance. The 10 ksps signaling rate has a symbol duration of 100 microseconds and the 10 microsecond delay spread is only one tenth of the symbol duration. In these cases, for the purposes of the system analysis, it may be assumed that signals received by the end-to-end relay at one instant will be relayed and transmitted at essentially the same time, as described earlier.

[0172] In other cases, there may be a significant difference in the propagation delay relative to the signaling interval (transmitted symbol duration) of the signals transmitted from the transmit

antenna elements **409** to the ANs **515**. The path that the signals take from each AN **515** through the distribution network **518** may contain significant delay variations. In these cases, delay equalization may be employed to match the path delays.

[0173] For end-to-end return link signals received through the distribution network **518** by the CPS **505**, signals may be time aligned by using a relay beacon signal transmitted from the end-to-end relay, for example a PN beacon as described earlier. Each AN **515** may time stamp the composite return signal using the relay beacon signal as a reference. Therefore, different ANs **515** may receive the same signal at different times, but the received signals in each AN **515** may be time stamped to allow the CPS **505** to time align them. The CPS **505** may buffer the signals so that beamforming is done by combining signals that have the same time stamp.

[0174] Returning to FIGS. **33** and **34**, delay equalization for the return link may be performed by de-multiplexing the composite return signals to the return time-slice beamformers **3016**. For example, each AN may split up the composite return signal into sets of samples associated with time-slice indices t , which may include interleaved samples of the composite return signal. The time-slice indices t may be determined based on the relay beacon signal. The ANs may send the subsets of samples multiplexed with the corresponding time-slice indices t (e.g., as a multiplexed composite return signal) to the return beamformer **531**, which may serve as synchronization timing information on the return link. The subsets of samples from each AN may be de-multiplexed (e.g., via switching) and one return time-slice beamformer **3016** may receive the subsets of samples from each AN for a time-slice index t (for one of multiple sub-bands, in some cases). By performing the matrix product of the return beam weight matrix and the subsets of samples from each of the M composite return signals associated with the time-slice index t , return time-slice beamformer **3016** may align the signals relayed by the end-to-end relay at the same time for applying the return beam weight matrix.

[0175] For the forward link, the beamformer **513** within the CPS **505** may generate a time stamp that indicates when each access node-specific forward signal transmitted by the ANs **515** is desired to arrive at the end-to-end relay **503**. Each AN **515** may transmit an access node beacon signal **2530**, for example a loopback PN signal. Each such signal may be looped-back and transmitted back to the ANs **515** by the end-to-end relay **503**. The ANs **515** may receive both the relay beacon signal and the relayed (looped-back) access node beacon signals from any or all of the ANs. The received timing of the access node beacon signal relative to receive timing of the relay beacon signal indicates when the access node beacon signal arrived at the end-to-end relay. Adjusting the timing of the access node beacon signal such that, after relay by the end-to-end relay, it arrives at the AN at the same time as the relay beacon signal arrives at the AN, forces the access node beacon signal to arrive at the end-to-end relay synchronized with the relay beacon. Having all ANs perform this function enables all access node beacon signals to arrive at the end-to-end relay synchronized with the relay beacon. The final step in the process is to have each AN transmit its access node-specific forward signals synchronized with its access node beacon signal. This can be done using timestamps as described subsequently. Alternatively, the CPS may manage delay equalization by sending the respective access node-specific forward signals offset by the respective time-domain offsets to the ANs (e.g., where the timing via the distribution network is deterministic). In some cases, the feeder-link frequency range may be different from the user-link frequency range. When the feeder-link downlink frequency range (e.g., a frequency range in V band) is non-overlapping with the user-link downlink frequency range (e.g., a frequency range in Ka band), and the ANs are within the user coverage area, the ANs may include antennas and receivers operable over the user-link downlink frequency range in order to receive the relayed access node beacon signals via the receive/transmit signal paths of the end-to-end relay. In such a case, the end-to-end relay can include a first relay beacon generator that generates a first relay beacon signal in the user-link downlink frequency range to support feeder link synchronization. The end-to-end relay can also include a second relay beacon generator that generates a second relay beacon signal in the feeder-

link downlink frequency range to support removal of feeder-link impairments from the return downlink signals.

[0176] FIG. 36 is an illustration of PN sequences used to align the timing of the system. The horizontal axis of the figure represents time. An AN.sub.1 PN sequence **2301** of chips **2303** is transmitted in the access node beacon signal from the first AN. The relative time of arrival of this sequence at the end-to-end relay is depicted by the PN sequence **2305**. There is a time shift of PN sequence **2305** with respect to AN.sub.1 PN sequence **2301**, due to the propagation delay from the AN to the end-to-end relay. A relay PN beacon sequence **2307** is generated within, and transmitted from, the end-to-end relay in a relay beacon signal. A PN chip of the relay PN beacon sequence **2307** at time T.sub.0 **2315** is aligned with a PN chip **2316** of the AN.sub.1 PN received signal **2305** at time T.sub.0. The PN chip **2316** of the AN.sub.1 PN received signal **2305** is aligned with the PN chip **2315** of the relay PN beacon **2307** when the AN.sub.1 transmit timing is adjusted by the proper amount. The PN sequence **2305** is looped back from the end-to-end relay and the PN sequence **2317** is received at AN.sub.1. A PN sequence **2319** transmitted from the end-to-end relay in the relay PN beacon is received at AN.sub.1. Note that the PN sequences **2317**, **2319** are aligned at AN.sub.1 indicating that they were aligned at the end-to-end relay.

[0177] FIG. 37 shows an example of an AN.sub.2 that has not properly adjusted the timing of the PN sequence generated in the AN.sub.2. Notice that the PN sequence **2311** generated by the AN.sub.2 is received at the end-to-end relay shown as sequence **2309** with an offset by an amount dt from the relay PN beacon PN sequence **2307**. This offset is due to an error of the timing used to generate the sequence in the AN.sub.2. Also, note that the arrival of the AN.sub.2 PN sequence **2321** at AN.sub.2 is offset from the arrival of the relay PN beacon PN sequence at AN.sub.2 **2323** by the same amount dt . The signal processing in AN.sub.2 will observe this error and may make a correction to the transmit timing by adjusting the timing by an amount dt to align the PN sequences **2321**, **2323**.

[0178] In FIGS. 36 and 37 the same PN chip rate has been used for the relay PN beacon and all of the AN (loopback) PN signals for ease of illustration of the concept. The same timing concepts can be applied with different PN chip rates. Returning to FIGS. 31 and 32, the time-slice indices t may be used for synchronizing the access node-specific forward signals received from each of the ANs at the end-to-end relay. For example, the time-slice indices t may be multiplexed with the access node-specific forward signals **516**. Each AN may transmit samples of the access node-specific forward signals with a particular time-slice index t aligned with corresponding timing information in the PN sequence of chips transmitted in the respective access node beacon signals. Because the respective access node beacon signals have been adjusted to compensate for the respective path delays and phase shifts between the ANs and the end-to-end relay, the samples associated with the time-slice index t will arrive at the end-to-end relay with timing synchronized and phase aligned correctly relative to each other.

[0179] In cases where ANs receive their own access node beacon signals, it is possible to loop back the access node beacon signals using the same end-to-end relay communication hardware that is also carrying the forward direction communication data. In these cases, the relative gains and/or phases of the transponders in the end-to-end relay can be adjusted as subsequently described.

[0180] FIG. 38 is a block diagram of an example AN **515**. AN **515** comprises receiver **4002**, receive timing and phase adjuster **4024**, relay beacon signal demodulator **2511**, multiplexer **4004**, network interface **4006**, controller **2523**, de-multiplexer **4060**, transmit timing and phase compensator **4020**, and transmitter **4012**. Network interface **4006** may be connected to, for example, CPS **505** via network port **4008**.

[0181] On the return link, receiver **4002** receives a return downlink signal **527**. The return downlink signal **527** may include, for example, a composite of return uplink signals relayed by the end-to-end relay (e.g., via multiple receive/transmit signal paths, etc.) and the relay beacon signal. Receiver **4002** may perform, for example, down-conversion and sampling. Relay beacon signal

demodulator **2511** may demodulate the relay beacon signal in the digitized composite return signal **907** to obtain relay timing information **2520**. For example, relay beacon signal demodulator **2511** may perform demodulation to recover the chip timing associated with the relay PN code and generate time stamps corresponding to the transmission time from the end-to-end relay for samples of the digitized composite return signal **527**. Multiplexer **4004** may multiplex the relay timing information **2520** with the samples of the digitized composite return signal (e.g., to form a multiplexed composite return signal) to be sent to the CPS **505** (e.g., via network interface **4006**). Multiplexing the relay timing information **2520** may include generating subsets of samples corresponding to time-slice indices t for sending to the CPS **505**. For example, multiplexer **4004** may output subsets of samples associated with each time slice index t for input to the return time-slice beamforming architecture described above with reference to FIGS. **33**, **34**, and **35**. Multiplexer **4004** may include an interleaver **4044** for interleaving samples for each subset of samples, in some cases.

[0182] On the forward link, network interface **4006** may obtain AN input signal **4014** (e.g., via network port **4008**). De-multiplexer **4060** may de-multiplex AN input signal **4014** to obtain access node-specific forward signal **516** and forward signal transmit timing information **4016** indicating transmission timing for the access node-specific forward signal **516**. For example, the access node-specific forward signal **516** may comprise the forward signal transmit timing information (e.g., multiplexed with data samples, etc.). In one example, the access node-specific forward signal **516** comprises sets of samples (e.g., in data packets), where each set of samples is associated with a time-slice index t . For example, each set of samples may be samples of the access node-specific forward signal **516** generated according to the forward time-slice beamforming architecture discussed above with reference to FIGS. **31**, **32** and **35**. De-multiplexer **4060** may include a de-interleaver **4050** for de-interleaving samples associated with time-slice indices t .

[0183] Transmit timing and phase compensator **4020** may receive and buffer access node-specific forward signal **516** and output forward uplink signal samples **4022** for transmission by the transmitter **4012** at an appropriate time as forward uplink signal **521**. The transmitter **4012** may perform digital-to-analog conversion and up-conversion to output the forward uplink signal **521**. Forward uplink signal samples **4022** may include the access node-specific forward signal **516** and an access node beacon signal **2530** (e.g., loopback PN signal), which may include transmit timing information (e.g., PN code chip timing information, frame timing information, etc.). Transmit timing and phase compensator **4020** may multiplex the access node-specific forward signal **516** with the access node beacon signal **2530** such that the forward signal transmit timing and phase information **4016** is synchronized to corresponding transmit timing and phase information in the access node beacon signal **2530**.

[0184] In some examples, generation of the access node beacon signal **2530** is performed locally at the AN **515** (e.g., in access node beacon signal generator **2529**). Alternatively, generation of the access node beacon signal **2530** may be performed in a separate component (e.g., CPS **505**) and sent to the AN **515** (e.g., via network interface **4006**). As discussed above, the access node beacon signal **2530** may be used to compensate the forward uplink signal **521** for path differences and phase shifts between the AN and the end-to-end relay. For example, the access node beacon signal **2530** may be transmitted in the forward uplink signal **521** and relayed by the end-to-end relay to be received back at receiver **4002**. The controller **2523** may compare relayed transmit timing and phase information **4026** obtained (e.g., by demodulation, etc.) from the relayed access node beacon signal with receive timing and phase information **4028** obtained (e.g., by demodulation, etc.) from the relay beacon signal. The controller **2523** may generate a timing and phase adjustment **2524** for input to the transmit timing and phase compensator **4020** to adjust the access node beacon signal **2530** to compensate for the path delay and phase shifts. For example, the access node beacon signal **2530** may include a PN code and frame timing information (e.g., one or more bits of a frame number, etc.). The transmit timing and phase compensator **4020** may, for example, adjust the frame

timing information for coarse compensation for the path delay (e.g., output frame timing information in the access node beacon signal such that the relayed access node beacon signal will have the relayed transmit frame timing information coarsely aligned with corresponding frame timing information in the relay beacon signal, changing which chip of the PN code is considered to be the LSB, etc.). Additionally or alternatively, the transmit timing and phase compensator **4020** may perform timing and phase adjustments to the forward uplink signal samples **4022** to compensate for timing or phase differences between the relayed transmit timing and phase information **4026** and receive timing and phase information **4028**. For example, where the access node beacon signal **2530** is generated based on a local oscillator, timing or phase differences between the local oscillator and the received relay beacon signal may be corrected by timing and phase adjustments to the forward uplink signal samples **4022**. In some examples, demodulation of the access node beacon signal is performed locally at the AN **515** (e.g., in access node beacon signal demodulator **2519**). Alternatively, demodulation of the access node beacon signal may be performed in a separate component (e.g., CPS **505**) and the relayed transmit timing and phase information **4026** may be obtained in other signaling (e.g., via network interface **4006**). For example, deep fading may make reception and demodulation of the AN's own relayed access node beacon signal difficult without transmission at higher power than other signaling, which may reduce the power budget for communication signals. Thus, combining reception of the relayed access node beacon signal from multiple ANs **515** may increase the effective received power and demodulation accuracy for the relayed access node beacon signal. Thus, demodulation of the access node beacon signal from a single AN **515** may be performed using downlink signals received at multiple ANs **515**. Demodulation of the access node beacon signal may be performed at the CPS **505** based on the composite return signals **907**, which may also include signal information for the access node beacon signals from most or all ANs **515**. If desired, end-to-end beamforming for the access node beacon signals can be performed taking into account the access node beacon uplinks (e.g., C.sub.r), relay loopback (e.g., E), and/or access node beacon downlinks (e.g., C.sub.t).

Feeder Link Impairment Removal

[0185] In addition to delay equalization of the signal paths to the end-to-end relay from all the ANs, the phase shifts induced by feeder links can be removed prior to beamforming. The phase shift of each of the links between the end-to-end relay and the M ANs will be different. The causes for different phase shifts for each link include, but are not limited to, the propagation path length, atmospheric conditions such as scintillation, Doppler frequency shift, and different AN oscillator errors. These phase shifts are generally different for each AN and are time varying (due to scintillation, Doppler shift, and difference in the AN oscillator errors). By removing dynamic feeder link impairments, the rate at which beam weights adapt may be slower than an alternative where the beam weights adapt fast enough to track the dynamics of the feeder link.

[0186] In the return direction, feeder downlink impairments to an AN are common to both the relay PN beacon and user data signals (e.g., return downlink signals). In some cases, coherent demodulation of the relay PN beacon provides channel information that is used to remove most or all of these impairments from the return data signal. In some cases, the relay PN beacon signal is a known PN sequence that is continually transmitted and located in-band with the communications data. The equivalent (or effective) isotropically radiated power (EIRP) of this in-band PN signal is set such that the interference to the communications data is not larger than a maximum acceptable level. In some cases, a feeder link impairment removal process for the return link involves coherent demodulation and tracking of the received timing and phase of the relay PN beacon signal. For example, relay beacon signal demodulator **2511** may determine receive timing and phase adjustments **2512** to compensate for feeder link impairment based on comparing the relay PN beacon signal with a local reference signal (e.g., local oscillator or PLL). The recovered timing and phase differences are then removed from the return downlink signal (e.g., by receive timing and

phase adjuster **4024**), hence removing feeder link impairments from the communications signal (e.g., return downlink signals **527**). After feeder link impairment removal, the return link signals from a beam will have a common frequency error at all ANs and thus be suitable for beamforming. The common frequency error may include, but is not limited to, contributions from the user terminal frequency error, user terminal uplink Doppler, end-to-end relay frequency translation frequency error and relay PN beacon frequency error.

[0187] In the forward direction, the access node beacon signal from each AN may be used to help remove feeder uplink impairments. The feeder uplink impairments will be imposed upon the forward link communications data (e.g., the access node-specific signal) as well as the access node beacon signal. Coherent demodulation of the access node beacon signal may be used to recover the timing and phase differences of the access node beacon signal (e.g., relative to the relay beacon signal). The recovered timing and phase differences are then removed from the transmitted access node beacon signal such that the access node beacon signal arrives in phase with the relay beacon signal.

[0188] In some cases, the forward feeder link removal process is a phase locked loop (PLL) with the path delay from the AN to the end-to-end relay and back within the loop structure. In some cases, the round-trip delay from the AN to the end-to-end relay and back to the AN can be significant. For example, a geosynchronous satellite functioning as an end-to-end relay will generate round-trip delay of approximately 250 milliseconds (ms). To keep this loop stable in the presence of the large delay, a very low loop bandwidth can be used. For a 250 ms delay, the PLL closed loop bandwidth may typically be less than one Hz. In such cases, high-stability oscillators may be used on both the satellite and the AN to maintain reliable phase lock, as indicated by block **2437** in FIG. **39** (see below).

[0189] In some cases, the access node beacon signal is a burst signal that is only transmitted during calibration intervals. During the calibration interval, communications data is not transmitted to eliminate this interference to the access node beacon signal. Since no communications data is transmitted during the calibration interval, the transmitted power of the access node beacon signal can be large, as compared to what would be required if it were broadcast during communication data. This is because there is no concern of causing interference with the communications data (the communications data is not present at this time). This technique enables a strong signal-to-noise ratio (SNR) for the access node beacon signal when it is transmitted during the calibration interval. The frequency of occurrence of the calibration intervals is the reciprocal of the elapsed time between calibration intervals. Since each calibration interval provides a sample of the phase to the PLL, this calibration frequency is the sample rate of this discrete time PLL. In some cases, the sample rate is high enough to support the closed loop bandwidth of the PLL with an insignificant amount of aliasing. The product of the calibration frequency (loop sample rate) and the calibration interval represents the fraction of time the end-to-end relay cannot be used for communications data without additional interference from the channel sounding probe signal. In some cases, values of less than 0.1 are used and in some cases, values of less than 0.01 are used.

[0190] FIG. **39** is a block diagram of an example AN transceiver **2409**. The input **2408** to the AN transceiver **2409** receives end-to-end return link signals received by the AN **515** (e.g., for one of a plurality of frequency sub-bands). The input **2408** is coupled to the input **2501** of a down converter (D/C) **2503**. The output of the D/C **2503** is coupled to an analog to digital converter (A/D) **2509**. The output of the A/D **2509** is coupled to an Rx time adjuster **2515** and/or Rx phase adjuster **2517**. Rx time adjuster **2515** and Rx phase adjuster **2517** may illustrate aspects of the receive timing and phase adjuster **4024** of FIG. **38**. The D/C **2503** is a quadrature down converter. Accordingly, the D/C **2503** outputs an in-phase and quadrature output to the A/D **2509**. The received signals may include communications signals (e.g., a composite of return uplink signals transmitted by user terminals), access node beacon signals (e.g., transmitted from the same AN and/or other ANs) and a relay beacon signal. The digital samples are coupled to a relay beacon signal demodulator **2511**.

The relay beacon signal demodulator **2511** demodulates the relay beacon signal. In addition, the relay beacon signal demodulator **2511** generates a time control signal **2513** and a phase control signal **2514** to remove feeder link impairments based on the received relay beacon signal. Such impairments include Doppler, AN frequency error, scintillation effects, path length changes, etc. By performing coherent demodulation of the relay beacon signal, a phase locked loop (PLL) may be used to correct for most or all of these errors. By correcting for the errors in the relay beacon signal, corresponding errors in the communication signals and access node beacon signals on the feeder link are corrected as well (e.g., since such errors are common to the relay beacon signal, the access node beacon signals and the communications signals). After feeder link impairment removal, the end-to-end return link communication signal from a user terminal **517** nominally have the same frequency error at each of the M ANs **515**. That common error includes the user terminal frequency error, the user link Doppler, the end-to-end relay frequency translation error, and the relay beacon signal frequency error.

[0191] The digital samples, with feeder link impairments removed, are coupled to a multiplexer **2518**, which may be an example of the multiplexer **4004** of FIG. **38**. The multiplexer **2518** associates (e.g., time stamps) the samples with the relay timing information **2520** from the relay beacon signal demodulator **2511**. The output of the multiplexer **2518** is coupled to the output port **2410** of the AN transceiver **2409**. The output port **2410** is coupled to the multiplexer **2413** and through the interface **2415** (see FIG. **40**) to the CPS **505**. The CPS **505** can then use the time stamps associated with the received digital samples to align the digital samples received from each of the ANs **515**. Additionally or alternatively, feeder link impairment removal may be performed at the CPS **505**. For example, digital samples of the end-to-end return link signals with the embedded relay beacon signal may be sent from the AN **515** to the CPS **505**, and the CPS **505** may use the synchronization timing information (e.g., embedded relay beacon signal) in each of the composite return signals to determine respective adjustments for the respective composite return signals to compensate for downlink channel impairment.

[0192] An access node beacon signal **2530** may be generated locally by an access node beacon signal generator **2529**. An access node beacon signal demodulator **2519** demodulates the access node beacon signal received by the AN **515** (e.g., after being relayed by the end-to-end relay and received at input **2408**). The relay beacon signal demodulator **2511** provides a received relay timing and phase information signal **2521** to a controller **2523**. The controller **2523** also receives a relayed transmit timing and phase information signal **2525** from the access node beacon signal demodulator **2519**. The controller **2523** compares the received relay timing and phase information with the relayed transmit timing and phase information and generates a coarse time adjust signal **2527**. The coarse time adjust signal **2527** is coupled to the access node beacon signal generator **2529**. The access node beacon signal generator **2529** generates the access node beacon signal **2530** with embedded transmit timing information to be transmitted from the AN **515** to the end-to-end relay **503**. As noted in the discussion above, the difference between the relay timing and phase information (embedded in the relay beacon signal) and the transmit time and phase information (embedded in the access node beacon signal) is used to adjust the transmit timing and phase information to synchronize the relayed transmit timing and phase information with the received relay timing and phase information. Coarse time is adjusted by the signal **2527** to the access node beacon signal generator **2529** and fine time is adjusted by the signal **2540** to the Tx time adjuster **2539**. With the relayed transmit timing and phase information **2525** from the access node beacon signal demodulator **2519** synchronized with the received relay timing and phase information **2521**, the access node beacon signal generator **2529** generates timestamps **2531** that assist in the synchronization of the access node beacon signal **2530** and the access node-specific forward signal from the CPS **505** that is transmitted. That is, data samples from the CPS **505** are received on input port **2423** together with timestamps **2535** that indicate when the associated data samples is desired to arrive at the end-to-end relay **503**. A buffer, time align and sum module **2537** buffers the data

samples coupled from the CPS **505** and sums them with the samples from the access node beacon signal generator **2529** based on the timestamps **2535**, **2531**. PN samples and communication data samples with identical times, as indicated by the time stamps, are summed together. In this example, the multiple beam signals ($x_{\text{sub.k}}(n) \cdot b_{\text{sub.k}}$) are summed together in the CPS **505** and the access node-specific forward signal comprising a composite of the multiple beam signals is sent to the AN by the CPS **505**.

[0193] When aligned properly by the ANs, the data samples arrive at the end-to-end relay **503** at the desired time (e.g., at the same time that the same data samples from other ANs arrive). A transmit time adjuster **2539** performs fine time adjustments based on a fine time controller output signal **2540** from the time controller module **2523**. A transmit phase adjuster **2541** performs phase adjustments to the signal in response to a phase control signal **2542** generated by the access node beacon signal demodulator **2519**. Transmit time adjuster **2539** and transmit phase adjuster **2541** may illustrate, for example, aspects of the transmit timing and phase compensator **4020** of FIG. **38**.

[0194] The output of the transmit phase adjuster **2541** is coupled to the input of a digital to analog converter (D/A) **2543**. The quadrature analog output from the D/A **2543** is coupled to an up-converter (U/C) **2545** to be transmitted by the HPA **2433** (see FIG. **40**) to the end-to-end relay **503**. An amplitude control signal **2547** provided by the access node beacon signal demodulator **2519** provides amplitude feedback to the U/C **2545** to compensate for items such as uplink rain fades.

[0195] In some cases, the PN code used by each AN for the access node beacon signal **2530** is different from that used by every other AN. In some cases, the PN codes in the access node beacon signals are each different from the relay PN code used in the relay beacon signal. Accordingly, each AN **515** may be able to distinguish its own access node beacon signal from those of the other ANs **515**. ANs **515** may distinguish their own access node beacon signals from the relay beacon signal.

[0196] As was previously described, the end-to-end gain from any point in the coverage area to any other point in the area is a multipath channel with L different paths that can result in very deep fades for some point to point channels. The transmit diversity (forward link) and receive diversity (return link) are very effective in mitigating the deep fades and enable the communications system to work. However for the access node beacon signals, the transmit and receive diversity is not present. As a result, the point-to-point link of a loopback signal, which is the transmission of the signal from an AN back to the same AN, can experience end-to-end gains that are much lower than the average. Values of 20 dB below the average can occur with a large number of receive/transmit signal paths (L). These few low end-to-end gains result in lower SNR for those ANs and can make link closure a challenge. Accordingly, in some cases, higher gain antennas are used at the ANs.

Alternatively, referring to the example transponder of FIG. **16**, a phase adjuster **418** may be included in each of the receive/transmit signal paths. The phase adjuster **418** may be individually adjusted by the phase shift controller **427** (for example, under control of a telemetry, tracking, and command (TT&C) link from an Earth-based control center). Adjusting the relative phases may be effective in increasing the end-to-end gains of the low-gain loopback paths. For example, an objective may be to choose phase shift settings to increase the value of the worst case loopback gain (gain from an AN back to itself). Note that the selection of phases generally does not change the distribution of the gains when evaluated for all points in the coverage area to all other points in the coverage area, but it can increase the gains of the low gain loopback paths.

[0197] To elaborate, consider the set of gains from each of M ANs **515** to all of the other ANs **515**. There are $M \cdot \text{sup.}2$ gains, of which, only M of them are loopback paths. Consider two gain distributions, the first is the total distribution of all paths ($M \cdot 2$) which can be estimated by compiling a histogram of all $M \cdot \text{sup.}2$ paths. For ANs distributed evenly over the entire coverage area, this distribution may be representative of the distribution of the end-to-end gain from any point to any other point in the coverage area. The second distribution is the loopback gain distribution (loopback distribution) which can be estimated by compiling a histogram of just the M loopback paths. In many cases, custom selection of the receive/transmit signal path phase settings

(and optionally gain settings) does not provide a significant change to the total distribution. This is especially the case with random or interleaved mappings of transmit to receive elements. However, in most cases, the loopback distribution can be improved with custom selection (as opposed to random values) of the phase (and optionally gain) settings. This is because the set of loopback gains consist of M paths (as opposed to $M_{\text{sup.2}}$ total paths) and the number of degrees of freedom in the phase and gain adjustments is L . Often times L is on the same order as M which enables significant improvement in low loopback gains with custom phase selection. Another way of looking at this is that the custom phase selection is not necessarily eliminating low end-to-end gains, but rather moving them from the set of loopback gains (M members in the set) to the set of non-loopback gains ($M_{\text{sup.2}} - M$ members). For non-trivial values of M , the larger set is often much larger than the former.

[0198] An AN **515** may process one or more frequency sub-bands. FIG. **40** is a block diagram of an example AN **515** in which multiple frequency sub-bands are processed separately. On the end-to-end return link **523** (see FIG. **5**), the AN **515** receives the return downlink signals **527** from the end-to-end relay **503** through an LNA **2401**. The amplified signals are coupled from the LNA **2401** to a power divider **2403**. The power divider **2403** splits the signal into multiple output signals. Each signal is output on one of the output ports **2405**, **2407** of the power divider **2403**. One of the output ports **2407** may be provided as a test port. The other ports **2405** are coupled to an input **2408** of a corresponding one of multiple AN transceivers **2409** (only one shown). The AN transceivers **2409** process the signals received on corresponding sub-bands. The AN transceiver **2409** performs several functions, discussed in detail above. The outputs **2410** of the AN transceivers **2409** are coupled to input ports **2411** of a sub-band multiplexer **2413**. The outputs are combined in the sub-band multiplexer **2413** and output to a distribution network interface **2415**. The interface **2415** provides an interface for data from/to AN **515** to/from the CPS **505** over the distribution network (see FIG. **5**). Processing frequency sub-bands may be advantageous in reducing performance requirements on the RF components used to implement the end-to-end relay and AN. For example, by splitting up 3.5 GHz of bandwidth (e.g., as may be used in a Ka-band system) into seven sub-bands, each sub-band is only 500 MHz wide. That is, each of the access node-specific forward signals may include multiple sub-signals associated with the different sub-bands (e.g., associated with different subsets of the forward user beam coverage areas), and the AN transceivers **2409** may upconvert the sub-signals to different carrier frequencies. This bandwidth splitting may allow for lower tolerance components to be used since amplitude and phase variations between different sub-bands may be compensated by separate beamforming weights, calibration, etc. for the different sub-bands. Of course, other systems may use a different number of sub-bands and/or test ports. Some cases may use a single sub-band and may not include all the components shown here (e.g., omitting power divider **2403** and mux **2413**).

[0199] On the end-to-end forward link **501**, data is received from the CPS **505** by the interface **2415**. The received data is coupled to an input **2417** of a sub-band de-multiplexer **2419**. The sub-band de-multiplexer **2419** splits the data into multiple data signals. The data signals are coupled from output ports **2421** of the sub-band de-multiplexer **2419** to input ports **2423** of the AN transceivers **2409**. Output ports **2425** of the AN transceivers **2409** are coupled to input ports **2427** of the summer module **2429**. The summer module **2429** sums the signals output from the seven AN transceivers **2409**. An output port **2431** of the summer module **2429** couples the output of the summer module **2429** to the input port **2433** of a high power amplifier (HPA) **2435**. The output of the HPA **2435** is coupled to an antenna (not shown) that transmits the signals output to the end-to-end relay **503**. In some cases, an ultra-stable oscillator **2437** is coupled to the AN transceivers **2409** to provide a stable reference frequency source.

Beam Weight Computation

[0200] Returning to FIG. **8** which is an example description of signals on the return link, a mathematical model of the end-to-end return link may be used to describe the link as:

$$\begin{aligned} [00001] \quad y &= \text{Bret} \cdot \text{Math. Ct} E(\text{Ar}x + n_{ul}) + n_{dl} \cdot \text{Math.} \\ &= \text{Bret}[\text{Hret}x + \text{Ct}E n_{ul} + n_{dl}] \end{aligned} \quad \text{EQ. 1}$$

where, [0201] x is the $K \times 1$ column vector of the transmitted signal. In some cases, the magnitude squared of every element in x is defined to be unity (equal transmit power). In some cases, this may not always be the case. [0202] y is the $K \times 1$ column vector of the received signal after beamforming. [0203] Ar is the $L \times K$ return uplink radiation matrix. The element $a_{\text{sub.lk}}$ contains the gain and phase of the path from a reference location located in beam K to the $l_{\text{sup.th}}$ (the letter “el”) receive antenna element **406** in the Rx array. In some cases, the values of the return uplink radiation matrix are stored in the channel data store **941** (see FIG. **30**). [0204] E is the $L \times L$ payload matrix. The element $e_{\text{sub.ij}}$ defines the gain and phase of the signal from the $j_{\text{sup.th}}$ antenna element **406** in the receive array to an $i_{\text{sup.th}}$ antenna element **409** in the transmit array. In some cases, aside from incidental crosstalk between the paths (resulting from the finite isolation of the electronics), the E matrix is a diagonal matrix. The matrix E can be normalized such that the sum of the magnitude squared of all elements in the matrix is L . In some cases, the values of the payload matrix are stored in the channel data store **941** (see FIG. **29**). [0205] Ct is the $M \times L$ return downlink radiation matrix. The element $c_{\text{sub.ml}}$ contains the gain and phase of the path from $l_{\text{sup.th}}$ (the letter “el”) antenna element in the Tx array to an $m_{\text{sup.th}}$ AN **515** from among the M ANs **515**. In some cases, the values of the return downlink radiation matrix are stored in the channel data store **941** (see FIG. **29**). [0206] Hret is the $M \times K$ return channel matrix, which is equal to the product $C_{\text{sub.t}} \times E \times A_{\text{sub.r}}$. [0207] $n_{\text{sub.ul}}$ is an $L \times 1$ noise vector of complex Gaussian noise. The covariance of the uplink noise is $E[n_{\text{sub.ul}} n_{\text{sub.ul}}^H] = 2\sigma_{\text{sub.ul}}^2 I_{\text{sub.L}}$. $I_{\text{sub.L}}$ is the $L \times L$ identity matrix. [0208] $\sigma_{\text{sub.ul}}^2$ is noise variance. $\sigma_{\text{sub.ul}}^2$ is experienced on the uplink, while $\sigma_{\text{sub.dl}}^2$ is experienced on the downlink. $n_{\text{sub.dl}}$ is an $M \times 1$ noise vector of complex Gaussian noise. The covariance of the downlink noise is $E[n_{\text{sub.dl}} n_{\text{sub.dl}}^H] = 2\sigma_{\text{sub.dl}}^2 I_{\text{sub.M}}$. $I_{\text{sub.M}}$ is the $M \times M$ identity matrix. [0209] Bret is the $K \times M$ matrix of end-to-end return link beam weights.

[0210] Examples are generally described above (e.g., with reference to FIGS. **6-11**) in a manner that assumes certain similarities between forward and return end-to-end multipath channels. For example, the forward and return channel matrices are described above with reference generally to M , K , E , and other models. However, such descriptions are intended only to simplify the description for added clarity, and are not intended to limit examples only to cases with identical configurations in the forward and return directions. For example, in some cases, the same transponders are used for both forward and return traffic, and the payload matrix E can be the same for both forward and return end-to-end beamforming (and corresponding beam weight computations), accordingly. In other cases, different transponders are used for forward and return traffic, and a different forward payload matrix (E_{fwd}) and a return payload matrix (E_{ret}) can be used to model the corresponding end-to-end multipath channels and to compute corresponding beam weights. Similarly, in some cases, the same M ANs **515** and K user terminals **517** are considered part of both the forward and return end-to-end multipath channels. In other cases, M and K can refer to different subsets of ANs **515** and/or user terminals **517**, and/or different numbers of ANs **515** and/or user terminals **517**, in the forward and return directions.

[0211] Beam weights may be computed in many ways to satisfy system requirements. In some cases, they are computed after deployment of the end-to-end relay. In some cases, the payload matrix E is measured before deployment. In some cases, beam weights are computed with the objective to increase the signal to interference plus noise (SINR) of each beam and can be computed as follows:

$$[00002] \quad \text{Bret} = (R^{-1} H)^H \quad \text{EQ. 2, 3} \quad R = 2 \quad {}_{\text{dl}}^2 I_M + 2 \quad {}_{\text{ul}}^2 C_t E E^H C_t^H + H H^H$$

where R is the covariance of the received signal and $(*)_{\text{sup.H}}$ is the conjugate transpose

(Hermetian) operator.

[0212] The k, m element of the $K \times M$ return beam weight matrix B_{ret} provides the weights to form the beam to the m .sup.th AN 515 from a user terminal in the k .sup.th user beam. Accordingly, in some cases, each of the return beam weights used to form return user beams are computed by estimating end-to-end return gains (i.e., elements of the channel matrix H_{ret}) for each of the end-to-end multipath channels (e.g., each of the end-to-end return multipath channels).

[0213] EQ. 2 holds true where R is the covariance of the received signal as provided in EQ. 3. Therefore, when all of the matrices of EQ. 1, 2 and 3 are known, the beam weights used to form end-to-end beams may be directly determined.

[0214] This set of beam weights reduces the mean squared error between x and y . It also increases the end-to-end signal to noise plus interference ratio (SINR) for each of the K end-to-end return link signals 525 (originating from each of the K beams).

[0215] The first term $2\sigma_{sub.dl.sup.2I.sub.M}$ in EQ. 3 is the covariance of the downlink noise (which is uncorrelated). The second term $2\sigma_{sub.dl.sup.2C.sub.tEE.sup.HC.sub.t.sup.H}$ in EQ. 3 is the covariance of the uplink noise (which is correlated at the ANs). The third term $HH_{sup.H}$ in EQ. 3 is the covariance of the signal. Setting the variance of the uplink noise to zero and ignoring the last term ($HH_{sup.H}$) results a set of weights that increases the signal to downlink noise ratio by phase-aligning the received signals on each of the M ANs 515. Setting the downlink noise variance to zero and ignoring the $3_{sup.rd}$ term results in a set of weights that increases the uplink SINR. Setting both the uplink and downlink noise variances to zero results in a de-correlating receiver that increases the carrier to interference (C/I) ratio.

[0216] In some cases, the beam weights are normalized to make the sum of the magnitude squared of any row of B_{ret} sum to unity.

[0217] In some cases, the solution to EQ. 2 is determined by a priori knowledge of the matrices A_r , C_t , and E as well as the variances of the noise vectors $n_{sub.ul}$ and $n_{sub.dl}$. Knowledge of the element values of the matrices can be obtained during measurements made during the manufacturing and testing of relevant components of the end-to-end relay. This may work well for systems where one does not expect the values in the matrices to change significantly during system operation. However, for some systems, especially ones operating in higher frequency bands, such expectations may not be present. In such cases, the matrices A_r , C_t , and E may be estimated subsequent to the deployment of a craft (such as a satellite) on which the end-to-end relay is disposed.

[0218] In some cases where a priori information is not used to set the weights, the solution to EQ. 2 may be determined by estimating the values of R and H . In some cases, designated user terminals 517 in the center of each user beam coverage area 519 transmit known signals x during calibration periods. The vector received at an AN 515 is:

$$[00003] \quad u = Hx + C_t E n_{ul} + n_{dl} \quad \text{EQ. 4}$$

[0219] In an example, the CPS 505 estimates the values of R and H based on the following relationships:

$$[00004] \quad \hat{R} = \text{Math. } uu^H \quad \text{EQ. 5} \quad \hat{H} = [\hat{p}_1, \hat{p}_2, \dots, \hat{p}_K] \quad \text{EQ. 6} \quad \hat{p}_K = \text{Math. } u\tilde{x}_k^* \quad \text{EQ. 7}$$

[0220] $\{\text{circumflex over (R)}\}$ is an estimate of the covariance matrix R , $\hat{H}$ is an estimate of channel matrix H and $\{\text{circumflex over (p)}\}_{sub.k}$ is an estimate of the correlation vector, $\{\text{tilde over (x)}\}_{sub.k}^*$ is the conjugate of the k .sup.th component of the transmitted vector with the frequency error introduced by the uplink transmission. In some cases, no return communication data is transmitted during the calibration period. That is, only calibration signals that are known to the ANs are transmitted on the end-to-end return link during the calibration period in order to allow the value of $\{\text{circumflex over (p)}\}_{sub.k}$ to be determined from the received vector u using the equation above. This, in turn allows the value of $\hat{H}$ to be determined. Both the covariance matrix estimate $\{\text{circumflex over (R)}\}$ and the channel matrix estimate $\hat{H}$ are determined based on the

signals received during the calibration period.

[0221] In some cases, the CPS **505** can estimate the covariance matrix $\{\text{circumflex over (R)}\}$ while communication data is present (e.g., even when x is unknown). This may be seen from the fact that 1 is determined based only on the received signal u . Nonetheless, the value of $\hat{H}$ is estimated based on signals received during a calibration period during which only calibration signals are transmitted on the return link.

[0222] In some cases, estimates of both the channel matrix $\hat{H}$ and the covariance matrix $\{\text{circumflex over (R)}\}$ are made while communication data is being transmitted on the return link. In this case, the covariance matrix $\{\text{circumflex over (R)}\}$ is estimated as noted above. However, the value of x is determined by demodulating the received signal. Once the value of x is known, the channel matrix may be estimated as noted above in EQ. 6 and EQ. 7.

[0223] The signal and interference components of the signal after beamforming are contained in the vector $\text{Bret } H x$. The signal and interference powers for each of the beams are contained in the $K \times K$ matrix $\text{Bret } H$. The power in the $k.\text{sup.th}$ diagonal element of $\text{Bret } H$ is the desired signal power from beam k . The sum of the magnitude squared of all elements in row k except the diagonal element is the interference power in beam k . Hence the C/I for beam k is:

$$[00005] \left(\frac{C}{I}\right)_k = \frac{\text{Math. } s_{kk} \cdot \text{Math. } ^2}{\sum_{j \neq k} \text{Math. } s_{kj} \cdot \text{Math. } ^2} \quad \text{EQ. 8}$$

where $s.\text{sub.}kj$ are the elements of $\text{Bret } H$. The uplink noise is contained in the vector $\text{Bret } C_t E_{n.\text{sub.}ul}$, which has a $K \times K$ covariance matrix of $2\sigma.\text{sub.}ul.\text{sup.2} \text{Bret } C_t E E.\text{sup.}H C_t.\text{sup.}H$. The $k.\text{sup.th}$ diagonal element of the covariance matrix contains the uplink noise power in beam k . The uplink signal to noise ratio for beam k is then computed as:

$$[00006] \left(\frac{S}{N_{ul}}\right)_k = \frac{\text{Math. } s_{kk} \cdot \text{Math. } ^2}{t_{kk}} \quad \text{EQ. 9}$$

[0224] where $t.\text{sub.}kk$ is the $k.\text{sup.th}$ diagonal element of the uplink covariance matrix. The downlink noise is contained in the vector $\text{Bret } n.\text{sub.}dl$, which has a covariance of $2\sigma.\text{sub.}dl.\text{sup.2} I.\text{sub.}k$ by virtue of the normalized beam weights. Hence the downlink signal to noise ratio is:

$$[00007] \left(\frac{S}{N_{dl}}\right)_k = \frac{\text{Math. } s_{kk} \cdot \text{Math. } ^2}{2\sigma_{dl}^2} \quad \text{EQ. 10}$$

[0225] The end-to-end SINR is the combination of EQ. 8-10:

$$[00008] \text{SINR}_k = \left[\left(\frac{C}{I}\right)_k^{-1} + \left(\frac{S}{N_{ul}}\right)_k^{-1} + \left(\frac{S}{N_{dl}}\right)_k^{-1} \right]^{-1} \quad \text{EQ. 11}$$

[0226] The above equations describe how to calculate the end-to-end SINR given the payload matrix E . The payload matrix may be constructed by intelligent choice of the gain and phases of each of the elements of E . The gain and phase of the diagonal elements of E that optimize some utility metric (which is generally a function of the K beam SINR's as computed above) may be selected and implemented by setting the phase shifter **418** in each of the L transponders **411**. Candidate utility functions include, but are not limited to, sum of $\text{SINR}.\text{sub.}k$ (total SINR), sum of $\text{Log}(1+\text{SINR}.\text{sub.}k)$ (proportional to total throughput) or total power in the channel matrix, H . In some cases, the improvement in the utility function by customizing the gains and phases is very small and insignificant. This is sometimes the case when random or interleaved mappings of antenna elements are used. In some cases, the utility function can be improved by a non-trivial amount by custom selection of the receive/transmit signal gain and phase.

[0227] Returning to FIG. **9**, a mathematical model of the end-to-end forward link **501** may be used to describe the link **501** as:

$$[00009] \begin{aligned} y &= A t E [C r B f w d x + n_{ul}] + n_{dl} \\ &= H f w d B f w d x + A E n_{ul} + n_{dl} \end{aligned} \quad \text{EQ. 12}$$

where, [0228] x is the $K \times 1$ column vector of the transmitted signal. The magnitude squared of

every element in $\mathbf{x}$ is defined to be unity (equal signal power). In some cases, unequal transmit power may be achieved by selection of the forward beam weights. [0229] $\mathbf{y}$ is the $K \times 1$ column vector of the received signal. [0230] $\mathbf{C}_r$ is the $L \times M$ forward uplink radiation matrix. The element $c_{\text{sub.lm}}$ contains the gain and phase of the path **2002** from $m_{\text{sup.th}}$ AN **515** to the $l_{\text{sup.th}}$ (letter “el”) receive antenna element **406** of the Rx array of antenna on the end-to-end relay **503**. In some cases, the values of the forward uplink radiation matrix are stored in the channel data store **921** (see FIG. **29**). [0231] $\mathbf{E}$ is the $L \times L$ payload matrix. The element $e_{\text{sub.ij}}$ defines the gain and phase of the signal from $j_{\text{sup.th}}$ receive array antenna element to the $i_{\text{sup.th}}$ antenna element of the transmit array. Aside from incidental crosstalk between the paths (resulting from the finite isolation of the electronics), the $\mathbf{E}$ matrix is a diagonal matrix. In some cases, the matrix $\mathbf{E}$ is normalized such that the sum of the magnitude squared of all elements in the matrix is L . In some cases, the values of the payload matrix are stored in the channel data store **921** (see FIG. **29**). [0232] $\mathbf{A}_t$ is the $K \times L$ forward downlink radiation matrix. The element $a_{\text{sub.kl}}$ contains the gain and phase of the path from antenna element L (letter “el”) in the Tx array of the end-to-end relay **503** to a reference location in user beam k . In some cases, the values of the forward downlink radiation matrix are stored in the channel data store **921** (see FIG. **29**). [0233] $\mathbf{H}_{\text{fwd}}$ is the $K \times M$ forward channel matrix, which is equal to the product $\mathbf{A}_{\text{sub.t}} \mathbf{E} \mathbf{C}_{\text{sub.r}}$. $\mathbf{n}_{\text{sub.ul}}$ is an $L \times 1$ noise vector of complex Gaussian noise. The covariance of the uplink noise is:

$E[\mathbf{n}_{\text{sub.ul}} \mathbf{n}_{\text{sub.ul}}^{\text{sup.H}}] = 2\sigma_{\text{sub.ul}}^2 \mathbf{I}_{\text{sub.L}}$, [0234] where $\mathbf{I}_{\text{sub.L}}$ is the $L \times L$ identity matrix. [0235] $\mathbf{n}_{\text{sub.dl}}$ is an $K \times 1$ noise vector of complex Gaussian noise. The covariance of the downlink noise is:

$E[\mathbf{n}_{\text{sub.dl}} \mathbf{n}_{\text{sub.dl}}^{\text{sup.H}}] = 2\sigma_{\text{sub.dl}}^2 \mathbf{I}_{\text{sub.K}}$, [0236] where $\mathbf{I}_{\text{sub.K}}$ is the $K \times K$ identity matrix. [0237] $\mathbf{B}_{\text{fwd}}$ is the $M \times K$ beam weight matrix of end-to-end forward link beam weights. [0238] The beam weights for user beam k are the elements in column k of $\mathbf{B}_{\text{fwd}}$. Unlike the return link, the C/I for beam k is not determined by the beam weights for beam k . The beam weights for beam k determine the uplink signal to noise ratio (SNR) and the downlink SNR, as well as the carrier (C) power in the C/I. However, the interference power in beam k is determined by the beam weights for all of the other beams, except for beam k . In some cases, the beam weight for beam k is selected to increase the SNR. Such beam weights also increase the C/I for beam k , since C is increased. However, interference may be generated to the other beams. Thus, unlike in the case of the return link, optimal beam weights are not computed on a beam-by-beam basis (independent of the other beams).

[0239] In some cases, beam weights (including the radiation and payload matrices used to compute them) are determined after deployment of the end-to-end relay. In some cases, the payload matrix $\mathbf{E}$ is measured before deployment. In some cases, one can compute a set of beam weights by using the interference created in the other beams by beam k and counting it as the interference in beam k . Although this approach may not compute optimum beam weights, it may be used to simplify weight computation. This allows a set of weights to be determined for each beam independent of all other beams. The resulting forward beam weights are then computed similar to the return beam weights:

$$[00010] \mathbf{B}_{\text{fwd}} = \mathbf{H}^H \mathbf{R}^{-1}, \text{ where } (\text{EQ. 13}) \quad \mathbf{R} = 2 \sum_{\text{dl}} \mathbf{I}_K + 2 \sum_{\text{ul}} \mathbf{A}_t \mathbf{E} \mathbf{E}^H \mathbf{A}_t^H + \mathbf{H} \mathbf{H}^H \quad (\text{EQ. 14})$$

[0240] The first term $2\sigma_{\text{sub.dl}}^2 \mathbf{I}_{\text{sub.K}}$ in EQ. 14 is the covariance of the downlink noise (uncorrelated). The second term $2\sigma_{\text{sub.ul}}^2 \mathbf{A}_t \mathbf{E} \mathbf{E}^H \mathbf{A}_t^H$ is the covariance of the uplink noise (which is correlated at the ANs). The third term $\mathbf{H} \mathbf{H}^H$ is the covariance of the signal. Setting the variance of the uplink noise to zero and ignoring the last term ($\mathbf{H} \mathbf{H}^H$) results in a set of weights that increases the signal to downlink noise ratio by phase aligning the received signals at the M ANs **515**. Setting the downlink noise variance to zero and ignoring the 3.sup.rd

term results in a set of weights that increases the uplink SNR. Setting both the uplink and downlink noise variances to zero results in a de-correlating receiver that increases the C/I ratio. For the forward link, the downlink noise and interference generally dominate. Therefore, these terms are generally useful in the beam weight computation. In some cases, the second term in EQ. 14 (the uplink noise) is insignificant compared to the first term (the downlink noise). In such cases, the second term can be ignored in co-variance calculations, further simplifying the calculation while still yielding a set of beam weights that increases the end-to-end SINR.

[0241] As with the return link, the beam weights may be normalized. For transmitter beam weights with equal power allocated to all K forward link signals, each column of Bfwd may be scaled such that the sum of the magnitude squared of the elements in any column will sum to unity. Equal power sharing will give each of the signals the same fraction of total AN power (total power from all ANs allocated to signal x.sub.k). In some cases, for forward links, an unequal power sharing between forward link signals is implemented. Accordingly, in some cases, some beam signals get more than an equal share of total AN power. This may be used to equalize the SINR in all beams or give more important beams larger SINR's than lesser important beams. To create the beam weights for unequal power sharing, the M×K equal power beam weight matrix, Bfwd, is post multiplied by a K×K diagonal matrix, P, thus the new Bfwd=Bfwd P. Let

$P = \text{diag}(\sqrt{\{p_{\text{sub}.k}\}})$,

then the squared valued of the k.sup.th diagonal element represents the power allocated to user signal x.sub.k. The power sharing matrix P is normalized such that the sum or the square of the diagonal elements equals K (the non-diagonal elements are zero).

[0242] In some cases, the solution to EQ. 13 is determined by a priori knowledge of the matrices At, Cr, and E, as well as the variances of the noise vectors n.sub.ul and n.sub.dl. In some cases, knowledge of the matrices can be obtained during measurements made during the manufacturing and testing of relevant components of the end-to-end relay. This can work well for systems where one does not expect the values in the matrices to change significantly from what was measured during system operation. However, for some systems, especially ones operating in higher frequency bands, this may not be the case.

[0243] In some cases where a priori information is not used to set the weights, the values of R and H for the forward link can be estimated to determine the solution to EQ. 13. In some cases, ANs transmit a channel sounding probe during calibration periods. The channel sounding probes can be many different types of signals. In one case, different, orthogonal and known PN sequences are transmitted by each AN. The channel sounding probes may be pre-corrected in time, frequency, and/or phase to remove the feeder link impairments (as discussed further below). All communication data may be turned off during the calibration interval to reduce the interference to the channel sounding probes. In some cases, the channel sounding probes can be the same signals as those used for feeder link impairment removal.

[0244] During the calibration interval, a terminal in the center of each beam may be designated to receive and process the channel sounding probes. The K×1 vector, u, of received signals during the calibration period is $u = Hx + A_t E n_{\text{sub}.ul} + n_{\text{sub}.dl}$ where x is the M×1 vector of transmitted channel sounding probes. In some cases, each designated terminal first removes the incidental frequency error (resulting from Doppler shift and terminal oscillator error), and then correlates the resulting signal with each of the M known, orthogonal PN sequences. The results of these correlations are M complex numbers (amplitude and phase) for each terminal and these results are transmitted back to the CPS via the return link. The M complex numbers calculated by the terminal in the center of the kl beam can be used to form the kl row of the estimate of the channel matrix, Hi. By using the measurements from all of K designated terminals, an estimate of the entire channel matrix is obtained. In many cases, it is useful to combine the measurement from multiple calibration intervals to improve the estimate of the channel matrix. Once the estimate of the

channel matrix is determined, an estimate of the covariance matrix, $\{\text{circumflex over (R)}\}$, can be determined from EQ. 14 using a value of 0 for the second term. This may be a very accurate estimate of the covariance matrix if the uplink noise (the second term in EQ. 14) is negligible relative to the downlink noise (the first term in EQ. 14). The forward link beam weights may then be computed by using the estimates of the channel matrix and covariance matrix in EQ. 13. Accordingly, in some cases, the computation of beam weights comprises estimating end-to-end forward gains (i.e., the values of the elements of the channel matrix H_{fwd}) for each of the end-to-end forward multipath channels between an AN 515 and a reference location in a user beam coverage area. In other cases, computation of beam weights comprises estimating end-to-end forward gains for $K \times M$ end-to-end forward multipath channels from M ANs 515 to reference locations located within K user beam coverage areas.

[0245] The signal and interference components of the signal after beamforming are contained in the vector $H B_{\text{fwd}} x$ (product of H , B_{fwd} , x). The signal and interference powers for each of the beams are contained in the $K \times K$ matrix $H B_{\text{fwd}}$. The power in the k .sup.th diagonal element of $H B_{\text{fwd}}$ is the desired signal power intended for beam k . The sum of the magnitude squared of all elements in row k except the diagonal element is the interference power in beam k . Hence the C/I for beam k is:

$$[00011] \left(\frac{C}{I}\right)_k = \frac{\text{Math. } s_{kk} \cdot \text{Math. } ^2}{\text{Math. }_{j \neq k} s_{kj} \cdot \text{Math. } ^2} \quad (\text{EQ. 15})$$

where s_{kj} are the elements of $H B_{\text{fwd}}$. The uplink noise is contained in the vector $A_t E n_{\text{sub.ul}}$, which has a $K \times K$ covariance matrix of $2\sigma_{\text{sub.ul}} \cdot 2A_t E E_{\text{sup}} H A_t \cdot \text{sub.t.} \cdot H$. The k .sup.th diagonal element of the covariance matrix contains the uplink noise power in beam k . The uplink signal to noise ratio for beam k is then computed as:

$$[00012] \left(\frac{S}{N_{\text{ul}}}\right)_k = \frac{\text{Math. } s_{kk} \cdot \text{Math. } ^2}{t_{kk}} \quad \text{EQ. 16}$$

where t_{kk} is the k .sup.th diagonal element of the uplink covariance matrix. The downlink noise is contained in the vector $n_{\text{sub.dl}}$, which has a covariance of $2\sigma_{\text{sub.dl}} \cdot 2I_{\text{sub.K}}$. Hence the downlink signal to noise ratio is:

$$[00013] \left(\frac{S}{N_{\text{dl}}}\right)_k = \frac{\text{Math. } s_{kk} \cdot \text{Math. } ^2}{2 \cdot \frac{2}{\text{dl}} I_K} \quad \text{EQ. 17}$$

[0246] The end-to-end SINR is the combination of EQ. 15-EQ. 17:

$$[00014] \text{SINR}_k = \left[\left(\frac{C}{I}\right)_k^{-1} + \left(\frac{S}{N_{\text{ul}}}\right)_k^{-1} + \left(\frac{S}{N_{\text{dl}}}\right)_k^{-1} \right]^{-1} \quad \text{EQ. 18}$$

[0247] The above equations describe how to calculate the end-to-end SINR given the payload matrix E . The payload matrix may be constructed by intelligent choice of the gain and phases of each of the elements of E . The gain and phase of the diagonal elements of E that optimize some utility metric (which is generally a function of the K beam SINR's as computed above) may be selected and implemented by setting the phase shifter 418 in each of the L transponders 411. Candidate utility functions include, but are not limited to, sum of $\text{SINR}_{\text{sub.k}}$ (total SINR), sum of $\text{Log}(1 + \text{SINR}_{\text{sub.k}})$ (proportional to total throughput) or total power in the channel matrix, H . In some cases, the improvement in the utility function by customizing the gains and phases is very small and insignificant. This is sometimes the case when random or interleaved mappings of antenna elements are used. In some cases, the utility function can be improved by a non-trivial amount by custom selection of the receive/transmit signal gain and phase.

Distinct Coverage Areas

[0248] Some examples described above assume that the end-to-end relay 503 is designed to service a single coverage area shared by both the user terminals 517 and the ANs 515. For example, some cases describe a satellite having an antenna subsystem that illuminates a satellite coverage area, and both the ANs 515 and the user terminals 517 are geographically distributed throughout the satellite coverage area (e.g., as in FIG. 27). The number of beams that can be formed in the satellite coverage area, and the sizes (beam coverage areas) of those beams can be affected by aspects of the

antenna subsystem design, such as number and arrangement of antenna elements, reflector size, etc. For example, realizing a very large capacity can involve deploying a large number (e.g., hundreds) of ANs 515 with sufficient spacing between the ANs 515 to allow for end-to-end beamforming. For example, as noted above with reference to FIG. 28, increasing the number of ANs 515 can increase system capacity, although with diminishing returns as the number increases. When one antenna subsystem supports both the user terminals 517 and the ANs 515, achieving such a deployment with sufficient spacing between ANs 515 can force a very wide geographical distribution of the ANs 515 (e.g., across the entire satellite coverage area, as in FIG. 27). Practically, achieving such a distribution may involve placing ANs 515 in undesirable locations, such as in areas with poor access to a high-speed network (e.g., a poor fiber infrastructure back to the CPS 505), multiple legal jurisdictions, in expensive and/or highly populated areas, etc. Accordingly, AN 515 placement often involves various tradeoffs.

[0249] Some examples of the end-to-end relay 503 are designed with multiple antenna subsystems, thereby enabling separate servicing of two or more distinct coverage areas from a single end-to-end relay 503. As described below, the end-to-end relay 503 can include at least a first antenna subsystem that services an AN area 3450, and at least a second antenna subsystem that services a user coverage area 3460. Because the user coverage area 3460 and AN area 3450 may be serviced by different antenna subsystems, each antenna subsystem can be designed to meet different design parameters, and each coverage area can be at least partially distinct (e.g., in geography, in beam size and/or density, in frequency band, etc.). For example, using such a multi-antenna subsystem approach can enable user terminals 517 distributed over one or more relatively large geographic areas 3460 (e.g., the entire United States) to be serviced by a large number of ANs 515 distributed over one or more relatively small geographic areas (e.g., a portion of the Eastern United States). For example, the AN area 3450 can be a fraction (e.g., less than one half, less than one quarter, less than one fifth, less than one tenth) of the user coverage area 3460 in physical area.

[0250] FIG. 41 is an illustration of an example end-to-end beamforming system 3400. The system 3400 is an end-to-end beamforming system that includes: a plurality of geographically distributed ANs 515; an end-to-end relay 3403; and a plurality of user terminals 517. The end-to-end relay 3403 can be an example of end-to-end relay 503 described herein. The ANs 515 are geographically distributed in an AN area 3450, the user terminals 517 are geographically distributed in a user coverage area 3460. The AN area 3450 and the user coverage area 3460 are both within the visible Earth coverage area of the end-to-end relay 3403, but the AN area 3450 is distinct from the user coverage area 3460. In other words, the AN area 3450 is not coextensive with the user coverage area 3460, but may overlap at least partially with the user coverage area 3460. However, the AN area 3450 may have a substantial (non-trivial) area (e.g., more than one-tenth, one-quarter, one-half, etc. of the AN area 3450) that does not overlap with the user coverage area 3460. For example, in some cases, at least half of the AN area 3450 does not overlap the user coverage area 3460. In some cases, the AN area 3450 and user coverage area 3460 may not overlap at all, as discussed with reference to FIG. 45C. As described above (e.g., in FIG. 5), the ANs 515 can exchange signals through a distribution network 518 with a CPS 505 within a ground segment 502, and the CPS 505 can be connected to a data source.

[0251] The end-to-end relay 3403 includes a separate feeder-link antenna subsystem 3410 and user-link antenna subsystem 3420. Each of the feeder-link antenna subsystem 3410 and the user-link antenna subsystem 3420 is capable of supporting end-to-end beamforming. For example, as described below, each antenna subsystem can have its own array(s) of cooperating antenna elements, its own reflector(s), etc. The feeder-link antenna subsystem 3410 can include an array 3415 of cooperating feeder-link constituent receive elements 3416 and an array 3415 of cooperating feeder-link constituent transmit elements 3419. The user-link antenna subsystem 3420 can include an array 3425 of cooperating user-link constituent receive elements 3426 and an array 3425 of cooperating user-link constituent transmit elements 3429. The constituent elements are

“cooperating” in the sense that the array of such constituent elements has characteristics making its respective antenna subsystem suitable for use in a beamforming system. For example, a given user-link constituent receive element **3426** can receive a superposed composite of return uplink signals **525** from multiple (e.g., some or all) user beam coverage areas **519** in a manner that contributes to forming of return user beams. A given user-link constituent transmit element **3429** can transmit a forward downlink signal **522** in a manner that superposes with corresponding transmissions from other user-link constituent transmit elements **3429** to form some or all forward user beams. A given feeder-link constituent receive element **3416** can receive a superposed composite of forward uplink signals **521** from multiple (e.g., all) ANs **515** in a manner that contributes to forming of forward user beams (e.g., by inducing multipath at the end-to-end relay **3403**). A given feeder-link constituent transmit element **3419** can transmit a return downlink signal **527** in a manner that superposes with corresponding transmissions from other feeder-link constituent transmit elements **3419** to contribute to forming of some or all return user beams (e.g., by enabling the ANs **515** to receive composite return signals that can be beam-weighted to form the return user beams).

[0252] The example end-to-end relay **3403** includes a plurality of forward-link transponders **3430** and a plurality of return-link transponders **3440**. The transponders can be any suitable type of bent-pipe signal path between the antenna subsystems. Each forward-link transponder **3430** couples a respective one of the feeder-link constituent receive elements **3416** with a respective one of the user-link constituent transmit elements **3429**. Each return-link transponder **3440** couples a respective one of the user-link constituent receive elements **3426** with a respective one of the feeder-link constituent transmit elements **3419**. Some examples are described as having a one-to-one correspondence between each user-link constituent receive element **3426** and a respective feeder-link constituent transmit element **3419** (or vice versa), or that each user-link constituent receive element **3426** is coupled with “one and only one” feeder-link constituent transmit element **3419** (or vice versa), or the like. In some such cases, one side of each transponder is coupled with a single receive element, and the other side of the transponder is coupled with a single transmit element. In other such cases, one or both sides of a transponder can be selectively coupled (e.g., by a switch, splitter, combiner, or other means, as described below) with one of multiple elements. For example, the end-to-end relay **3403** can include one feeder-link antenna subsystem **3410** and two user-link antenna subsystems **3420**; and each transponder can be coupled, on one side, to a single feeder-link element, and selectively coupled, on the other side, either to a single user-link element of the first user-link antenna subsystem **3420** or to a single user-link element of the second user-link antenna subsystem **3420**. In such selectively coupled cases, each side of each transponder can still be considered at any given time (e.g., for a particular signal-related transaction) as being coupled with “one and only one” element, or the like.

[0253] For forward communications, transmissions from the ANs **515** can be received (via feeder uplinks **521**) by the feeder-link constituent receive elements **3416**, relayed by the forward-link transponders **3430** to the user-link constituent transmit elements **3429**, and transmitted (via user downlinks **522**) by the user-link constituent transmit elements **3429** to user terminals **517** in the user coverage area **3460**. For return communications, transmissions from the user terminals **517** can be received (via user uplink signals **525**) by user-link constituent receive elements, relayed by the return-link transponders **3440** to the feeder-link constituent transmit elements **3419**, and transmitted by the feeder-link constituent transmit elements **3419** to ANs **515** in the AN area **3450** (via feeder downlink signals **527**). The full signal path from an AN **515** to a user terminal **517** via the end-to-end relay **3403** is referred to as the end-to-end forward link **501**; and the full signal path from a user terminal **517** to an AN **515** via the end-to-end relay **3403** is referred to as the end-to-end return link **523**. As described herein, the end-to-end forward link **501** and the end-to-end return link **523** can each include multiple multipath channels for forward and return communications.

[0254] In some cases, each of the plurality of geographically distributed ANs **515** has an end-to-end beam-weighted forward uplink signal **521** output. The end-to-end relay **3403** comprises an array

3415 of cooperating feeder-link constituent receive elements **3416** in wireless communication with the distributed ANs **515**, an array **3425** of cooperating user-link constituent transmit elements **3429** in wireless communication with the plurality of user terminals **517**, and a plurality of forward-link transponders **3430**. The forward-link transponders **3430** may be “bent-pipe” (or non-processing) transponders, so that each transponder outputs a signal that corresponds to the signal it receives with little or no processing. For example, each forward-link transponder **3430** can amplify and/or frequency translate its received signal, but may not perform more complex processing (e.g., there is no analog-to-digital conversion, demodulation and/or modulation, no on-board beamforming, etc.). In some cases, each forward-link transponder **3430** accepts an input at a first frequency range (e.g., 30 GHz LHCP) and outputs at a second frequency range (e.g., 20 GHz RHCP), and each return-link transponder **3440** accepts an input at the first frequency range (e.g., 30 GHz RHCP) and outputs at the second frequency range (e.g., 20 GHz LHCP). Any suitable combination of frequency and/or polarization can be used, and the user-link and feeder-link can use the same or different frequency ranges. As used herein, a frequency range refers to a set of frequencies used for signal transmission/reception and may be a contiguous range or include multiple non-contiguous ranges (e.g., such that a given frequency range may contain frequencies from more than one frequency band, a given frequency band may contain multiple frequency ranges, etc.). Each forward-link transponder **3430** is coupled between a respective one of the feeder-link constituent receive elements **3416** and a respective one of the user-link constituent transmit elements **3419** (e.g., with a one-to-one correspondence). The forward-link transponders **3430** convert superpositions of a plurality of beam-weighted forward uplink signals **521** received via the feeder-link constituent receive elements **3416** into forward downlink signals **522**. Transmission of the forward downlink signals **522** by the user-link constituent transmit elements **3429** contributes to forming a forward user beam servicing at least some of the plurality of user terminals **517** (e.g., which may be grouped into one or more user beam coverage areas **519** for transmissions via corresponding beamformed forward user beams). As described herein, the forward uplink signals **521** can be end-to-end beam-weighted and synchronized (e.g., phase-synchronized, and, if desired, time-synchronized) prior to transmission from the ANs **515**, which can enable the desired superposition of those signals **521** at the feeder-link constituent receive elements **3416**.

[0255] The transmission of the forward uplink signals **521** contributes to forming the forward user beam in the sense that the beamforming is end-to-end, as described herein; the beamforming is a result of multiple steps, including computing and applying appropriate weights to the forward uplink signals **521** prior to transmission to the relay from the ANs **515**, inducing multipath reception by the multiple forward-link transponders **3430** of the end-to-end relay **3403**, and transmitting the forward downlink signals **522** from multiple user-link constituent transmit elements **3429**. Still, for the sake of simplicity, some descriptions can refer to the forward beam as being formed by superposition of the transmitted forward downlink signals **522**. In some cases, each of the plurality of user terminals **517** is in wireless communication with the array **3425** of cooperating user-link constituent transmit elements **3429** to receive a composite (e.g., a superposition) of the transmitted forward downlink signals **522**.

[0256] In some cases, the end-to-end relay **3403** further includes an array **3425** of user-link constituent receive elements **3426** in wireless communication with the user terminals **517**, an array **3415** of cooperating feeder-link constituent transmit elements **3419** in wireless communication with the distributed ANs **515**, and a plurality of return-link transponders **3440**. The return-link transponders **3440** can be similar or identical to the forward-link transponders **3430** (e.g., bent-pipe transponders), except that each is coupled between a respective one of the user-link constituent receive elements **3426** and a respective one of the feeder-link constituent transmit elements **3419**. Receipt of return uplink signals **525** via the array of cooperating user-link constituent receive element **3426** allows the formation of return downlink signals **527** in the return-link transponders **3440**. In some cases, each return downlink signal **527** is a respective superposition of return uplink

signals 525 received by a user-link constituent receive element 3426 from multiple user terminals 517 (e.g., from one or more user beam coverage areas 519). In some such cases, each of the plurality of user terminals 517 is in wireless communication with the array of cooperating user-link constituent receive elements 3426 to transmit a respective return uplink signal 525 to multiple of the user-link constituent receive elements 3426.

[0257] In some cases, the return downlink signals 527 are transmitted by the feeder-link constituent transmit elements 3419 to the geographically distributed ANs 515. As described herein, each AN 515 can receive a superposed composite of the return downlink signals 527 transmitted from the feeder-link constituent transmit elements 3419. The superposed composite may be an example of superposition 1706 described with reference to FIG. 6. The received return downlink signals 527 (which may be referred to as composite return signals) can be coupled to a return beamformer 531, which can combine, synchronize, beam weight, and perform any other suitable processing. For example, the return beamformer 531 can weight the received superpositions 1706 of the return downlink signals 527 (i.e., apply return beam weights to the composite return signals) prior to combining the signals. The return beamformer 531 can also synchronize the composite return signals 1706 prior to combining the signals to account at least for respective path delay differences between the end-to-end relay 3403 and the ANs 515. In some cases, the synchronizing can be according to a received beacon signal (received by one or more, or all, of the ANs 515).

[0258] Because of the end-to-end nature of the beamforming, proper application of return beam weights by the return beamformer 531 enables formation of the return user beams, even though the return beamformer 531 may be coupled to the feeder-link side of the end-to-end multipath channels, and the user beams may be formed at the user-link side of the end-to-end multipath channels. Accordingly, the return beamformer 531 can be referred to as contributing to the forming of the return user beams (a number of other aspects of the system 3400 also contribute to the end-to-end return beamforming, such as the inducement of multipath by the return-link transponders 3440 of the end-to-end relay 3403). Still, the return beamformer 531 can be referred to as forming the return user beams for the sake of simplicity.

[0259] In some cases, the end-to-end relay 3403 further includes a feeder-link antenna subsystem 3410 to illuminate an AN area 3450 within which the ANs 515 are distributed. The feeder-link antenna subsystem 3410 comprises the array 3415 of cooperating feeder-link constituent receive elements 3416. In some cases, the end-to-end relay 3403 also includes a user-link antenna subsystem 3420 to illuminate a user coverage area 3460 within which the plurality of user terminals 517 is geographically distributed (e.g., in a plurality of user beam coverage areas 519). The user-link antenna subsystem 3420 comprises the array 3425 of cooperating user-link constituent transmit elements 3429. In some cases, the user-link antenna subsystem 3420 includes a user-link receive array and a user-link transmit array (e.g., separate, half-duplex arrays of cooperating user-link constituent elements). The user-link receive array and the user-link transmit array can be spatially interleaved (e.g., to point to a same reflector), spatially separated (e.g., to point at receive and transmit reflectors, respectively), or arranged in any other suitable manner (e.g., as discussed with reference to FIG. 62). In other cases, the user-link antenna subsystem 3420 includes full-duplex elements (e.g., each user-link constituent transmit element 3429 shares radiating structure with a respective user-link constituent receive element 3426). Similarly, in some cases, the feeder-link antenna subsystem 3410 includes a feeder-link receive array and a feeder-link transmit array, which may be spatially related in any suitable manner and may directly radiate, point to a single reflector, point to separate transmit and receive reflectors, etc. In other cases, the feeder-link antenna subsystem 3410 includes full-duplex elements. The feeder-link antenna subsystem 3410 and the user-link antenna subsystem 3420 can have the same or different aperture sizes. In some cases, the feeder-link antenna subsystem 3410 and the user-link antenna subsystem 3420 operate in a same frequency range (e.g., a frequency range within the K/Ka band, etc.). In some cases, the feeder-link antenna subsystem 3410 and the user-link antenna subsystem 3420

operate in different frequency ranges (e.g., feeder-link uses V/W band, the user-link uses K/Ka band, etc.). In some cases, the feeder-link antenna subsystem **3410** and/or the user-link antenna subsystem **3420** may operate in multiple frequency ranges (e.g., feeder-link uses V/W band and K/Ka-band, as described below with reference to FIG. **64A**, **64B**, **65A** or **65B**).

[0260] In examples, such as those illustrated by FIG. **41**, the AN area **3450** is distinct from the user coverage area **3460**. The AN area **3450** can be a single, contiguous coverage area, or multiple disjoint coverage areas. Similarly (and independently of whether the AN area **3450** is single or multiple), the user coverage area **3460** can be a single, contiguous coverage area, or multiple disjoint coverage areas. In some cases, the AN area **3450** is a subset of the user coverage area **3460**. In some cases, at least half of the user coverage area **3460** does not overlap the AN area **3450**. As described below, in some cases, the feeder-link antenna subsystem **3410** further comprises one or more feeder-link reflectors, and the user-link antenna subsystem **3420** further comprises one or more user-link reflectors. In some cases, the feeder-link reflector is significantly larger (e.g., at least twice the physical area, at least five times, ten times, fifty times, eighty times, etc.) than the user-link reflector. In some cases, the feeder-link reflector is approximately the same physical area (e.g., within 5%, 10%, 25%) as the user-link reflector.

[0261] In some cases, the system **3400** operates in the context of ground network functions, as described with reference to FIG. **5**. For example, the end-to-end relay **3403** communicates with ANs **515**, which communicate with a CPS **505** via a distribution network **518**. In some cases, the CPS **505** includes a forward beamformer **529** and/or a return beamformer **531**, for example, as described with reference to FIG. **29**. As described above, the forward beamformer **529** can participate in forming forward end-to-end beams by applying computed forward beam weights (e.g., supplied by a forward beam weight generator **918**) to forward uplink signals **521**; and the return beamformer **531** can participate in forming return end-to-end beams by applying computed return beam weights (e.g., supplied by a return beam weight generator **935**) to return downlink signals **527**. As described above, the end-to-end forward beam weights and/or the set of end-to-end return beam weights can be computed according to estimated end-to-end gains for end-to-end multipath channels, each end-to-end multipath channel communicatively coupling a respective one of the distributed ANs **515** with a respective location in the user coverage area **3460** (e.g., a user terminal **517** or any suitable reference location) via a respective plurality of the forward-link bent-pipe transponders **3430** and/or via a respective plurality of the return-link bent-pipe transponders **3440**. In some cases, though not shown, the end-to-end relay **3403** includes a beacon signal transmitter. The beacon signal transmitter can be implemented as described above with reference to the beacon signal generator and calibration support module **424** of FIG. **15**. In some cases, the generated beacon signal can be used so that the plurality of distributed ANs **515** is in time-synchronized wireless communication with the end-to-end relay **3403** (e.g., with the plurality of feeder-link constituent receive elements **3416** according to the beacon signal).

[0262] In some cases, the system **3400** includes a system for forming a plurality of forward user beams using end-to-end beamforming. Such cases include means for transmitting a plurality of forward uplink signals **521** from a plurality of geographically distributed locations, wherein the plurality of forward uplink signals **521** is formed from a weighted combination of a plurality of user beam signals, and wherein each user beam signal corresponds to one and only one user beam. For example, the plurality of geographically distributed locations can include a plurality of ANs **515**, and the means for transmitting the plurality of forward uplink signals **521** can include some or all of a forward beamformer **529**, a distribution network **518**, and the geographically distributed ANs **515** (in communication with the end-to-end relay **3403**). Such cases can also include means for relaying the plurality of forward uplink signals **521** to form a plurality of forward downlink signals **522**. Each forward downlink signal **522** is created by amplifying a unique superposition of the plurality of forward uplink signals **521**, and the plurality of forward downlink signals **522** superpose to form the plurality of user beams, wherein each user beam signal is dominant within

the corresponding user beam coverage area **519**. For example, the means for relaying the plurality of forward uplink signals **521** to form the plurality of forward downlink signals **522** can include the end-to-end relay **3403** (in communication with one or more user terminals **517** in user beam coverage areas **519**) with its collocated plurality of signal paths, which can include forward-link transponders **3430** and return-link transponders **3440**.

[0263] Some such cases include first means for receiving a first superposition of the plurality of forward downlink signals **522** and recovering a first one of the plurality of user beam signals. Such first means can include a user terminal **517** (e.g., including a user terminal antenna, and a modem or other components for recovering user beam signals from the forward downlink signals). Some such cases also include second means (e.g., including a second user terminal **517**) for receiving a second superposition of the plurality of forward downlink signals **522** and recovering a second one of the plurality of user beam signals. For example, the first means for receiving is located within a first user beam coverage area **519**, and the second means for receiving is located within a second user beam coverage area **519**.

[0264] FIG. **42** is an illustration of an example model of signal paths for signals carrying return data on the end-to-end return link **523**. The example model can operate similarly to the model described with reference to FIGS. **6-8**, except that the end-to-end relay **3403** includes return-link signal paths **3502** dedicated for return-link communications. Each return-link signal path **3502** can include a return-link transponder **3440** coupled (e.g., selectively coupled) between a user-link constituent receive element **3426** and a feeder-link constituent transmit element **3419**. Signals originating with user terminals **517** in K user beam coverage areas **519** are transmitted (as return uplink signals **525**) to the end-to-end relay **3403**, received by an array of L user-link constituent receive elements **3426**, communicated through L return-link signal paths **3502** (e.g., via L return-link transponders **3440**) to L corresponding feeder-link constituent transmit elements **3419**, and transmitted by each of the L feeder-link constituent transmit elements **3419** to some or all of the M ANs **515** (similar to what is shown in FIG. **7**). In this way, the multiple return-link signal paths **3502** (e.g., the return-link transponders **3440**) induce multipath in the return-link communications. For example, the output of each return-link signal path **3502** is a return downlink signal **527** corresponding to a received composite of the return uplink signals **525** transmitted from multiple of the user beam coverage areas **519**, and each return downlink signal **527** is transmitted to some or all of the M ANs **515** (e.g., geographically distributed over an AN area **3450**). Accordingly, each AN **515** may receive a superposition **1706** of some or all of the return downlink signals **527**, which may then be communicated to a return beamformer **531**. As described above, there are L (or up to L) different ways for a signal to get from a user terminal **517** located in a user beam coverage area **519** to a particular AN **515**. The end-to-end relay **3403** thereby creates L paths between a user terminal **517** and an AN **515**, referred to collectively as an end-to-end return multipath channel **1908** (e.g., similar to FIG. **8**).

[0265] The end-to-end return multipath channels can be modeled in the same manner described above. For example, A_r is the $L \times K$ return uplink radiation matrix, C_t is the $M \times L$ return downlink radiation matrix, and E_{ret} is the $L \times L$ return payload matrix for the paths from the user-link constituent receive elements **3426** to the feeder-link constituent transmit elements **3419**. As described above, the end-to-end return multipath channel from a user terminal **517** in a particular user beam coverage area **519** to a particular AN **515** is the net effect of the L different signal paths induced by L unique return-link signal paths **3502** through the end-to-end relay **3403**. With K user beam coverage areas **519** and M ANs **515**, there can be $M \times K$ induced end-to-end return multipath channels in the end-to-end return link **523** (via the end-to-end relay **3403**), and each can be individually modeled to compute a corresponding element of an $M \times K$ return channel matrix H_{ret} ($C_{sub.t} \times E_{ret} \times A_r$). As noted above (e.g., with reference to FIGS. **6-8**), not all ANs **515**, user beam coverage areas **519**, and/or return-link transponders **3440** have to participate in the end-to-end return multipath channels. In some cases, the number of user beams K is greater than the number of

transponders **L** in the signal path of the end-to-end return multipath channel; and/or the number of ANs **515** **M** is greater than the number of return-link transponders **3440** **L** in the signal path of the end-to-end return multipath channel. As described with reference to FIG. 5, the CPS **505** can enable forming of return user beams by applying return beam weights to the received downlink return signals **527** (the received signals, after reception by the AN **515** are referred to as composite return signals **907**, as explained further below). The return beam weights can be computed based on the model of the $M \times K$ signal paths for each end-to-end return multipath channel that couples the user terminals **517** in one user beam coverage area **519** with one of the plurality of ANs **515**.

[0266] FIG. 43 is an illustration of an example model of signal paths for signals carrying forward data on the end-to-end forward link **501**. The example model can operate similarly to the model described with reference to FIGS. 9-11, except that the end-to-end relay **3403** includes forward-link signal paths **3602** dedicated for forward-link communications. Each forward-link signal path **3602** can include a forward-link transponder **3430** coupled between a feeder-link constituent receive element **3416** and a user-link constituent transmit element **3429**. As described above, each forward uplink signal **521** is beam weighted (e.g., at a forward beamformer **529** in the CPS **505** of the ground segment **502**) prior to transmission from an AN **515**. Each AN **515** receives a unique forward uplink signal **521** and transmits the unique forward uplink signal **521** via one of M uplinks (e.g., in a time-synchronized manner). The forward uplink signals **521** are received from geographically distributed locations (e.g., from the ANs **515**) by some or all of the forward-link transponders **3430** in a superposed manner that creates composite input forward signals **545**. The forward-link transponders **3430** concurrently receive respective composite input forward signals **545**, though with slightly different timing due to differences in the locations of each receiving feeder-link constituent receive element **3416** associated with each forward-link transponder **3430**. For example, even though each feeder-link constituent receive element **3416** can receive a composite of the same plurality of forward uplink signals **521**, the received composite input forward signals **545** can be slightly different. The composite input forward signals **545** are received by L forward-link transponders **3430** via respective feeder-link constituent receive elements **3416**, communicated through the L forward-link transponders **3430** to L corresponding user-link constituent transmit elements **3429**, and transmitted by the L user-link constituent transmit elements **3429** to one or more of the K user beam coverage areas **519** (e.g., as forward downlink signals **522**, each corresponding to a respective one of the received composite input forward signals **545**). In this way, the multiple forward-link signal paths **3602** (e.g., forward-link transponders **3430**) induce multipath in the forward-link communications. As described above, there are L (or up to L) different ways for a signal to get from an AN **515** to a particular user terminal **517** in a user beam coverage area **519**. The end-to-end relay **3403** thereby induces multiple (e.g., up to L) signal paths **3602** between one AN **515** and one user terminal **517** (or one user beam coverage area **519**), which may be referred to collectively as an end-to-end forward multipath channel **2208** (e.g., similar to FIG. 10).

[0267] The end-to-end forward multipath channels **2208** can be modeled in the same manner described above. For example, C_r is the $L \times M$ forward uplink radiation matrix, A_t is the $K \times L$ forward downlink radiation matrix, and E_{fwd} is the $L \times L$ forward payload matrix for the paths from the feeder-link constituent receive elements **3416** to the user-link constituent transmit elements **3429**. In some cases, the forward payload matrix E_{fwd} and return payload matrix E_{ret} may be different to reflect differences between the forward-link signal paths **3602** and the return-link signal paths **3502**. As described above, the end-to-end forward multipath channel from a particular AN **515** to a user terminal **517** in a particular user beam coverage area **519** is the net effect of the L different signal paths induced by L unique forward-link signal paths **3602** through the end-to-end relay **3403**. With K user beam coverage areas **519** and M ANs **515**, there can be $M \times K$ induced end-to-end forward multipath channels in the end-to-end forward link **501**, and each can be individually modeled to compute a corresponding element of an $M \times K$ forward channel matrix H_{fwd}

(At×E_{fwd}×Cr). As noted with reference to the return direction, not all ANs **515**, user beam coverage areas **519**, and/or forward-link transponders **3430** have to participate in the end-to-end forward multipath channels. In some cases, the number of user beams K is greater than the number of forward-link transponders **3430** L in the signal path of the end-to-end forward multipath channel; and/or the number of ANs **515** M is greater than the number of forward-link transponders **3430** L in the signal path of the end-to-end forward multipath channel. As described with reference to FIG. 5, an appropriate beam weight may be computed for each of the plurality of end-to-end forward multipath channels by the CPS **505** to form the forward user beams. Using multiple transmitters (ANs **515**) to a single receiver (user terminal **517**) can provide transmit path diversity to enable the successful transmission of information to any user terminal **517** in the presence of the intentionally induced multipath channel.

[0268] FIGS. **41-43** describe end-to-end relays **3403** implemented with separate forward-link transponders **3430** and return-link transponders **3440**. FIGS. **44A** and **44B** show an illustration of an example forward signal path **3700** (like the forward signal path **3602** of FIG. **43**) and return signal path **3750** (like the return signal path **3502** of FIG. **42**), respectively. As described above, the forward signal path **3700** includes a forward-link transponder **3430** coupled between a feeder-link constituent receive element **3416** and a user-link constituent transmit element **3429**. The return signal path **3750** includes a return-link transponder **3440** coupled between a user-link constituent receive element **3426** and a feeder-link constituent transmit element **3419**. In some cases, each forward-link transponder **3430** and each return-link transponder **3440** is a cross-pole transponder. [0269] FIG. **63A** illustrates an example frequency spectrum allocation **6300** in accordance with various embodiments of the present disclosure. Example frequency spectrum allocation **6300** of FIG. **63A** illustrates two frequency ranges **6325a** and **6330a**. Though illustrated as being separated, frequency ranges **6325a** and **6330a** may alternatively be adjacent (e.g., one contiguous range). As illustrated in FIG. **63A**, the forward-link transponder **3430** receives a forward uplink signal **6340a** (e.g., which may be an example of forward uplink signal **521** of FIG. **41**) at an uplink frequency range **6330a** with left-hand circular polarization (LHCP) and outputs a forward downlink signal **6345a** (e.g., which may be an example of forward downlink signal **522** of FIG. **41**) at a downlink frequency range **6325a** with right-hand circular polarization (RHCP); and each return-link transponder **3440** receives a return uplink signal **6350a** (e.g., which may be an example of return uplink signal **525** of FIG. **41**) at the uplink frequency range **6330a** with right-hand circular polarization (RHCP) and outputs a return downlink signal **6355a** (e.g., which may be an example of return downlink signal **527** of FIG. **41**) at the downlink frequency range **6325a** with left-hand circular polarization (LHCP). One such case (i.e., following the polarizations described in the preceding example) is illustrated by following only the solid lines of FIGS. **44A** and **44B**, and another such case (i.e., following opposite polarizations from those described in the preceding example) is illustrated by following only the dashed lines of FIGS. **44A** and **44B**.

[0270] In other cases, some or all transponders can provide a dual-pole signal path pair. For example, following both the solid and dashed lines of FIGS. **44A** and **44B**, the forward-link transponders **3430** and the return-link transponders **3440** can receive forward uplink signals **521** at the same or different uplink frequency with both polarizations (LHCP and RHCP) and can both output forward downlink signals **522** at the same or different downlink frequency with both polarizations (RHCP and LHCP). Such cases can use any suitable type of interference mitigation techniques (e.g., using time division, frequency division, spatial separation, etc.) and can enable multiple systems to operate in parallel. One such frequency-division implementation is shown in the example frequency allocation **6301** of FIG. **63B**. In example frequency allocation **6301**, each forward-link transponder **3430** receives a forward uplink signal **6340b** over a first portion of uplink frequency range **6330b** (e.g., using both polarizations) and outputs a forward downlink signal **6345b** over a first portion of a downlink frequency range **6325b** (e.g., using both polarizations); and each return-link transponder **3440** receives a return uplink signal **6350b** over a second portion

of the uplink frequency range **6330b** (e.g., using both polarizations) and outputs a return downlink signal **6355a** over a second portion of the downlink frequency range **6325b** (e.g., using both polarizations). In some cases, the bandwidths of the first portions and second portions of the frequency ranges **6330b** and **6325b** may be equal. In other examples, the bandwidths of the first portions and second portions may be different. As an example, when traffic flows through end-to-end relay **3403** predominantly in the forward direction (represented by ETE forward link **501** in FIG. **41**), the bandwidths of the first portions of frequency ranges **6330b** and **6325b** used for forward link communications may be larger (e.g., significantly larger) than the bandwidths of the second portions used for return link communications.

[0271] In some cases, the end-to-end relay **3403** includes a large number of transponders, such as 512 forward-link transponders **3430** and 512 return-link transponders **3440** (e.g., 1,024 transponders total). Other implementations can include smaller numbers of transponders, such as 10, or any other suitable number. In some cases, the antenna elements are implemented as full-duplex structures, so that each receive antenna element shares structure with a respective transmit antenna element. For example, each illustrated antenna element can be implemented as two of four waveguide ports of a radiating structure adapted for both transmission and reception of signals. In some cases, only the feeder-link elements, or only the user-link elements, are full duplex. Other implementations can use different types of polarization. For example, in some implementations, the transponders can be coupled between a receive antenna element and transmit antenna element of the same polarity.

[0272] Both the example forward-link transponder **3430** and return-link transponder **3440** can include some or all of LNAs **3705**, frequency converters and associated filters **3710**, channel amplifiers **3715**, phase shifters **3720**, power amplifiers **3725** (e.g., traveling wave tube amplifiers (TWTAs), solid state power amplifiers (SSPAs), etc.) and harmonic filters **3730**. In dual-pole implementations, as shown, each pole has its own signal path with its own set of transponder components. Some implementations can have more or fewer components. For example, the frequency converters and associated filters **3710** can be useful in cases where the uplink and downlink frequencies are different. As one example, each forward-link transponder **3430** can accept an input at a first frequency range and can output at a second frequency range; and each return-link transponder **3440** can accept an input at the first frequency range and can output at the second frequency range.

[0273] In some cases, multiple sub-bands are used (e.g., seven 500 MHz sub-bands, as described above). For example, in some cases, transponders can be provided that operate over the same sub-bands as used in a multiple sub-band implementation of the ground network, effectively to enable multiple independent and parallel end-to-end beamforming systems through a single end-to-end relay (each end-to-end beamforming system operating in a different sub-band). In such cases, each transponder can include multiple frequency converters and associated filters **3710**, and/or other components, dedicated to handling one or more of the sub-bands. The use of multiple frequency sub-bands may allow relaxed requirements on the amplitude and phase response of the transponder, as the ground network may separately determine beam weights used in each of the sub-bands, effectively calibrating out passband amplitude and phase variation of the transponders. For example, with separate forward and return transponders, and using 7 sub-bands, a total of 14 different beam weights may be used for each beam (i.e., 7 sub-bands*2 directions (forward and return)). In other cases, a wide bandwidth end-to-end beamforming system may use multiple sub-bands in the ground network, but pass one or more (or all) sub-bands through wideband transponders (e.g., passing 7 sub-bands, each 500 MHz wide, through a 3.5 GHz bandwidth transponders). In some cases, each transponder path includes only a LNA **3705**, a channel amplifier **3715**, and a power amplifier **3725**. Some implementations of the end-to-end relay **3403** include phase shift controllers and/or other controllers that can individually set the phases and/or other characteristics of each transponder as described above.

[0274] The antenna elements can transmit and/or receive signals in any suitable manner. In some cases, the end-to-end relay **3403** has one or more array fed reflectors. For example, the feeder-link antenna subsystem **3410** can have a feeder-link reflector for both transmit and receive, or a separate feeder-link transmit reflector and feeder-link receive reflector. In some cases, the feeder-link antenna subsystem **3410** can have multiple feeder-link reflectors for transmission or reception, or both. Similarly, the user-link antenna subsystem **3420** can have a user-link reflector for both transmit and receive, or a separate user-link transmit reflector and user-link receive reflector. In some cases, the user-link antenna subsystem **3420** can have multiple user-link reflectors for transmission or reception, or both. In one example case, the feeder-link antenna subsystem **3410** comprises an array of radiating structures, and each radiating structure includes a feeder-link constituent receive element **3416** and a feeder-link constituent transmit element **3419**. In such a case, the feeder-link antenna subsystem **3410** can also include a feeder-link reflector that illuminates the feeder-link constituent receive elements **3416** and is illuminated by the feeder-link constituent transmit elements **3419**. In some cases, the reflector is implemented as multiple reflectors, which may be of different shapes, sizes, orientations, etc. In other cases, the feeder-link antenna subsystem **3410** and/or the user-link antenna subsystem **3420** is implemented without reflectors, for example, as a direct radiating array.

[0275] As discussed above, achieving a relatively uniform distribution of ANs **515** across a given user coverage area **3460** may involve placing ANs **515** in undesirable locations. Thus, the present disclosure describes techniques to enable the ANs **515** to be geographically distributed within an AN area **3450** that is smaller (sometimes significantly) than the user coverage area **3460**. For example, in some cases the AN area **3450** may be less than half, less than one quarter, less than one-fifth, or less than one-tenth the physical area of the user coverage area **3460**. In addition, multiple AN areas **3450** may be used concurrently or may be activated for use at different times. As discussed herein, these techniques include the use of different sized reflectors, compound reflector(s), selectively coupled transponders, different user link and feeder link antenna subsystems, etc.

[0276] As noted above, separating the feeder-link antenna subsystem **3410** and the user-link antenna subsystem **3420** can enable servicing of one or more AN areas **3450** that are distinct from one or more user coverage areas **3460**. For example, the feeder-link antenna subsystem **3410** can be implemented with a reflector having an appreciably larger physical area than the reflector of the user coverage area **3460**. The larger reflector can permit a large number of ANs **515** to be geographically distributed in an appreciably smaller AN area **3450**, such as in a small subset of the user coverage area **3460**. Some examples are shown in FIGS. **45A-45G**. Alternatively, an AN area **3450** that is a subset of the user coverage area may be deployed using a single antenna subsystem for both the feeder-link and user-link by using different frequency ranges for the feeder-link and user-links. For example, an AN area **3450** that is one-quarter the area of a user coverage area **3460** may be deployed using a feeder-link carrier frequency that is approximately double the user-link carrier frequency. In one example, the user-link may use a frequency range (or ranges) in the K/Ka bands (e.g., around 30 GHz) while the feeder-link uses frequency range(s) in the V/W bands (e.g., around 60 GHz). In this case, the AN area **3450** will be concentric with the user coverage area **3460**.

[0277] FIG. **45A** shows an example of an end-to-end relay **3403** (e.g., a satellite) visible Earth coverage area **3800**. In the example end-to-end relay **3403**, the feeder-link antenna subsystem **3410** includes an 18-meter feeder-link reflector, and the user-link antenna subsystem **3420** includes a 2-meter user-link reflector (e.g., the feeder-link reflector area is about eighty times larger than the user-link reflector area). Each antenna subsystem also includes an array of 512 cooperating constituent receive/transmit elements. The example end-to-end relay **3403** can include 512 forward-link transponders **3430** (e.g., forming 512 forward signal paths **3700** as shown in FIG. **44A**) and 512 return-link transponders **3440** (e.g., forming 512 return signal paths **3750** as shown

in FIG. 44B). From a geostationary orbital position of the end-to-end relay 3403, the user-link antenna subsystem 3420 illuminates user coverage area 3460 that extends substantially over the visible Earth coverage area 3800 while the feeder-link reflector illuminates AN area 3450 that is a fraction of the user coverage area 3460. Although the AN area 3450 is a small subset of the large user coverage area 3460, a large system capacity including a large number of user beams can be supported using end-to-end beamforming with a large number of ANs 515 in the AN area 3450 (e.g., used cooperatively in an AN cluster). For example, hundreds of cooperating ANs 515 may be geographically distributed within AN area 3450 shown in FIG. 45A as a shaded region in the eastern United States. In one example, 597 ANs 515 are geographically distributed within AN area 3450.

[0278] FIG. 46A shows the visible earth coverage with end-to-end beamforming applied between the ANs 515 in the AN area 3450 and the user coverage area 3460. The user coverage area 3460 includes 625 user beam coverage areas 519 providing service to user terminals 517 within the visible Earth coverage area 3800.

[0279] FIG. 45B shows an example of an end-to-end relay 3403 (e.g., a satellite) Continental United States (CONUS) coverage area 3900. The example end-to-end relay 3403 is similar to the example shown in FIG. 45A, except that the feeder-link antenna subsystem 3410 uses an 18-meter feeder-link reflector while the user-link antenna subsystem 3420 includes a 5-meter user-link reflector (e.g., the area of the feeder-link reflector is about thirteen times larger than the area of the user-link reflector). The AN area 3450 (e.g., the area containing the cooperating AN cluster) is the same as that of FIG. 45A: a region that is a small subset of the user coverage area 3460 in the eastern United States having e.g., 597 ANs 515 distributed therein.

[0280] FIG. 46B shows the CONUS coverage area 3900 with end-to-end beamforming applied between the ANs 515 in the AN area 3450 and the user coverage area 3460. The user coverage area 3460 includes 523 user beam coverage areas 519 providing service to user terminals 517 within the CONUS coverage area.

[0281] Various geographical and relative locations of the AN cluster are supported by the present disclosure. As described herein, an end-to-end relay 3403 like those illustrated in FIGS. 49A and 49B can provide communications service between one or more user coverage areas 3460 and ANs 515 located in one or more AN areas 3450. In some examples, such as the example illustrated in FIG. 45B, the AN area 3450 may overlap or be located entirely within the user coverage area 3460. Additionally or alternatively, an AN area 3450 may be non-overlapping with a user coverage area 3460 as illustrated in FIG. 45C. In some cases, such an arrangement may require the use of a special loopback mechanism, which is discussed below with reference to FIGS. 55A-55C.

[0282] As another example of a possible geographic arrangement, the AN cluster (e.g., the AN area 3450) may at least partially overlap with a low demand area of the user coverage area 3460. An example is shown in FIG. 45D, where the AN area 3450 is located in a low demand area of user coverage area 3460. In some cases, a low demand area may be determined based on the demand for the communication service being below a demand threshold. For example, the low demand area may have an average demand that is less than a fraction (e.g., one-half, one-quarter, etc.) of the average demand across other served areas of user coverage area 3460. Such a deployment may support increased system capacity in higher demand areas (e.g., by allowing portions of the frequency spectrum associated with feeder-link communications in the low demand area to be used for user beams in the higher demand areas). That is, a given system bandwidth (which may be a contiguous or multiple non-contiguous frequency ranges) may be mostly or fully utilized for serving user beams in areas outside the low demand area, and may be allocated mostly to feeder-link communications within the low demand area, with the user beams in the low demand area being allocated a smaller portion (e.g., less than half) of the system bandwidth. Thus, in some cases, the user-link communications in higher demand areas may use at least a portion of the same frequency bandwidth used for feeder-link communications in a low demand area in which the

access node area **3450** is located. In this example, the AN area **3450** is contained completely within user coverage area **3460**, although the two may only partially overlap in some cases.

[0283] In some cases, the AN cluster may be located within (e.g., on the surface of) an aquatic body (e.g., a lake, sea, or ocean). An example is shown in FIG. **45E**, which shows a user coverage area **3460** including the United States and an AN area **3450** located off the eastern coast of the United States. In some cases, the AN area may at least partially overlap with a landmass (e.g., some ANs **515** may not be located within the aquatic body). Thus, the example discussed with respect to FIG. **45E** includes a scenario in which only one AN **515** is located within the aquatic body, all ANs **515** are located within the aquatic body, or some intermediate number of ANs **515** are located within the aquatic body. Benefits of locating parts or all of an AN cluster on an aquatic body include availability of large areas for the AN cluster in proximity to land masses where user coverage is desired, flexibility in placement of ANs **515** within the AN area **3450**, and reduced competition for spectrum rights. For example, regulatory considerations such as interference and band-sharing with other services may be reduced when an AN cluster is not located over a particular country or landmass.

[0284] ANs **515** located within the aquatic body may be located on fixed or floating platforms. Examples of fixed platforms used for ANs **515** include fixed oil platforms, fixed offshore wind turbines, or other platforms installed on pilings. Examples of floating platforms include barges, buoys, offshore oil platforms, floating offshore wind turbines, and the like. Some fixed or floating platforms may already have power sources, while other fixed or floating platforms dedicated for use in an AN cluster may be configured with power generation (e.g., a generator, solar power generation, wind turbine, etc.). Distribution of access node specific forward signals **521** from a beamformer **529** to the ANs **515** and composite return signals **1706** from the ANs **515** to the beamformer **531** may be provided via a distribution network **518** that includes wired or wireless links between the beamformer(s) or a distribution platform and the ANs **515**. In some cases, the distribution network **518** may include a submarine cable coupled with the beamformer(s) and ANs **515** distributed within the aquatic body as discussed with reference to FIG. **45G**. The submarine cable may also provide a power source. The distribution network may additionally or alternatively include wireless RF links (e.g., microwave backhaul links) or free space optical links. In some examples, the beamformer(s), a distribution point for the beamformer(s), or the distribution network **518** as a whole may be located within the aquatic body. For example, FIG. **58** shows a CPS **505** disposed on an offshore (e.g., fixed or floating) platform **5805** that communicates traffic to a terrestrial network node and is coupled to ANs **515** in the aquatic body via distribution network **518**.

[0285] In some cases, at least some ANs **515** in the AN cluster may be mobile (e.g., may be located on moveable platforms). For example, ANs **515** within an aquatic body may be located on boats or barges that may be controlled to relocate position as illustrated by floating platform **5805** in FIG. **58**. Similarly, terrestrial ANs **515** may be located on vehicular platforms while airborne ANs **515** may be located on mobile platforms such as aircraft, balloons, drones, and the like. In some examples, mobile ANs **515** may be used to optimize distribution of ANs **515** within the AN area **3450**. For example, ANs **515** may be relocated for better geographic distribution within the AN area **3450**, or ANs **515** may be relocated upon failure of one or more ANs **515** (e.g., to redistribute the available ANs **515**). The beamforming weights may be recalculated for the new positions and the ANs **515** may resynchronize transmit timing and phase to adjust to the new positions, as described above.

[0286] In some examples, the AN area **3450** may be relocated using mobile ANs **515** (e.g., one or more ANs **515** in the AN cluster may be located on mobile platforms). An example is shown in FIG. **45F**, which shows an initial AN area **3450a** including multiple ANs **515** geographically distributed within the AN area **3450a**. For various reasons, the AN cluster may be relocated to be within new AN area **3450b**. For example, a mobile AN cluster may be used to adapt to changes in

position of the end-to-end relay **3403**. In one example, an orbital position or orientation of a satellite end-to-end relay **3403** changes due to a change in deployment to a new orbital slot or because of orbital drift or alignment, and the change in AN area **3450** adapts to the new orbital position or orientation. The mobile ANs **515** may move to new positions within the new AN area **3450b**. Additionally, while the mobile AN cluster is displayed as being located within an aquatic body, some or all of the ANs **515** may be located on land (e.g., mobile ANs **515** need not be located in an aquatic body). In some cases, one or more of the ANs **515** may be located on an airborne craft (e.g., a plane, a balloon, a drone, etc.). Also, while the current example describes first and second AN areas **3450a** and **3450b** that are similar in size at different locations, the AN areas **3450** at the different locations may be (e.g., significantly) different (e.g., due to a difference in slant range or adaptation of an antenna assembly on the end-to-end relay). As an example, the first and second AN areas **3450a** and **3450b** may have the same (or similar) center points but significantly different physical sizes (e.g., through a combination of orbit slot shift and repointing of the end-to-end relay antenna).

[0287] As an example, the AN cluster may initially be located at a first location **3450a**. While at the first location **3450a**, each AN **515** of the AN cluster may receive an access node-specific forward signal for transmission via end-to-end relay **3403** to one or more of the user terminals in user coverage area **3460**. In aspects, the access node-specific forward signal may be received from a forward beamformer **529** via a distribution network **518**, which may be a free space optical link or any other suitable link. As discussed above, the access node-specific forward signals may be appropriately weighted by the forward beamformer **529** before reception at the AN **515**. While at the first location **3450a**, each AN **515** may synchronize a forward uplink signal **521** for reception at the end-to-end relay **3403** so that the forward uplink signal **521** is time and phase aligned with other forward uplink signals **521** from other ANs **515** in the AN cluster. Synchronization may be accomplished using any of the techniques described herein (e.g., using relay beacons).

[0288] Subsequently, the AN cluster (or portions thereof) may move to a second location **3450b**. The movement may be in response to some stimulus (e.g., a change in location of the end-to-end relay, weather patterns, etc.). At the second location **3450b**, the ANs **515** of the AN cluster may obtain weighted access node-specific forward signals (e.g., generated using an updated beam weight matrix determined based on the new locations of the ANs **515** within the new AN area **3450b**), synchronize transmissions, and transmit forward uplink signals **521** to end-to-end relay **3403**. While described as being performed at the second location, one or more of these steps may be performed prior to reaching the second location.

[0289] In some cases, the location and shape of the AN cluster may be configured to take advantage of existing network infrastructure. For example, as shown in FIG. 45G, the AN area **3450** may be located near an existing submarine cable **4551** (e.g., fiber-optic cable used in Internet backbone communications, etc.). The submarine cable **4551** may also provide a power source. The distribution network **518** (e.g., between ANs) may additionally or alternatively include wireless RF links (e.g., microwave backhaul links) or free space optical links. In some examples, the beamformer(s), a distribution point for the beamformer, or the distribution network **518** as a whole may be located within the aquatic body. As shown in FIG. 45G, one or more of the AN areas **3450** may be shaped (e.g., using an appropriately shaped reflector, etc.) so as to minimize the total distance between the ANs **515** and the submarine cable **4551**. The example of FIG. 45G shows an elliptically shaped AN area **3450**, though any suitable shape may be used. Further, while only one AN area **3450** is displayed in FIG. 45G, multiple AN areas **3450** may exist (e.g., located along the same submarine cable **4551** or different submarine cables **4551**). The multiple AN areas **3450** may be disjoint or overlap at least partially.

Multiple Coverage Areas

[0290] In the example end-to-end relays **3403** described above, the user-link antenna subsystem **3420** is described as a single antenna subsystem (e.g., with a single user-link reflector), and the

feeder-link antenna subsystem **3410** is described as a single antenna subsystem (e.g., with a single feeder-link reflector). In some cases, the user-link antenna subsystem **3420** can include one or more antenna subsystems (e.g., two or more sub-arrays of constituent antenna elements) associated with one or more user-link reflectors, and the feeder-link antenna subsystem **3410** can include one or more antenna subsystems associated with one or more feeder-link reflectors. For example, some end-to-end relays **3403** can have a user-link antenna subsystem **3420** that includes a first set of user-link constituent receive/transmit elements associated with a first user-link reflector (e.g., each element is arranged to illuminate, and/or be illuminated by, the first user-link reflector) and a second set of user-link constituent receive/transmit elements associated with a second user-link reflector. In some cases, the two user-link reflectors are approximately the same physical area (e.g., within 5%, 10%, 25%, etc.) of each other. In some cases, one user-link reflector is significantly larger (e.g., 50% larger, at least twice the physical area, etc.) than the other. Each set of the user-link constituent receive/transmit elements, and its associated user-link reflector, can illuminate a corresponding, distinct user coverage area **3460**. For example, the multiple user coverage areas can be non-overlapping, partially overlapping, fully overlapping (e.g., a smaller user coverage could be contained within a larger user coverage area), etc. In some cases, the multiple user coverage areas can be active (illuminated) at the same time. Other cases, as described below, can enable selective activation of the different portions of user-link constituent receive/transmit elements, thereby activating different user coverage areas at different times. Similarly, selective activation of different portions of feeder-link constituent receive/transmit elements can activate different AN areas **3450** at different times. Switching between multiple coverage areas may be coordinated with the CPS **505**. For example, beamforming calibration, beam weight calculation and beam weight application may occur in two parallel beamformers, one for each of two different coverage areas. The usage of appropriate weights in the beamformers can be timed to correspond to the operation of the end-to-end relay. For example, switching between multiple coverage areas may be coordinated to occur at a time-slice boundary if time-slice beamformers are employed.

[0291] FIGS. **47A** and **47B** show an example forward signal path **4000** and return signal path **4050**, respectively, each having selective activation of multiple user-link antenna subsystems **3420**.

Forward signal path **4000** (and other forward signal paths described herein) may be an example of forward signal path **3602** described with reference to FIG. **43**. Return signal path **4050** (and other return signal paths described herein) may be an example of return signal path **3502** described with reference to FIG. **42**. For example, each forward signal path **4000** may have a transponder **3430** coupled between constituent antenna elements. In FIG. **47A**, the forward-link transponder **3430b** is similar to the one described with reference to FIG. **44A**, except that the output side of the forward-link transponder **3430b** is selectively coupled to one of two user-link constituent transmit elements **3429**, each part of a separate user-link antenna subsystem **3420** (e.g., each part of a separate array **3425** of cooperating user-link constituent transmit elements **3429**). As described above, the forward-link transponder **3430b** can include some or all of LNAs **3705a**, frequency converters and associated filters **3710a**, channel amplifiers **3715a**, phase shifters **3720a**, power amplifiers **3725a**, and harmonic filters **3730a**.

[0292] The forward-link transponder **3430b** of FIG. **47A** further includes switches **4010a** (forward-link switches) that selectively couple the transponder either to a first user-link constituent transmit element **3429a** (of a first user-link antenna element array **3425a**) via a first set of power amplifiers **3725a** and harmonic filters **3730a**, or to a second user-link constituent transmit element **3429b** (of a second user-link antenna element array **3425b**) via a second set of power amplifiers **3725a** and harmonic filters **3730a**. For example, in a first switch mode, the forward-link transponder **3430b** effectively forms a signal path between a feeder-link constituent receive element **3416** and a first user-link constituent transmit element **3429a**; and in a second switch mode, the forward-link transponder **3430b** effectively forms a signal path between the same feeder-link constituent receive element **3416** and a second user-link constituent transmit element **3429b**. The switches **4010a** can

be implemented using any suitable switching means, such as an electromechanical switch, a relay, a transistor, etc. Though shown as switches **4010a**, other implementations can use any other suitable means for selectively coupling the input of the forward-link transponder **3430** to multiple outputs. For example, the power amplifiers **3725a** can be used as switches (e.g., providing high gain when “on,” and zero gain (or loss) when “off”). Switches **4010a** may be examples of switches that selectively couple one input to one of two or more outputs.

[0293] In FIG. **47B**, the return-link transponder **3440b** functionally mirrors the forward-link transponder **3430** of FIG. **47A**. Rather than selectively coupling the output side of the transponder, as in the forward-link case of FIG. **47A**, the input side of the return-link transponder **3440b** is selectively coupled to one of two user-link constituent receive elements **3426**. Again, each user-link constituent receive element **3426** can be part of a separate array of cooperating user-link constituent receive elements **3426**, which may be part of the same user-link antenna subsystem **3420**, or different user-link antenna subsystems **3420**). As described above (e.g., in FIG. **44B**), the return-link transponder **3440** can include some or all of LNAs **3705b**, frequency converters and associated filters **3710b**, channel amplifiers **3715b**, phase shifters **3720b**, power amplifiers **3725b**, and harmonic filters **3730b**.

[0294] The return-link transponder **3440b** of FIG. **47B** further includes switches **4010b** (return-link switches) that selectively couple the transponder either to a first user-link constituent receive element **3426a** (of a first user-link antenna element array **3425a**) via a first set of LNAs **3705b**, or to a second user-link constituent receive element **3426b** (of a second user-link antenna element array **3425b**) via a second set of LNAs **3705b**. For example, in a first switch mode, the return-link transponder **3440b** effectively forms a signal path between a first user-link constituent receive element **3426a** and a feeder-link constituent transmit element **3419**; and in a second switch mode, the return-link transponder **3440b** effectively forms a signal path between a second user-link constituent receive element **3426b** and the same feeder-link constituent transmit element **3419**. The switches **4010b** can be implemented using any suitable switching means, such as an electromechanical switch, a relay, a transistor, etc. Though shown as switches **4010b**, other implementations can use any other suitable means for selectively coupling the output of the forward-link transponder **3440b** to multiple inputs. For example, the power amplifiers **3705b** can be used as switches (e.g., providing high gain when “on,” and zero gain (or loss) when “off”). Switches **4010b** may be examples of switches that selectively couple one of two or more inputs to a single output.

[0295] Examples of the end-to-end relay **3403** can include a switch controller **4070** to selectively switch some or all of the switches **4010** (or other suitable selective coupling means) according to a switching schedule. For example, the switching schedule can be stored in a storage device on-board the end-to-end relay **3403**. In some cases, the switching schedule effectively selects which user-link antenna element array **3425** to activate (e.g., which set of user beams to illuminate) in each of a plurality of time intervals (e.g., timeslots). In some cases, the switching allocates equal time to the multiple user-link antenna element arrays **3425** (e.g., each of two arrays is activated for about half the time). In other cases, the switching can be used to realize capacity-sharing goals. For example, one user-link antenna element array **3425** can be associated with higher-demand users and can be allocated a greater portion of time in the schedule, while another user-link antenna element array **3425** can be associated with lower-demand users and can be allocated a smaller portion of time in the schedule.

[0296] FIGS. **48A** and **48B** show an example of end-to-end relay **3403** coverage areas **4100** and **4150** that include multiple, selectively activated user coverage areas **3460a** and **3460b**, respectively. The example end-to-end relay **3403** is similar to the relay in FIGS. **38** and **39** except for the presence of different antenna subsystems. In this example, the user-link antenna subsystem **3420** includes two 9-meter user-link reflectors, and the transponders are configured to selectively activate only half of the user beam coverage areas **519** at any given time (e.g., the transponders are

implemented as in FIGS. 47A and 47B). For example, during a first time interval, as shown in FIG. 48A, the user coverage area 3460a includes 590 active user beam coverage areas 519. The active user beam coverage areas 519 effectively cover the western half of the United States. The AN area 3450 (the AN cluster) is the same as that of FIGS. 38 and 39: a region in the eastern United States having e.g., 597 ANs 515 distributed therein. During the first time interval, the AN area 3450 does not overlap with the active user coverage area 3460a. During a second time interval, as shown in FIG. 48B, the user coverage area 3460b includes another 590 active user beam coverage areas 519. The active user beam coverage areas 519 in the second time interval effectively cover the eastern half of the United States. The AN area 3450 does not change. However, during the second time interval, the AN area 3450 is fully overlapped by (is a subset of) the active user coverage area 3460b. Capacity may be flexibly allocated to various regions (e.g., between eastern and western user coverage areas 3460) by dynamically adjusting the ratio of time allocated to the corresponding user-link antenna subsystems 3420.

[0297] While the previous example illustrates two similarly sized user coverage areas 3460, other numbers of user coverage areas 3460 can be provided (e.g., three or more) and can be of differing sizes (e.g., earth coverage, continental U.S. only, U.S. only, regional only, etc.). In cases with multiple user coverage areas 3460, the user coverage areas 3460 can have any suitable geographic relationship. In some cases, first and second user coverage areas 3460 partially overlap (e.g., as shown in FIGS. 48A and 48B). In other cases, a second user coverage area 3460 can be a subset of a first user coverage area 3460 (e.g., as shown in FIGS. 46A and 46B). In other cases, the first and second user coverage areas 3460 do not overlap (e.g., are disjoint).

[0298] In some cases, it can be desirable for traffic of particular geographic regions to terminate in their respective regions. FIG. 50A illustrates a first AN area 3450a in North America used to provide communications service to a first user coverage area 3460a in North America, and a second AN area 3450b to provide communications service to a second user coverage area 3460b in South America. In some cases, the ANs within the first AN area 3450a exchange signals with a first CPS (e.g., located within or proximate to AN area 3450a), and the ANs within the second AN area 3450b exchange signals with a second CPS (e.g., located within or proximate to AN area 3450b) that is separate and distinct from the first CPS. For example, the first AN The end-to-end relay 3403 as shown in FIGS. 49A and 49B may support multiple user coverage areas with multiple AN areas as illustrated in FIG. 50A. Each combination of AN area and user coverage area may employ frequency allocations 6300 or 6301 as shown in FIG. 63A or 63B.

[0299] FIG. 49A shows an example forward signal path 4900 of an end-to-end relay 3403 for supporting multiple user coverage areas with multiple AN areas 3450. The example forward signal path 4900 has a first forward-link transponder 3430c coupled between a first feeder-link constituent receive element 3416a of a first feeder-link antenna element array 3415a and a first user-link constituent transmit element 3429a of a first user-link antenna element array 3425a. In addition, the example forward signal path 4900 has a second forward-link transponder 3430c coupled between a second feeder-link constituent receive element 3416b of a second feeder-link antenna element array 3415b and a second user-link constituent transmit element 3429b of a second user-link antenna element array 3425b. As described above, each of the forward-link transponders 3430 can include some or all of LNAs 3705a, frequency converters and associated filters 3710a, channel amplifiers 3715a, phase shifters 3720a, power amplifiers 3725a, and harmonic filters 3730a.

[0300] FIG. 49B shows an example return signal path 4950 of an end-to-end relay 3403 for supporting multiple user coverage areas with multiple AN areas 3450. The example return signal path 4950 has a first return-link transponder 3440c coupled between a first user-link constituent receive element 3426a of a first user-link antenna element array 3425a and a first feeder-link constituent transmit element 3419a of a first feeder-link antenna element array 3415a. In addition, the example return signal path 4950 has a second return-link transponder 3440c coupled between a second user-link constituent receive element 3426b of a second user-link antenna element array

3425b and a second feeder-link constituent transmit element **3419b** of a second feeder-link antenna element array **3415b**. As described above, each of the return-link transponders **3440** can include some or all of LNAs **3705b**, frequency converters and associated filters **3710b**, channel amplifiers **3715b**, phase shifters **3720b**, power amplifiers **3725b**, and harmonic filters **3730b**.

[0301] In some cases, feeder-link antenna element arrays **3415a** and **3415b** are part of separate feeder-link antenna subsystems **3410**. Alternatively, a single feeder-link antenna subsystem **3410** may include both feeder-link antenna element arrays **3415a** and **3415b** (e.g., via use of a single reflector as described in more detail below with reference to FIGS. **56A** and **56B**). Similarly, user-link antenna element arrays **3425a** and **3425b** may be part of the same or separate user-link antenna subsystems **3420**. The forward signal path **4900** and return signal path **4950** of FIGS. **49A** and **49B** may be used to support multiple independent end-to-end beamforming systems using a single end-to-end relay payload. For example, end-to-end beamforming between the first AN area **3450a** and the first user coverage area **3460a** shown in FIG. **50A** may be supported by one beamformer and distribution system, while a separate and independent beamformer and distribution system supports end-to-end beamforming between the second AN area **3450b** and the second user coverage area **3460b**. FIGS. **49A** and **49B** illustrate examples where the constituent receive elements may be the same as the constituent transmit elements, and therefore only show one polarization in each direction. However, other examples may employ different constituent receive elements and constituent transmit elements, and may use multiple polarizations in each direction.

[0302] FIGS. **47A** and **47B** describe signal path selection on the user-link side. However, some cases alternatively or additionally include signal path switching on the feeder-link side. FIG. **51A** shows an example forward signal path **5100** having selective activation of multiple user-link antenna element arrays **3425** (which may be part of the same or different user-link antenna subsystems **3420**) and multiple feeder-link antenna element arrays **3415** (which may be part of the same or different feeder-link antenna subsystems **3410**). The signal path has a forward-link transponder **3430d** coupled between constituent antenna elements. As described above, the forward-link transponder **3430d** can include some or all of LNAs **3705a**, frequency converters and associated filters **3710a**, channel amplifiers **3715a**, phase shifters **3720a**, power amplifiers **3725a**, and harmonic filters **3730a**. The input side of the forward-link transponder **3430d** is selectively coupled to one of two feeder-link constituent receive elements **3416** (e.g., using switches **4010b** or any other suitable path selection means). Each feeder-link constituent receive element **3416** can be part of a separate feeder-link antenna element array **3415** (e.g., each part of a separate array of cooperating feeder-link constituent receive elements **3416**). The output side of the forward-link transponder **3430d** is selectively coupled to one of two user-link constituent transmit elements **3429** (e.g., using switches **4010a** or any other suitable path selection means). Each user-link constituent transmit element **3429** can be part of a separate user-link antenna element array **3425** (e.g., each part of a separate array of cooperating user-link constituent transmit elements **3429**). One or more switching controllers **4070** (not shown) can be included in the end-to-end relay **3403** for selecting between some or all of the four possible signal paths enabled by the forward-link transponder **3430d**. For example, the switching controller **4070** may operate the forward link transponder **3430d** according to one of several switch modes, which may be determined according to which AN areas **3450** are used to support user coverage areas **3460**. In one example, the switching controller **4070** applies a first switch mode for switches **4010** to couple the forward link transponders **3430d** between the first feeder-link antenna element array **3415a** and the first user-link antenna element array **3425a**, and applies second switch mode for switches **4010** to couple the forward link transponders **3430d** between the second feeder-link antenna element array **3415b** and the second user-link antenna element array **3425b**. Alternatively, a first switch mode for switches **4010** may couple the forward link transponders **3430d** between the first feeder-link antenna element array **3415a** and the second user-link antenna element array **3425b**, and a second switch mode for switches **4010** may couple the forward link transponders **3430d** between the second feeder-link

antenna element array **3415b** and the first user-link antenna element array **3425a**.

[0303] FIG. **51B** shows an example return signal path **5150** having selective activation of multiple user-link antenna element arrays **3425** (e.g., which may be part of the same or different user-link antenna subsystems **3420**) and multiple feeder-link antenna element arrays **3415** (e.g., which may be part of the same or different feeder-link antenna subsystems **3410**). The signal path has a return-link transponder **3440d** coupled between constituent antenna elements. As described above, the return-link transponder **3440d** can include some or all of LNAs **3705b**, frequency converters and associated filters **3710b**, channel amplifiers **3715b**, phase shifters **3720b**, power amplifiers **3725b**, and harmonic filters **3730b**. The input side of the return-link transponder **3440d** is selectively coupled to one of two user-link constituent receive elements **3426a**, **3426b** (e.g., using switches **4010b** or any other suitable path selection means). Each user-link constituent receive element **3426a**, **3426b** can be part of a separate user-link antenna element array **3425a**, **3425b** (e.g., each part of a separate array of cooperating user-link constituent receive elements **3426**). The output side of the return-link transponder **3440d** is selectively coupled to one of two feeder-link constituent transmit elements **3419a** or **3419b** (e.g., using switches **4010a** or any other suitable path selection means). Each feeder-link constituent transmit element **3419a** or **3419b** can be part of a separate feeder-link antenna element array **3415a** or **3415b** (e.g., each part of a separate array of cooperating feeder-link constituent transmit elements **3419**). One or more switching controllers **4070** (not shown) can be included in the end-to-end relay **3403** for selecting between some or all of the four possible signal paths enabled by the return-link transponder **3440d**. For example, the switching controller **4070** may operate the return-link transponder **3440d** according to one of several switch modes, which may be determined according to which AN areas **3450** are used to support user coverage areas **3460**. In one example, the switching controller **4070** applies a first switch mode for switches **4010** to couple the return-link transponders **3440d** between the first user-link antenna element array **3425a** and the first feeder-link antenna element array **3415a**, and applies second switch mode for switches **4010** to couple the return-link transponders **3440d** between the second user-link antenna element array **3425b** and the second feeder-link antenna element array **3415b**. Alternatively, a first switch mode for switches **4010** may couple the return-link transponders **3440d** between the first user-link antenna element array **3425a** and the second feeder-link antenna element array **3415b**, and a second switch mode for switches **4010** may couple the return-link transponders **3440d** between the second user-link antenna element array **3425b** and the first feeder-link antenna element array **3415a**.

[0304] The transponders of FIGS. **47A**, **47B**, **51A**, and **51B** are intended only to illustrate a few of many possible cases of end-to-end relays **3403** employing path selection. Further, some cases can include path selection between more than two user-link antenna element arrays **3425** or user-link antenna subsystems **3420** and/or more than two feeder-link antenna element arrays **3415** or feeder-link antenna subsystems **3410**.

[0305] The end-to-end relay **3403** as shown in FIGS. **51A** and **51B** may support multiple user coverage areas **3460** with multiple AN areas **3450**. As discussed above, it can be desirable for traffic of particular geographic regions to terminate in their respective regions. For example, an end-to-end relay **3403** with or without paired transponders like those illustrated in FIGS. **51A** and **51B** can utilize a first AN area **3450a** in North America to provide communications service to a first user coverage area **3460a** in North America, and utilize a second AN area **3450b** to provide communications service to a second user coverage area **3460b** in South America as illustrated in FIG. **50A**. Using path selection (e.g., switching) in the transponders, a single end-to-end relay **3403** (e.g., a single satellite) can service traffic associated with the North American user coverage area **3460a** using ANs **515** in the North American AN area **3450a** (or using ANs **515** in the South American AN area **3450b**), and service traffic associated with the South American user coverage area **3460b** using ANs **515** in the South American AN area **3450b** (or using ANs **515** in the North American AN area **3450a**). Capacity may be flexibly allocated to various regions (e.g., between

North and South American user coverage areas **3460**) by dynamically adjusting the ratio of time allocated to the corresponding antenna sub-systems.

[0306] FIG. **50B** illustrates a second possible deployment having multiple AN areas **3450** and multiple user coverage areas **3460**. For example, the deployment shown in FIG. **50B** may be supported by the end-to-end relay **3403** illustrated by FIGS. **51A** and **51B**. As shown in FIG. **50B**, an end-to-end relay **3403** with path selection in the transponders services traffic in a first user coverage area **3460a** with a first AN area **3450a** and services traffic in a second user coverage area **3460b** with a second AN area **3450b**. Because the first AN area **3450a** does not overlap with the first user coverage area **3460a**, the same or overlapping portions of bandwidth may be used for uplink or downlink communications between the end-to-end relay **3403** and user terminals or ANs. Additionally, in the present example, because AN area **3450a** or **3450b** and its corresponding user coverage area **3460a** or **3460b**, respectively, do not overlap, a special loopback mechanism may be employed to synchronize transmissions from the ANs **515**. Example loopback mechanisms in the form of loopback transponders are discussed with reference to FIGS. **55A**, **55B**, and **55C**. Referring to FIG. **63A** for example, a system may have a total of 3.5 GHz of uplink bandwidth **6330a** and 3.5 GHz of downlink bandwidth **6325a** available. In a first switch configuration, the full 3.5 GHz uplink bandwidth (e.g., using both of two orthogonal polarizations) may be used concurrently for return uplink transmissions **525** from the first user coverage area **3460a** and forward uplink transmissions **521** from the AN area **3450a**. Similarly, the full 3.5 GHz downlink bandwidth (e.g., using both of two orthogonal polarizations) may be used concurrently for forward downlink transmissions **522** to the first user coverage area **3460a** and return downlink transmissions **527** to the first AN area **3450a**. The full uplink and downlink bandwidth may also be used in a second switching configuration for the second user coverage area **3460b** and second AN area **3450b**. While the case of two AN areas **3450** and two user coverage areas **3460** is discussed with respect to FIG. **50B** for the sake of simplicity, any suitable number of AN areas **3450** and user coverage areas **3460** may be possible. Further, aspects discussed above with respect to a single AN cluster (e.g., mobility, location in an aquatic body, etc.) may be applicable to one or both of the AN clusters in the present example.

[0307] The above example describes AN area **3450a** as servicing a non-overlapping user coverage area **3460a**. As an alternative example, AN area **3450a** may service user coverage area **3460b** (e.g., a user coverage area **3460** may contain its associated AN area **3450** or some portion thereof). A similar example is generally discussed with reference to FIG. **50A** in the context of a first AN area **3450a** located in North America (e.g., which may correspond to AN area **3450a** of FIG. **50B**) servicing a user coverage area **3460a** located in North America while a second AN area **3450b** located in South America services a user coverage area **3460b** located in South America. However, FIG. **50B** shows that user coverage areas **3460** served by different AN areas **3450** may also overlap to provide an aggregate user coverage area for a particular region. In this instance, the user coverage areas **3460** may be used in different time intervals using the switching transponders illustrated by FIGS. **51A** and **51B**. Alternatively, the user coverage areas **3460a** and **3460b** may be serviced concurrently by access node areas **3450a** and **3450b** (either with access node area **3450a** servicing user coverage area **3460a** while access node area **3450b** services user coverage area **3460b** or with access node area **3450b** servicing user coverage area **3460a** while access node area **3450a** services user coverage area **3460b**) using the multiple transponder paths shown in FIGS. **49A** and **49B**. In this case, the uplink and downlink resources used for user beams in user coverage areas **3460a** and **3460b** may be orthogonal (different frequency resources, different polarizations, etc.), or user beams in user coverage areas **3460a** and **3460b** may use the same resources (the same frequency range and polarization), with interference mitigated using interference mitigation techniques such as adaptive coding and modulation (ACM), interference cancellation, space-time coding, and the like.

[0308] As a third example, in some cases AN areas **3450a** and **3450b** combine to service user

coverage area **3460b** (or user coverage area **3460a**). In this case, a special loopback mechanism may not be necessary since a subset of the ANs **515** are contained within the user coverage area **3460**. In some cases, the ANs **515** of AN areas **3450a** and **3450b** may be considered cooperating in the sense that forward uplink signals **521** from each of the AN areas **3450** may combine to service a single user beam coverage area **519**. Alternatively, the ANs **515** of AN area **3450a** may service a first subset of the user beam coverage areas **519** of user coverage area **3460b** while the ANs **515** of AN area **3450b** may service a second subset of the user beam coverage areas **519** of user coverage area **3460b**. In some cases of this example, there may be some overlap between the first and second subsets of user beam coverage areas **519** (e.g., such that the AN areas **3450** may be considered cooperating in some user beam coverage areas **519** and non-cooperating in others). As a further example, AN area **3450a** may service user coverage area **3460b** at a first time interval (or set of time intervals) and AN area **3450b** may service user coverage area **3460b** at a second time interval (or set of time intervals). In some examples, the AN areas **3450a** and **3450b** may cooperate to serve user coverage area **3460b** during the first time interval(s) and may cooperate to serve user coverage area **3460a** during the second time interval(s).

[0309] In general, features of the end-to-end relay **3403** described in FIG. **41** enable servicing of at least one user beam coverage area **3460** using ANs **515** geographically distributed within at least one AN area **3450** that is a different physical area than the user beam coverage area **3460**. In some cases, AN cluster(s) can provide high capacity to a large user coverage area **3460**. FIGS. **45A-45F**, **46A**, **46B**, **48A**, **48B**, **50A**, and **50B** show various examples of such AN cluster implementations. Deploying large numbers of ANs **515** in a relatively small geographic area can provide a number of benefits. For example, it can be easier to ensure that more (or even all) of the ANs **515** are deployed closer to a high-speed network (e.g., in a region with good fiber connectivity back to the CPS **505**), within borders of a single country or region, on accessible areas, etc., with less deviation from an ideal AN **515** distribution. Implementing distinct coverage area servicing with path selection (e.g., as in FIGS. **47A** and **47B**) can provide additional features. For example, as described above, a single AN cluster (and a single end-to-end relay **3403**) can be used to selectively service multiple user coverage areas **3460**. Similarly, a single end-to-end relay **3403** can be used to distinguish and service traffic by region.

[0310] In some cases, the distinct coverage area servicing with path selection can enable various interference management and/or capacity management features. For example, turning back to FIGS. **48A** and **48B**, four categories of communications links can be considered: forward-link communications from the AN cluster to the western active user coverage area **3460a** ("Link A"); forward-link communications from the AN cluster to the eastern active user coverage area **3460b** ("Link B"); return-link communications from the western active user coverage area **3460a** to the AN cluster ("Link C"); and return-link communications from the eastern active user coverage area **3460b** to the AN cluster ("Link D"). In a first time interval, the eastern user coverage area **3460b** is active, so that communications are over Link B and Link D. Because there is full overlap between the AN area **3450** and the eastern user coverage area **3460b**, Links B and D potentially interfere. Accordingly, during the first time interval, Link B can be allocated a first portion of the bandwidth (e.g., 2 GHz), and Link D can be allocated a second portion of the bandwidth (e.g., 1.5 GHz). In a second time interval, the western user coverage area **3460a** is active, so that communications are over Link A and Link C. Because there is no overlap between the AN area **3450** and the western user coverage area **3460a**, Link A and Link C can use the full bandwidth (e.g., 3.5 GHz) of the end-to-end relay **3403** during the second time interval. For example, during the first time interval, the forward uplink signals **521** can be received using a first frequency range, and the return uplink signals **525** can be received using a second frequency range different from the first frequency range; and during the second time interval, the forward uplink signals **521** and the return uplink signals **525** can be received using a same frequency range (e.g., the first, second, or other frequency range). In some cases, there can be frequency reuse during both the first and second time intervals,

with other interference mitigation techniques used during the first time interval. In some cases, the path selection timing can be selected to compensate for such a difference in bandwidth allocation during different time intervals. For example, the first time interval can be longer than the second time interval, so that Links B and D are allocated less bandwidth for more time to at least partially compensate for allocating Links A and C more bandwidth for a shorter time. Other alternative frequency allocations are discussed below.

[0311] In some cases, first return uplink signals **525** are received during the first time interval by the plurality of cooperating user-link constituent receive elements **3426a** from a first portion of the plurality of user terminals **517** geographically distributed over some or all of a first user coverage area **3460** (e.g., the eastern user coverage area **3460b**), and second return uplink signals **525** are received during the second time interval by the plurality of cooperating user-link constituent receive elements **3426b** from a second portion of the plurality of user terminals **517** geographically distributed over some or all of a second user coverage area **3460** (e.g., the western user coverage area **3460a**). When the AN area **3450** (the AN cluster) is a subset of the first user coverage area **3460b** (e.g., as illustrated in FIG. **48B**), the AN **515** timing can be calibrated with the end-to-end relay **3403** during the first time frame (e.g., when there is overlap between the user coverage area **3460b** and the AN area **3450**).

[0312] As described above, some cases can include determining a respective relative timing adjustment for each of the plurality of ANs **515**, such that associated transmissions from the plurality of ANs **515** reach the end-to-end relay **3403** in synchrony (e.g., with sufficiently coordinated timing relative to the symbol duration, which is typically a fraction of the symbol duration such as 10%, 5%, 2% or other suitable value). In such cases, the forward uplink signals **521** are transmitted by the plurality of ANs **515** according to the respective relative timing adjustments. In some such cases, a synchronization beacon signal (e.g., a PN signal generated by a beacon signal generator, as described above) is received by at least some of the plurality of ANs **515** from the end-to-end relay **3403**, and the respective relative timing adjustments are determined according to the synchronization beacon signal. In other such cases, some or all of the ANs **515** can receive loopback transmissions from the end-to-end relay **3403**, and the respective relative timing adjustments are determined according to the loopback transmissions. The various approaches to calibrating the ANs **515** can depend on the ability of the ANs **515** to communicate with the end-to-end relay **3403**. Accordingly, some cases can calibrate the ANs **515** only during time intervals during which appropriate coverage areas are illuminated. For example, loopback transmissions via the user-link antenna subsystem **3420** can only be used in time intervals during which there is some overlap between the AN area **3450** and the user coverage area **3460** (e.g., the ANs **515** communicate over a loopback beam which can use both a feeder-link antenna subsystem **3410** and a user-link antenna subsystem **3420** of the end-to-end relay **3403**). In some cases, proper calibration can further rely on some overlap between the feeder downlink frequency range and the user downlink frequency range.

[0313] As discussed above, an end-to-end relay **3403** with or without selectively coupled transponders like those illustrated in FIGS. **49A**, **49B**, **51A** and **51B** can service user terminals within a first user coverage area **3460** using ANs **515** within a first AN area **3450** that is overlapping with the first user coverage area **3460** (e.g., both in North America), and service user terminals within a second user coverage area **3460** using ANs **515** within a second AN area **3450** that is overlapping with the second user coverage area **3460** (e.g., both in South America). Alternatively, an end-to-end relay **3403** like that of FIGS. **51A** and **51B** can service user terminals within a first user coverage area **3460** using ANs **515** within a first AN area **3450** that is non-overlapping with the first user coverage area **3460** and service user terminals within a second user coverage area **3460** using ANs **515** within a second AN area **3450** that is non-overlapping with the second user coverage area **3460**, as shown in FIG. **50B**. As also shown in FIG. **50B**, the first and second user coverage areas **3460** may be configured to at least partially overlap with each other to

provide contiguous coverage to a given region (e.g., CONUS region, visible Earth coverage region, etc.). Other similar implementations are also possible.

[0314] The system discussed with reference to FIG. 50B may, for example, include a forward beamformer 529 that generates access node-specific forward signal for each of the pluralities of ANs 515 within AN areas 3450. Each of the plurality of ANs 515 within a given AN area 3450 may obtain an access node-specific forward signal from the forward beamformer 529 (e.g., via a distribution network 518) during a time window in which the given AN cluster is active, and transmit a corresponding forward uplink signal 521 to the end-to-end relay 3403. The time window in which the given AN cluster is active may include one or more time-slices, if a time-slice beamformer architecture is employed as described above.

[0315] As described above, the system may include a means for pre-correcting the forward uplink signals 521 to compensate for, e.g., path delays, phase shifts, etc. between the respective ANs and the end-to-end relay 3403. In some cases, the pre-correction may be performed by the forward beamformer 529. Additionally or alternatively, the pre-correction may be performed by the ANs 515 themselves. As an example, each of the ANs 515 may transmit an access node beacon signal to end-to-end relay 3403 and receive signaling from end-to-end relay 3403 including a relay beacon signal and the relayed access node beacon signal (e.g., relayed from end-to-end relay 3403). In this example, each AN 515 may adjust its respective forward uplink signal 521 (e.g., may adjust timing and/or phase information associated with the signal transmission) based on the relayed access node beacon signal. As an example, the AN 515 may adjust the forward uplink signal 521 to time and phase align the relayed access node beacon signal with the received relay beacon signal. In some cases, the signaling described in this example (e.g., the access node beacon signal, the relay beacon signal, and the relayed access node beacon signal) may be received or transmitted via a feeder-link antenna subsystem 3410, as described above. Thus, in some cases, though not shown, the end-to-end relay 3403 includes a beacon signal transmitter. The beacon signal transmitter can be implemented as described above with reference to the beacon signal generator and calibration support module 424 of FIG. 15.

[0316] While portions of the above description have discussed techniques for end-to-end beamforming between a single active AN area 3450 (e.g., selected between two or more AN areas 3450) and a single active user coverage area 3460 (e.g., selected between two or more user coverage areas 3460), in some cases it may be desirable to have multiple distinct AN areas 3450 concurrently (e.g., cooperatively) used to provide service to a single user coverage area 3460. An example of such a system is displayed with respect to FIG. 50C, which includes AN areas 3450a and 3450b as well as user coverage area 3460.

[0317] With reference to FIG. 50C, an example system may include multiple AN clusters (e.g., two relatively dense AN clusters). Each AN cluster may contain multiple ANs 515 geographically distributed within the respective AN area 3450, where each AN 515 is operable to transmit a respective pre-corrected forward uplink signal 521 to the end-to-end relay 3403. The multiple AN clusters may be used cooperatively for providing service to user terminals 517 within the user coverage area 3460. Multiple AN clusters may be employed cooperatively using a variety of techniques. In one example, an end-to-end relay 3403 may employ a feeder-link antenna subsystem 3410 having a single feeder-link antenna element array 3415 and a compound reflector that illuminates the multiple AN clusters.

[0318] FIG. 57 illustrates a feeder-link antenna subsystem 3410c having a single feeder-link antenna element array 3415 and a compound reflector 5721. Each of multiple regions of the compound reflector 5721 may have a focal point 1523 (which may be the same or a different distance from the compound reflector). A first example is illustrated in FIG. 57 in which the compound reflector 5721 has a single focal point (or region) 1523a. The feeder-link antenna element array 3415 may be positioned at a defocused point of the compound reflector. As illustrated, the feeder-link antenna element array 3415 is located inside the focal point 1523a (i.e.,

is closer to the compound reflector **5721** than the focal point **1523a**). Alternatively, the feeder-link antenna element array **3415** may be located outside the focal point **1523a** (i.e., the feeder-link antenna element array **3415** may be farther from the compound reflector **5721** than the focal point **1523a**). A second example is illustrated in FIG. 57 in which the compound reflector **5721** has two focal points (or regions) **1523b** and **1523c**. In the present example, the feeder-link antenna element array **3415** is illustrated as being located inside the focal points **1523b** and **1523c**. Alternatively, the feeder-link antenna element array **3415** may be located outside the focal points **1523b** and **1523c**. In yet another embodiment, the feeder-link antenna element array **3415** may be located inside one focal point (e.g. focal point **1523b**) and outside another focal point (e.g., focal point **1523c**). In some cases, focal point **1523b** may be associated with a top portion of the compound reflector **5721** while focal point **1523c** is associated with a bottom portion of the compound reflector **5721**. Alternatively, focal point **1523b** may be associated with a bottom portion of the compound reflector **5721** while focal point **1523c** is associated with a top portion of the compound reflector **5721**. The feeder-link antenna element array **3415** may include feeder-link constituent transmit elements **3419** and feeder-link constituent receive elements **3416**, which in some cases may be the same antenna elements (e.g., with different polarizations or frequencies used for transmitting and receiving, etc.). [0319] In the transmit direction, the output of the feeder-link constituent transmit elements **3419** may reflect from the reflector **5721** to form a first beam group **5705a** that illuminates a first AN area **3450** (e.g., AN area **3450a** of FIG. 50C) and a second beam group **5705b** that reflects a second AN area **3450** (e.g., AN area **3450b** of FIG. 50C). Although not shown, in a receive direction signals from a first AN area **3450a** and from a second AN area **3450b** may be reflected to feeder-link constituent receive elements **3416** of the feeder-link antenna element array **3415** using compound reflector **5721**.

[0320] Returning to FIG. 50C, the multiple AN areas **3450** may be used independently or together (e.g., cooperatively). For example, ANs of only one of AN areas **3450a** or **3450b** may be activated at a given time, and beamforming coefficients may be generated for forming user beam coverage areas **519** within user coverage area **3460** from the ANs **515** of the active AN cluster. Alternatively, beamforming coefficients may be generated for forming user beams within user coverage area **3460** using both AN clusters concurrently (e.g., cooperatively). In the forward direction, a forward beamformer **529** may apply the beamforming coefficients (e.g., by a matrix product between forward beam signals and a forward beam weight matrix) to obtain a plurality of access-node specific forward signals for ANs **515** within both clusters to generate the desired forward user beams. In the return direction, the return beamformer **531** may obtain the composite return signals from ANs **515** within both clusters and apply a return beam weight matrix to form the return beam signals associated with the return user beams.

[0321] In some cases, AN areas **3450a** and **3450b** may be non-overlapping (e.g., disjoint). Alternatively, AN areas **3450a** and **3450b** may be (e.g., at least partially) overlapping. Further, at least one of AN areas **3450a** and **3450b** may be at least partially overlapping with user coverage area **3460**. Alternatively, at least one of the AN areas **3450a** and **3450b** may be non-overlapping (e.g., disjoint) with user coverage area **3460**. As discussed above, in some cases at least one of the ANs **515** in one or both of AN areas **3450a** or **3450b** may be disposed on a mobile platform and/or located in an aquatic body.

[0322] Referring to FIG. 50B or 50C, each of multiple AN areas **3450** may be illuminated using a separate feeder-link antenna element array **3415**. In some cases, the separate feeder-link antenna element arrays **3415** may be used concurrently (e.g., multiple AN areas **3450** may be used cooperatively) to support service provided to a single user coverage area **3460**. With reference again to FIG. 50C, an end-to-end relay **3403** may have separate feeder-link antenna element arrays **3415** illuminating each of AN areas **3450a** and **3450b**. In some examples, the end-to-end relay **3403** may have separate feeder-link antenna subsystems **3410**, where each feeder-link antenna subsystem **3410** includes a feeder-link antenna element array **3415** and a reflector. FIG. 56A shows

an end-to-end relay **3403** having a feeder-link antenna subsystem **3410a** that includes a first feeder-link antenna element array **3415a** that illuminates the first AN area **3450a** via a first reflector **5621a** and a second feeder-link antenna element array **3415b** that illuminates the second AN area **3450b** via a second reflector **5621b**. The first and second feeder-link antenna element arrays **3415a** and **3415b** may each include feeder-link constituent receive elements **3416** and feeder-link constituent transmit elements **3419**. FIG. **56B** shows a feeder-link antenna subsystem **3410b** that includes a first feeder-link antenna element array **3415a** and a second feeder-link antenna element array **3415b** that illuminate corresponding AN areas **3450** via a single reflector **5621**. As illustrated in FIG. **56B**, the feeder-link element arrays **3415** may be located in defocused positions in relation to the focal point **1523** of reflector **5621**. Although the feeder-link element arrays **3415** are displayed as being located beyond the focal point **1523** of reflector **5621**, they may alternatively be located closer to the reflector **5621** than the focal point **1523**.

[0323] Similarly, multiple user coverage areas **3460** may be implemented using separate user-link antenna element arrays **3425** with either separate reflectors (similar to FIG. **56A**) or a single reflector (similar to FIG. **56B**). Thus, the multiple AN areas **3450** and multiple user coverage areas **3460** in FIG. **50B** may be deployed using any combination of a single feeder-link reflector or multiple feeder-link reflectors and a single user-link reflector or multiple user-link reflectors. In another example, a deployment similar to that shown in FIG. **50B** may be achieved with reflectors shared between feeder-links and user-links using different feeder-link and user-link frequency bands. For example, a single antenna element array may have feeder-link constituent elements and user-link constituent elements (e.g., in an interleaved pattern such as that shown in FIG. **62**). The feeder-link may use a frequency range that is higher (e.g., more than 1.5 or 2 times higher) to provide a higher gain with a common reflector. In one example, the user-link may use a frequency range (or ranges) in the K/Ka bands (e.g., around 30 GHz) while the feeder-link uses frequency range(s) in the V/W bands (e.g., around 60 GHz). Because of the narrower beamwidth at higher frequencies, the AN area **3450** sharing the common antenna element array (and thus reflector) will be a smaller area (and concentric with) the user coverage area. Thus, one antenna subsystem including a single antenna element array and reflector may be used to illuminate user coverage area **3450a** and AN area **3450b** while a second antenna subsystem including a single antenna element array and reflector may be used to illuminate user coverage area **3450b** and AN area **3450a**. In yet another example for a deployment similar to FIG. **50B**, a single antenna subsystem may include a single reflector and two antenna element arrays as shown in FIG. **56B**, where each antenna element array includes feeder-link constituent elements and user-link constituent elements.

[0324] Referring again to FIG. **56B**, in some cases, the first feeder-link antenna element array **3415a** may be coupled with a first subset of the multiple receive/transmit signal paths associated with the end-to-end relay **3403** while the second feeder-link antenna element array **3415b** may be coupled with a second subset of the multiple receive/transmit signal paths. Thus, a first set of forward uplink signals **521** from the AN cluster having AN area **3450a** may be carried via a first subset of the multiple receive/transmit signal paths associated with the end-to-end relay **3403**. Additionally, a second set of forward uplink signals **521** from the AN cluster having AN area **3450b** may be carried via a second subset of the multiple receive/transmit signal paths. In some cases, the first and second sets of forward uplink signals may both contribute to forming a forward user beam associated with at least one of the multiple forward user beam coverage areas **519** in user coverage area **3460**.

[0325] FIGS. **52A** and **52B** show example forward and return receive/transmit signal paths for cooperative use of multiple AN clusters, where each AN cluster is associated with a separate feeder-link antenna element array **3415**. Referring first to FIG. **52A**, an example forward signal path **5200** is shown. Forward signal path **5200** includes a first forward link transponder **3430e** coupled between a feeder-link constituent receive element **3416a** of a first feeder-link antenna element array **3415a** and a first user-link constituent transmit element **3429** of a user-link antenna

element array **3425** and a second forward link transponder **3430e** coupled between a feeder-link constituent receive element **3416b** of a second feeder-link antenna element array **3415b** and a second user-link constituent transmit element **3429** of the same user-link antenna element array **3425**. An end-to-end relay **3403** may have a first set of forward link transponders **3420** coupled as shown by the first forward link transponder **3430e** and a second set of forward link transponders **3430** coupled as shown by the second forward link transponder **3430e**. Thus, the feeder-link constituent receive elements **3416a** of the first feeder-link antenna element array **3415a** may be coupled via a first set of forward link transponders **3430e** to a first subset of user-link constituent transmit elements **3429** of a user-link antenna element array **3425** while the feeder-link constituent receive elements **3416b** of the second feeder-link antenna element array **3415b** may be coupled via a second set of forward link transponders **3430e** to a second subset of user-link constituent transmit elements **3429** of the same user-link antenna element array **3425**. The first and second sets of user-link constituent transmit elements **3429** may be spatially interleaved (e.g., alternated in rows and/or columns, etc.) within the user-link antenna element array **3425** (e.g., as shown in FIG. 62).

[0326] FIG. 52B illustrates an example return signal path **5250**. Return signal path **5250** includes a first return link transponder **3440e** coupled between a user-link constituent receive element **3426a** of a user-link antenna element array **3425** and a first feeder-link constituent transmit element **3419a** of a first feeder-link antenna element array **3415a**. Return signal path **5250** also includes a second return link transponder **3440e** coupled between a user-link constituent receive element **3426b** of the same user-link antenna element array **3425** and a second feeder-link constituent transmit element **3419b** of a second feeder-link antenna element array **3415b**. An end-to-end relay **3403** may have a first set of return link transponders **3440** coupled as shown by the first return link transponder **3440e** and a second set of return link transponders **3440** coupled as shown by the second return link transponder **3440e**. Thus, a first subset of the user-link constituent receive elements **3426a** of the user-link antenna element array **3425** may be coupled via a first set of return link transponders **3440e** to feeder-link constituent transmit element **3419a** of a first feeder-link antenna element array **3415a** while a second subset of the user-link constituent receive elements **3426b** of the same user-link antenna element array **3425** may be coupled via a second set of return link transponders **3440e** to feeder-link constituent transmit element **3419b** of a second feeder-link antenna element array **3415b**. As discussed above, the user-link constituent receive elements **3426** and user-link constituent transmit elements **3429** may be the same physical antenna elements. Similarly, the feeder-link constituent receive elements **3416** and feeder-link constituent transmit elements **3419** of a given feeder-link antenna element array **3415** may be the same physical antenna elements.

[0327] The first and second sets of user-link constituent receive elements **3426** may be spatially interleaved (e.g., alternated in rows and/or columns, etc.) within the user-link antenna element array **3425**. FIG. 62 shows an example antenna element array **6200** with spatially interleaved subsets of constituent antenna elements **6205**. Although each constituent antenna element **6205** is shown as a circular antenna element and the interleaved subsets are shown as being arranged in alternating rows, the constituent antenna elements **6205** may be any shape (e.g., square, hexagonal, etc.) and arranged in any suitable pattern (e.g., alternating rows or columns, a checkerboard, etc.). Each constituent antenna element **6205** may be an example of a user-link constituent receive element **3416** or a user-link constituent transmit element **3419**, or both (e.g., an element used for both transmit and receive).

[0328] With reference to FIGS. 52A and 52B where the user-link antenna element array **3425** is implemented as the antenna element array **6200** of FIG. 62, the first set of forward link transponders **3430e** may each have its output coupled with one the first set of user-link antenna elements **6205a** while the second set of forward link transponders **3430e** may each have its output coupled with one of the second set of user-link antenna elements **6205b**. In addition, the first set of return link transponders **3440e** may each have its input coupled with one the first set of user-link antenna elements **6205a** while the second set of return link transponders **3440e** may each have its

input coupled with one of the second set of user-link antenna elements **6205b**.

[0329] In some cases, the end-to-end relay **3403** includes a large number of transponders, such as 512 forward-link transponders **3430** and 512 return-link transponders **3440** (e.g., 1,024 transponders total). Thus, the first set of forward link transponders **3430e** of FIG. 52A may include 256 transponders and the second set of forward link transponders **3430e** may include 256 transponders.

[0330] In some cases, support for the use of multiple AN clusters is provided through characteristics of the transponders associated with the end-to-end relay **3403**. Additionally or alternatively, support for the use of multiple AN clusters may be provided using one or more appropriately designed reflectors. Some example transponders are described above (e.g., with respect to FIGS. 49A, 49B, 51A, 51B, 52A and 52B). Further examples of transponder designs are discussed below. It should be understood that techniques described with reference to any one the example forward link transponders **3430** and return link transponders **3440** may in some cases be applicable to any other example transponder. Further, the components of the transponders may be rearranged in any suitable fashion without deviating from the scope of the disclosure.

[0331] Only a single polarization of the receive/transmit paths (e.g., a cross-pole transponder) is shown in FIGS. 49A, 49B, 52A and 52B for clarity. For example, the forward-link transponder **3430** receives a forward uplink signal **521** at an uplink frequency with left-hand circular polarization (LHCP) and outputs a forward downlink signal **522** at a downlink frequency with right-hand circular polarization (RHCP); and each return-link transponder **3440** receives a return uplink signal **525** at the uplink frequency with right-hand circular polarization (RHCP) and outputs a return downlink signal **527** at the downlink frequency with left-hand circular polarization (LHCP). In other cases, some or all transponders can provide a dual-pole signal path pair. For example, the forward-link transponders **3430** and the return-link transponders **3440** can receive uplink signals at the same or different uplink frequency with both polarizations (LHCP and RHCP) and can both output downlink signals at the same or different downlink frequency with both polarizations (RHCP and LHCP). For example, such cases can enable multiple systems to operate in parallel using any suitable type of interference mitigation techniques (e.g., using time division, frequency division, etc.). In some cases, the end-to-end relay **3403** includes a large number of transponders, such as 512 forward-link transponders **3430** and 512 return-link transponders **3440** (e.g., 1,024 transponders total). Other implementations can include smaller numbers of transponders, such as 10, or any other suitable number. In some cases, the antenna elements are implemented as full-duplex structures, so that each receive antenna element shares structure with a respective transmit antenna element. For example, each illustrated antenna element can be implemented as two of four waveguide ports of a radiating structure adapted for both transmission and reception of signals. In some cases, only the feeder-link elements, or only the user-link elements, are full duplex. Other implementations can use different types of polarization. For example, in some implementations, the transponders can be coupled between a receive antenna element and transmit antenna element of the same polarity.

[0332] Both the example forward-link transponder **3430** and return-link transponder **3440** can include some or all of LNAs **3705**, frequency converters and associated filters **3710**, channel amplifiers **3715**, phase shifters **3720**, power amplifiers **3725** (e.g., TWTAs, SSPAs, etc.) and harmonic filters **3730**. In dual-pole implementations, as shown, each pole has its own signal path with its own set of transponder components. Some implementations can have more or fewer components. For example, the frequency converters and associated filters **3710** can be useful in cases where the uplink and downlink frequencies are different. As one example, each forward-link transponder **3430** can accept an input at a first frequency range and can output at a second frequency range; and each return-link transponder **3440** can accept an input at the first frequency range and can output at the second frequency band. Additionally or alternatively, each forward-link transponder **3430** can accept an input at a first frequency range and can output at a second

frequency range; and each return-link transponder **3440** can accept an input at the second frequency range and can output at the first frequency range.

[0333] As an example, the transponders of FIGS. **52A** and **52B** may be implemented in a system similar to that of FIG. **50C**. In this example, some or all of the ANs **515** in AN area **3450a** may transmit forward uplink signals **521** in coordination with some or all of the ANs **515** in AN area **3450b**. The forward uplink signals from the two AN clusters may thus combine to serve user terminals in user coverage area **3460**. In this example, some AN clusters may affect only some user link antenna elements (e.g., some AN clusters may be associated with a subset of feeder link constituent receive elements **3416** which may be coupled to a corresponding subset of user link constituent transmit elements **3429**). Although the above example discusses the use of two clusters, other embodiments using more clusters are also possible.

[0334] Another example forward signal path **5300** is shown in FIG. **53A**. Forward signal path **5300** may include some combination of LNAs **3705a**, frequency converters and associated filters **3710a**, channel amplifiers **3715a**, phase shifters **3720a**, power amplifiers **3725a** (e.g., TWTAs, SSPAs, etc.) and harmonic filters **3730a**. The input side of the forward-link transponder **3430f** is selectively coupled to one of feeder-link constituent receive elements **3416a** or **3416b** (e.g., using a switch **4010b**, or any other suitable path selection means). Each feeder-link constituent receive element **3416a** or **3416b** can be part of a separate feeder-link antenna element array **3415** (e.g., each part of a separate array **3415** of cooperating feeder-link constituent receive elements **3416**). The output side of the forward-link transponder **3430f** is coupled to a user-link constituent transmit element **3429** of a user-link antenna element array **3425** (e.g., which is part of a user-link antenna element subsystem **3420**). One or more switching controllers **4070** (not shown) can be included in the end-to-end relay **3403** for selecting between some or all of the possible signal paths enabled by the forward-link transponder **3430f**. Thus, where the example transponder **3430b** of FIG. **47A** allows, for example, selective coupling between a single feeder-link constituent receive element **3416** and multiple user-link constituent transmit elements **3429**, the example transponder **3430f** of FIG. **53A** allows, for example, selective coupling between multiple feeder-link constituent receive elements **3416a**, **3416b** and a single user-link constituent transmit element **3429**.

[0335] An example return signal path **5350** is shown in FIG. **53B**. Return signal path **5350** may include some combination of LNAs **3705b**, frequency converters and associated filters **3710b**, channel amplifiers **3715b**, phase shifters **3720b**, power amplifiers **3725b** (e.g., TWTAs, SSPAs, etc.) and harmonic filters **3730b**. The output side of the return-link transponder **3440f** is selectively coupled to one of feeder-link constituent transmit elements **3419a** or **3419b** (e.g., using a switch **4010a**, or any other suitable path selection means). Each feeder-link constituent transmit element **3419a** or **3419b** can be part of a separate feeder-link antenna element array **3415** (e.g., each part of a separate array **3415** of cooperating feeder-link constituent transmit elements **3419**). The input side of the return-link transponder **3440f** is coupled to a user-link constituent receive element **3426** of a user-link antenna element array **3425** (e.g., which is part of a user-link antenna element subsystem **3420**). One or more switching controllers **4070** (not shown) can be included in the end-to-end relay **3403** for selecting between some or all of the possible signal paths enabled by the return-link transponder **3440f**. Thus, where the example return link transponder **3440b** of FIG. **47B** allows, for example, selective coupling between a single feeder-link constituent transmit element **3419** and multiple user-link constituent receive elements **3426**, the example transponder **3440f** of FIG. **53B** allows, for example, selective coupling between a single user-link constituent receive element **3426** and multiple feeder-link constituent transmit elements **3419**.

[0336] As an example, the forward link transponder **3430f** of FIG. **53A** may be implemented in a system similar to that of FIG. **50C**. In this example, some or all of the ANs **515** in AN area **3450a** may transmit forward uplink signals **521** during a first time interval. Some or all of the ANs **515** in AN area **3450b** may transmit forward uplink signals **521** during a second time interval. Using some appropriate path selection means (e.g., a switch), the forward link transponder **3430f** can receive

input from AN area **3450a** (e.g., via the first array of cooperating feeder-link constituent receive elements **3416a**) during the first time interval and from AN area **3450b** (e.g., via the second array of cooperating feeder-link constituent receive elements **3416b**) during the second time interval. In some such scenarios, each AN area **3450** may include a full complement of ANs **515** (e.g., such that each AN area **3450** can provide appropriate beamforming over the entire user coverage area **3460**).

[0337] As an example, the return-link transponder **3440f** of FIG. **53B** may be implemented in a system similar to that of FIG. **50C**. In this example, some or all of the ANs **515** in AN area **3450a** may receive return downlink signals **527** during a first time interval. Some or all of the ANs **515** in AN area **3450b** may receive return downlink signals **527** during a second time interval. Using some appropriate path selection means (e.g., a switch), the return link transponder **3440f** can output to AN area **3450a** (e.g., via the first array of cooperating feeder-link constituent transmit elements **3419a**) during the first time interval and to AN area **3450b** (e.g., via the second array of cooperating feeder-link constituent transmit elements **3419b**) during the second time interval. In some such scenarios, each AN area **3450** may include a full complement of ANs **515** (e.g., such that the single AN area **3450** can provide appropriate beamforming over the entire user coverage area **3460**).

[0338] FIGS. **54A** and **54B** illustrate forward and return link transponders **3430g** and **3440g**, respectively. These transponders are similar to those of FIGS. **51A** and **51B** except that the components have been rearranged such that the switch **4010a** follows the harmonic filter(s) **3730**. As discussed above, other rearrangements of components may be possible. In some cases, this example arrangement may require fewer power amplifiers **3725** and/or harmonic filters **3730**. Similarly to FIGS. **51A** and **51B**, such an arrangement may enable selective association between AN clusters and user coverage areas **3460**. This selective association may allow flexible allocation of capacity between two (or more) user coverage areas **3460** as well as frequency reuse between user and feeder links (e.g., which may increase the capacity of the system).

[0339] As discussed above with reference to FIG. **46B**, in some cases there may not be overlap between the AN area **3450** and the user coverage area **3460**, which may require the use of a separate loopback mechanism from that discussed above. In some cases, the separate loopback mechanism may include the use of a loopback transponder **5450**, such as that shown in FIG. **55A**, **55B**, or **55C**. In some embodiments, the loopback transponder **5450** may receive AN loopback beacons (e.g., AN loopback beacons transmitted from each AN), which may be examples of the access node beacon signals **2530** discussed with reference to FIG. **38**. The loopback transponder **5450** may retransmit the access node beacon signals **2530** and transmit a satellite beacon (e.g., which may be generated using a relay beacon generator **426** as described above). In some of the following examples, the input side of the loopback transponder **5450** is coupled to a feeder-link antenna element. Alternatively, the input side of the loopback transponder **5450** may be coupled to a loopback antenna element that is separate and distinct from the feeder-link antenna element array(s). Similarly, in some of the following examples, the output side of the loopback transponder **5450** is coupled to a feeder-link antenna element or a user-link antenna element. Alternatively, the output side of the loopback transponder **5450** may be coupled to a loopback antenna element distinct from the feeder-link antenna element array(s) and the user-link antenna element array(s), which may be the same or different than the loopback antenna element coupled to the input side of the loopback transponder **5450**.

[0340] Referring to FIG. **55A**, loopback transponder **5450a** may include some combination of LNAs **3705c**, frequency converters and associated filters **3710c**, channel amplifiers **3715c**, phase shifters **3720c**, power amplifiers **3725c** (e.g., TWTAs, SSPAs, etc.) and harmonic filters **3730c**. Further, as illustrated in FIG. **55B**, in the case where an end-to-end relay **3403** has multiple feeder-link antenna element arrays **3415**, the input side of the loopback transponder **5450** may be selectively coupled to one of a first feeder-link constituent receive element **3416a** of a first feeder-

link antenna element array **3415a** or a second feeder-link constituent receive element **3416b** of a second feeder-link antenna element array **3415b** (e.g., using a switch **4010b**, or any other suitable path selection means). FIG. 55A shows the output side of example loopback transponder **5450a** coupled to a feeder-link constituent transmit element **3419**. FIG. 55B shows the output side of example loopback transponder **5450b** selectively coupled (e.g., using a switch **4010a**, or any other suitable path selection means) to either feeder-link constituent transmit element **3419a** or feeder-link constituent transmit element **3419b**, which may be components of a same feeder-link antenna element array **3415** or different feeder-link antenna element arrays. That is, feeder-link constituent transmit element **3419b** may be a component of the same antenna element array **3415** as feeder-link constituent transmit element **3419a** and/or feeder-link constituent receive element **3416b**. As illustrated, feeder-link constituent transmit element **3419b** is part of the same antenna element array **3415b** as feeder-link constituent receive element **3416b**. Similarly, the input side of loopback transponder **5450b** may be selectively coupled (e.g., using a switch **4010b**, or any other suitable path selection means) to either feeder-link constituent receive element **3416a** or **3416b**, which may be components of a same or different feeder-link antenna element arrays **3415**. The loopback transponder **5450b** of FIG. 55B may be employed in cases where the end-to-end relay **3403** supports the selective use of one of multiple access node areas **3450** (e.g., as discussed in some examples illustrated by FIG. 50B). Thus, switch **4010a** may be set to a first position to provide the output of loopback transponder **5450b** to feeder-link constituent transmit element **3419a** when a first access node area **3450** is active and to a second position to provide the output of loopback transponder **5450b** to the feeder-link constituent transmit element **3419b** when a second access node area **3450** is active. In some cases, there may be two or more feeder-link constituent transmit elements **3419**, and each can be part of a separate feeder-link antenna element array **3415** (e.g., for support of selective use of one access node area **3450** from two or more access node areas **3450**). Referring to FIG. 55B, one or more switching controllers **4070** (not shown) can be included in the end-to-end relay **3403** for selecting between some or all of the possible signal paths enabled by the loopback transponder **5450b**. In some cases, a feeder-link constituent receive element **3416** and a feeder-link constituent transmit element **3419** may be associated with the same physical structures, as described above. In some cases, the ANs **515** may be able to synchronize transmissions based on a comparison of the retransmitted access node beacon signals **2530** and the satellite beacon (e.g., the transmissions from ANs **515** within one or more AN clusters may be time and phase aligned based on the comparison).

[0341] In some cases, the feeder-link frequency range may be different from the user-link frequency range. When the feeder-link downlink frequency range is non-overlapping with the user-link downlink frequency range, the transponders that translate from the feeder-link uplink frequency range to the user-link downlink frequency range (e.g., using a frequency converter **3710**) cannot be used to relay the access node beacon signals (e.g., because the ANs cannot receive and process the user-link downlink frequency range). In such cases, the loopback transponder **5450** may solve the issue by translating the access node beam signals from the feeder-link uplink frequency range to the feeder-link downlink frequency range. For example, feeder-link communications (e.g., forward uplink signals **521** and return downlink signals **527**) may be in a first frequency range (e.g., a frequency range within V/W band), and user-link communications (e.g., forward downlink signals **522** and return uplink signals **525**) may be in a second frequency range (e.g., a frequency range within K/Ka band). Thus, even where the AN area **3450** overlaps the user coverage area **3460**, the ANs **515** may not be able to receive AN loopback signals relayed via the receive/transmit signal paths (e.g., forward transponders **3430** and/or return transponders **3440**) of the end-to-end relay **3403**.

[0342] FIG. 55C shows an example loopback transponder **5450c** that receives all AN loopback signals in the feeder-link uplink frequency range and relays the AN loopback signals in the feeder-link downlink frequency range. Loopback transponder **5450c** may be used in any of the above

access node cluster deployments where the access node area **3450** does not overlap with the user coverage area **3460** (e.g., at least some of the deployments discussed with FIG. **45C**, **45E**, **45F**, **45G** or **50B**). The feeder-link uplink frequency range and the feeder-link downlink frequency range may be part of the same band (e.g., K/Ka band, V band, etc.) or different bands. The AN loopback signals may be received via antenna element **3455**, which may be part of a feeder-link antenna element array **3415**, or may be a separate loopback antenna element. The relayed AN loopback signals may be transmitted via the same antenna element **3455** as shown, or a different antenna element, in some cases. The loopback transponder **5450c** includes loopback frequency converter **5460**, which may convert the AN loopback signals from one carrier frequency within the feeder-link uplink frequency range to a different carrier frequency within the feeder-link downlink frequency range. Loopback transponder **5450c** may additionally contain one or more of LNAs **3705c**, channel amplifiers **3715** (not illustrated), phase shifters **3720** (not illustrated), power amplifiers **3725c**, and harmonic filters (not illustrated).

[0343] Referring again to the example end-to-end beamforming system **3400** of FIG. **41**, aspects of system **3400** may be modified to support cooperative operation of multiple AN clusters that use different frequency ranges. FIGS. **59A** and **59B** illustrate examples of possible geographic coverage areas for multiple access node areas **3450**, each operating over a different frequency range, to be used cooperatively in end-to-end beamforming for a user coverage area **3460**. In the example illustrated in FIG. **59A**, AN area **3450a** may be associated with Ka-band transmissions while AN area **3450b** may be associated with V-band transmissions. As shown in FIG. **59A**, AN areas **3450a** and **3450b** may be disjoint. In some cases, the AN area **3450b** associated with V-band transmissions may be smaller (e.g., may cover a smaller geographic area) than the AN area **3450a** associated with Ka-band transmissions. In some cases AN area **3450a** and AN area **3450b** may be illuminated by separate feeder-link antenna element arrays **3415**. For example, AN area **3450a** may be illuminated by the first feeder-link antenna element array **3415a** and AN area **3450b** may be illuminated by the second feeder-link antenna element array **3415b** of the feeder-link antenna subsystem **3410b** shown in FIG. **56B**. As with the example of the first AN cluster in access node area **3450a** operating in Ka-band while the second AN cluster in access node area **3450b** is operating in V-band, the access node area **3450b** may be sized according to the difference in gain provided by the single reflector (e.g., which may be an example of the reflector **5621** of FIG. **56B**) in the different frequency ranges. Alternatively, the separate feeder-link antenna element arrays **3415** illuminating AN area **3450a** and AN area **3450b** may be illuminated by separate reflectors (e.g., which may be examples of reflectors **5621** discussed with reference to FIG. **56A**) which may be the same or different sizes. Alternatively, AN area **3450a** and AN area **3450b** may be illuminated by the same feeder-link antenna element array **3415** having multiple sets of feeder-link antenna elements **3416**, **3419** with a compound reflector **5721** as shown in FIG. **57**. The different frequency ranges for different AN clusters may provide higher isolation of different subsets of feeder link elements within a single feeder-link antenna element array, which may result in higher system capacity than multiple AN clusters operating in the same frequency range.

[0344] FIG. **59B** illustrates an alternative arrangement of multiple AN clusters using separate frequency ranges used cooperatively. As illustrated in FIG. **59B** the two AN clusters may at least partially overlap (or one may be completely contained within the other as shown). FIG. **59B** may illustrate examples where a single feeder-link antenna element array **3415** may illuminate AN area **3450a** and AN area **3450b** (e.g., simultaneously receive or transmit signals to both coverage areas over different frequency ranges). In some cases, a given AN **515** (e.g., one located within AN area **3450b**) may be associated with multiple AN clusters and communicate over feeder links in multiple frequency ranges (e.g., which may be contained in different frequency bands).

[0345] FIGS. **60A** and **60B** illustrate example receive/transmit signal paths supporting cooperating AN clusters operating in different frequency ranges in accordance with aspects of the present disclosure. Forward receive/transmit signal path **6000** of FIG. **60A** includes forward-link

transponders **3430h** coupled between feeder-link constituent receive elements **3416a** and user-link constituent transmit elements **3429a** and forward-link transponders **3430i** coupled between feeder-link constituent receive elements **3416b** and user-link constituent transmit elements **3429b**. As described above, the various user-link antenna elements may be part of different user-link antenna element arrays **3425**, which may be positioned to provide for non-overlapping access node areas **3450** as shown in FIG. **59A** or overlapping access node areas **3450** as shown in FIG. **59B**. Alternatively, the various user-link antenna elements may be part of the same feeder-link antenna element array **3415**, in which case the access node areas **3450** will overlap as shown in FIG. **59B**. The feeder-link constituent receive elements **3416a** and feeder-link constituent receive elements **3416b** may be interleaved within the same feeder-link antenna element array **3415** as illustrated in FIG. **62**.

[0346] As described above, the forward-link transponder **3430h** can include some or all of LNAs **3705a**, frequency converters and associated filters **3710h**, channel amplifiers **3715a**, phase shifters **3720a**, power amplifiers **3725a**, and harmonic filters **3730a**. Similarly, forward-link transponder **3430i** can include some or all of LNAs **3705a**, frequency converters and associated filters **3710i**, channel amplifiers **3715a**, phase shifters **3720a**, power amplifiers **3725a**, and harmonic filters **3730a**. In some cases, frequency converter **3710h** may be operable to convert signals from a first feeder-link uplink frequency range to a user-link downlink frequency range while frequency converter **3710i** is operable to convert signals from a second feeder-link uplink frequency range to the same user-link downlink frequency range.

[0347] Return receive/transmit signal path **6050** of FIG. **60B** includes return-link transponder **3440h** coupled between a user-link constituent receive element **3426a** and a corresponding feeder-link constituent transmit element **3419a** and return-link transponder **3440i** coupled between a user-link constituent receive element **3426b** and a corresponding feeder-link constituent transmit element **3419b**. As described above, the return-link transponder **3440h** can include some or all of LNAs **3705b**, frequency converters and associated filters **3710j**, channel amplifiers **3715b**, phase shifters **3720b**, power amplifiers **3725b**, and harmonic filters **3730b**. Similarly, return-link transponder **3440i** can include some or all of LNAs **3705b** frequency converters and associated filters **3710k**, channel amplifiers **3715b**, phase shifters **3720b**, power amplifiers **3725b**, and harmonic filters **3730b**. In some cases, frequency converter **3710j** may be operable to convert signals from a user-link uplink frequency range to a first feeder-link downlink frequency range (e.g., which may be the same range as the first feeder-link uplink frequency range described with reference to FIG. **60A**) while frequency converter **3710k** is operable to convert signals from the user-link uplink frequency range to a second feeder-link downlink frequency range (e.g., which may be the same range as the second feeder-link uplink frequency range described with reference to FIG. **60A**).

[0348] As described above, the various user-link antenna elements may be part of the same or different user-link antenna element arrays **3425** and the various feeder-link antenna elements may be part of the same or different feeder-link antenna element arrays **3415**. The feeder-link constituent transmit elements **3419a** and feeder-link constituent transmit elements **3419b** may be interleaved within the same feeder-link antenna element array **3415** as illustrated in FIG. **62**. Where the frequencies supported for the feeder links by the forward-link transponders **3430h** and **3430i** and return-link transponders **3440h** and **3440i** are substantially different (e.g., one being different by more than $1.5\times$ from the other, etc.), the different subsets of elements **6205a**, **6205b** of the antenna element array **6200** may be sized appropriately for the different supported frequency ranges (e.g., constituent antenna elements **6205b** supporting a higher frequency range than constituent antenna elements **6205a** may have smaller waveguides/horns, etc.).

[0349] FIG. **64A** illustrates an example frequency spectrum allocation **6400** with four frequency ranges displayed (frequency ranges **6425a**, **6430a**, **6435a**, and **6436a**). In the illustrated example, frequency ranges **6425a** and **6430a** are frequency ranges within the K/Ka-bands (e.g., between 17

GHz and 40 GHz) while frequency ranges **6435a** and **6436a** are within the V/W bands (e.g., between 40 GHz and 110 GHz). FIG. **64A** may illustrate operation of multiple AN clusters operating over different frequency ranges as shown in FIGS. **59A** and **59B**.

[0350] As one example, frequency spectrum allocation **6400** may be used in the scenario illustrated in FIG. **59A** using an end-to-end relay **3403** having forward and return receive/transmit signal paths **6000** and **6050** as shown in FIGS. **60A** and **60B**. In this example, forward uplink signals **6440a** from AN area **3450a** may be transmitted over frequency range **6430a** (e.g., using RHCP) while forward uplink signals **6440b** from AN area **3450b** may be transmitted over frequency range **6436a** (e.g., using RHCP). The first set of forward uplink signals **6440a** may be received by feeder-link constituent receive elements **3416a** while the second set of forward uplink signals **6440b** may be received by feeder-link constituent receive elements **3416b**. For the sake of simplicity, signals may be illustrated by their span over portions or all of a frequency range (e.g., forward uplink signal **6440a** shows the frequency span of an example of forward uplink signal **521** within frequency range **6430a**). In some cases, a given signal may span one or more frequency ranges. As discussed with reference to FIG. **60A**, the two sets of forward uplink signals **6440** are frequency converted by forward link transponders **3430h** and **3430i** (e.g., they are downconverted to the same frequency range **6425a** in the Ka-band). Subsequently, the outputs of the forward-link transponders **3430h** are transmitted by user-link constituent transmit elements **3429a** as a first set of forward downlink signals **6445a** while the outputs of the forward-link transponders **3430i** are transmitted by user-link constituent transmit elements **3429b** as a second set of forward downlink signals **6445b**. In the present example, these user-link constituent transmit elements **3429a**, **3429b** belong to the same user-link antenna element array **3425** and illuminate the same user coverage area **3460**.

Accordingly, the ANs **515** in access node areas **3450a** and **3450b** may be referred to as cooperating in that some fraction of ANs **515** in each area combine to serve the same user coverage area **3460**. That is, at least one beamformed forward user beam providing service to user terminals **517** within the corresponding user beam coverage area **519** is formed from forward uplink signals **6440a** from at least a subset of the ANs **515** in the first access node area **3450a** and from forward uplink signals **6440b** from at least a subset of the ANs **515** in the second access node area **3450b**.

[0351] Frequency spectrum allocation **6400** also illustrates an example of frequency allocation for return-link transmissions for the scenario illustrated in FIG. **59A** using an end-to-end relay **3403** having forward and return receive/transmit signal paths **6000** and **6050** as shown in FIGS. **60A** and **60B**. Return uplink signals **6450** (e.g., LHCP signals) originating from user terminals **517** distributed throughout the user coverage area **3460** may be transmitted over frequency range **6430a** (e.g., using LHCP) and received by user-link constituent receive elements **3426a** and **3426b** of FIG. **60B**, where the user-link constituent receive elements **3426a** and **3426b** belong to the same user-link antenna element array **3425**. As described with reference to FIG. **60B**, the return uplink signals **6450** may be fed to return-link transponders **3440h** and **3440i** and frequency converted to appropriate frequency ranges **6425a** (e.g., using RHCP) and **6435a** (e.g., using LHCP), respectively. The frequency converted signals **6455a** and **6455b** may then be transmitted by feeder-link constituent transmit elements **3419a** and **3419b** (e.g., which belong to separate feeder-link antenna element arrays **3415a** and **3415b**, respectively) to ANs **515** in access node areas **3450b** and **3450a**, respectively. It should be understood that the frequency allocation **6400** is one example and various other frequency allocations may be used. For example, the return uplink signals **6450** may be in a different frequency range (e.g., a different frequency range within the K/Ka band) from the forward uplink signals **6440a** and the forward downlink signals **6445** may be in a different frequency range (e.g., a different frequency range within the K/Ka band) from return downlink signals **6455a**. This may, for example, allow the use of dual-pole transponders in the forward and return receive/transmit signal paths **6000** and **6050**. Additionally or alternatively, the forward uplink signals **6440b** may be allocated within a different frequency range (e.g., a different frequency range within the V band) from the return downlink signals **6455b**, as illustrated. Other

arrangements of the forward uplink/downlink and return/uplink downlink signals within the different frequency ranges may also be considered. For example, the return uplink signals may be allocated within the same frequency range as the forward downlink signals (e.g., using an orthogonal polarization). Additionally or alternatively, the forward uplink signals **6440a** from the ANs in the first access node area **3450a** may be allocated within the same frequency range as the return downlink signals **6455a** (e.g., using an orthogonal polarization). Coupling of forward and return receive/transmit signal paths **6000** and **6050** to the various user-link and feeder-link constituent transmit/receive elements may be selected according to the desired frequency range allocation.

[0352] In some examples of a single feeder-link antenna element array **3415** supporting multiple AN clusters such as the multiple AN clusters illustrated in FIG. **59B**, each feeder-link constituent receive element **3416** and feeder-link constituent transmit element **3419** may be coupled with multiple forward link transponders **3430**. FIGS. **61A** and **61B** illustrate example receive/transmit signal paths supporting cooperating AN clusters operating in different frequency ranges in accordance with aspects of the present disclosure. Forward receive/transmit signal path **6100** of FIG. **61A** include multiple forward-link transponders **3430** coupled between a feeder-link constituent receive element **3416** and multiple user-link constituent transmit elements **3429**. In some examples, a feeder-link constituent receive element **3416** receives a composite of forward uplink signals **521** from ANs **515** in multiple AN areas **3450**. Following receipt by a feeder-link constituent receive element **3416**, the forward uplink signals may be split (e.g., using a splitter **6005**) and the split signals may serve as inputs to forward-link transponders **3430j** and **3430k**. In some examples, the splitter **6005** splits signals based on frequency ranges (e.g., such that received forward uplink signals occupying a first frequency range are fed to forward-link transponder **3430j** and received forward uplink signals occupying a second frequency range are fed to forward-link transponder **3430k**). In such a scenario, the splitter **6005** may alternatively be an example of a filter. Accordingly, frequency converters **3710d** and **3710e** may be operable to accept inputs at different frequency ranges and output signals at the same frequency range for superposition in the user downlink signals **522**.

[0353] A return receive/transmit signal path **6150** is shown in FIG. **61B** in which return-link transponders **3440** couple multiple user-link constituent receive elements **3426a** and **3426b** to a single user-link constituent transmit element **3419**. User-link constituent receive elements **3426a** and **3426b** may be parts of the same user-link antenna element array **3425** or separate user-link antenna element arrays **3425a** and **3425b** (as shown). User-link constituent receive element **3426a** may act as input to return-link transponder **3440j** while user-link constituent receive element **3426b** may act as input to return-link transponder **3440k**. The outputs of the return-link transponders **3440** may be fed to signal combiner **6010** before being transmitted by feeder-link constituent transmit element **3419** to ANs **515** in the AN areas **3450**. In some cases, components of receive/transmit signal paths **6000** and **6050** may be rearranged (or omitted) e.g., such that signal combiner **6010** may follow harmonic filters **3430b**, splitter **6005** may precede LNAs **3705a**, etc.

[0354] FIG. **64B** illustrates an example frequency spectrum allocation **6401** with four frequency ranges displayed (frequency ranges **6425b**, **6430b**, **6435b**, and **6436b**). In the illustrated example, frequency ranges **6425b** and **6430b** are frequency ranges within the K/Ka-bands (e.g., between 17 GHz and 40 GHz) while frequency ranges **6435b** and **6436b** are within the V/W bands (e.g., between 40 GHz and 110 GHz). For example, frequency ranges **6425b**, **6430b**, **6435b**, and **6436b** may be the same as frequency ranges **6425a**, **6430a**, **6435a**, and **6436a** illustrated in FIG. **64A**. FIG. **64B** may illustrate operation of multiple AN clusters operating over different frequency ranges as shown in FIG. **59A** or **59B**.

[0355] As one example, frequency spectrum allocation **6401** may be used in the scenario illustrated in FIG. **59B** using an end-to-end relay **3403** having forward and return receive/transmit signal paths **6100** and **6150** as shown in FIGS. **61A** and **61B**. In this example, forward uplink signals **6440c**

from AN area **3450a** may be transmitted over frequency range **6430b** (e.g., using RHCP) while forward uplink signals **6440d** from AN area **3450b** may be transmitted over frequency range **6436b** (e.g., using RHCP). The first set of forward uplink signals **6440c** may be received by feeder-link constituent receive elements **3416a** while the second set of forward uplink signals **6440d** may be received by feeder-link constituent receive elements **3416b** of forward receive/transmit signal paths **6100**. As discussed with reference to FIG. **61A**, the two sets of forward uplink signals **6440** are frequency converted by forward link transponders **3430j** and **3430k** (e.g., they are downconverted to the same frequency range **6425b** in the Ka-band). Subsequently, the outputs of the forward-link transponders **3430j** are transmitted by user-link constituent transmit elements **3429a** as a first set of forward downlink signals **6445c** while the outputs of the forward-link transponders **3430k** are transmitted by user-link constituent transmit elements **3429b** as a second set of forward downlink signals **6445d**. In the present example, these user-link constituent transmit elements **3429a**, **3429b** belong to the same user-link antenna element array **3425** and illuminate the same user coverage area **3460**. Accordingly, the ANs **515** in access node areas **3450a** and **3450b** may be referred to as cooperating in that some fraction of ANs **515** in each area combine to serve the same user coverage area **3460**. That is, at least one beamformed forward user beam providing service to user terminals **517** within the corresponding user beam coverage area **519** is formed from forward uplink signals **6440c** from at least a subset of the ANs **515** in the first access node area **3450a** and from forward uplink signals **6440d** from at least a subset of the ANs **515** in the second access node area **3450b**.

[0356] Frequency spectrum allocation **6401** also illustrates an example of frequency allocation for return-link transmissions for the scenario illustrated in FIG. **59B** using an end-to-end relay **3403** having forward and return receive/transmit signal paths **6100** and **6150** as shown in FIGS. **61A** and **61B**. Return uplink signals **6450a** originating from user terminals **517** distributed throughout the user coverage area **3460** may be transmitted over frequency range **6425b** (e.g., using RHCP) and received by user-link constituent receive elements **3426a** and **3426b** of FIG. **61B**, where the user-link constituent receive elements **3426a** and **3426b** belong to the same user-link antenna element array **3425**. As described with reference to FIG. **61B**, the return uplink signals **6450** may be fed to return-link transponders **3440j** and **3440k** and frequency converted to appropriate frequency ranges **6430b** (e.g., using LHCP) and **6435b** (e.g., using LHCP), respectively. The frequency converted signals may then be combined (e.g., summed, etc.) by signal combiner **6010** and transmitted by feeder-link constituent transmit elements **3419** to ANs **515** in access node areas **3450a** and **3450b**. It should be understood that the frequency allocation **6401** is one example and various other frequency allocations may be used. For example, the return uplink signals **6450a** may be in a different frequency range (e.g., a different frequency range within the K/Ka band) than the forward downlink signals **6445c** and **6445d**. Similarly, the forward uplink signals **6440c** may be in a different frequency range (e.g., a different frequency range within the K/Ka band) than return downlink signals **6455a** and the forward uplink signals **6440d** may be allocated within a different frequency range (e.g., a different frequency range within the V/W bands as illustrated) than the return downlink signals **6455d**. This may, for example, allow the use of dual-pole transponders in the forward and return receive/transmit signal paths **6100** and **6150**. Coupling of forward and return receive/transmit signal paths **6100** and **6150** to the various user-link and feeder-link constituent transmit/receive elements may be selected according to the desired frequency range allocation.

[0357] In some cases, the available bandwidths in a given band (e.g., K band, Ka band, etc.) for feeder-link transmissions and user-link transmissions may be unequal (e.g., significantly different). Additionally or alternatively, the available bandwidths for uplink and downlink transmissions within a given band may be (e.g., significantly) unequal. As an example, a regulatory body may specify what portions of a frequency spectrum are available for various types of transmissions.

[0358] FIGS. **65A** and **65B** show example frequency spectrum allocations **6500** and **6501** with three frequency ranges (frequency ranges **6520a**, **6525a**, and **6530a**) used for the forward link and three frequency ranges (frequency ranges **6520b**, **6525b**, and **6530b**) used for the return link. In the

illustrated example, frequency ranges **6520a**, **6520b**, **6525a**, and **6525b** are frequency ranges within the K/Ka-bands (e.g., between 17 GHz and 40 GHz) while frequency ranges **6530a** and **6530b** are within the V/W bands (e.g., between 40 GHz and 110 GHz). FIGS. **65A** and **65B** may illustrate operation of multiple AN clusters operating over different frequency ranges as shown in FIG. **59A** or **59B**.

[0359] Referring to FIG. **65A**, forward uplink signals **6540a** from AN area **3450a** may be transmitted over frequency range **6525a** (e.g., using RHCP) while forward uplink signals **6540b** from AN area **3450b** may be transmitted over frequency range **6530a** (e.g., using RHCP). As discussed with reference to FIG. **60A** or **61A**, the two sets of forward uplink signals **6540** are frequency converted by forward link transponders **3430** to the frequency range **6520a**. In the example illustrated in FIG. **65A**, the combined bandwidth of frequency ranges **6525a** and **6530a** equals the bandwidth of frequency range **6520a**. Thus, forward uplink signals **6540a** are frequency converted (e.g., via frequency converters in the forward link transponders of forward receive/transmit signal paths **6000** or **6100**) to forward downlink signals **6545** spanning a first portion **6521a** of frequency range **6520a** while forward uplink signals **6540b** are frequency converted (e.g., via frequency converters in the forward link transponders of forward receive/transmit signal paths **6000** or **6100**) to forward downlink signals **6545** spanning a second portion **6521b** of frequency range **6520a**. A given beamformed user beam in the user coverage area **3460** may span all of frequency range **6520a**, in which case the user beam is formed from both forward uplink signals **6540a** and **6540b**. Where each user beam formed by forward downlink signals **6545** uses a subset of frequency range **6520a**, some user beams may be formed by first portion **6521a** of frequency range **6520a** and some user beams may be formed by second portion **6521b** of frequency range **6520a**. Additionally or alternatively, in some cases some user beams may be formed by cooperative superposition of forward downlink signals **6545** associated with frequency range **6521a** and forward downlink signals **6545** associated with frequency range **6521b** (e.g., frequency ranges **6521a** and **6521b** may partially overlap to enable cooperatively forming user beams in user coverage area **3460** with forward uplink signals **6540** from different AN clusters). In another example, one or both of frequency ranges **6525a** or **6530a** may have the same bandwidth as frequency range **6520a** (e.g., or the combined bandwidth of frequency ranges **6525a** and **6530a** may exceed the bandwidth of frequency range **6520a**), and thus up to all forward user beams may be formed by cooperative superposition of forward downlink signals associated with frequency ranges **6521a** and **6521b**.

[0360] FIG. **65B** shows example return link allocations where at least one access node area **3450** utilizes frequency ranges within a different band than is used for the user coverage area **3460**. Specifically, the user terminals **517** may transmit return uplink signals **6550** over a frequency range **6520b** (e.g., within K/Ka bands), which may be received via two sets of user-link constituent receive elements **3416** as shown in either FIG. **60B** or **61B**, and frequency converted (e.g., via frequency converters in the return link transponders **3440** of return receive/transmit signal paths **6050** or **6150**) to a first set of return downlink signals **6555a** in frequency range **6525b** and a second set of return downlink signals **6555b** in frequency range **6530b**. The first and second sets of return downlink signals **6555a**, **6555b** may be transmitted from the same feeder-link constituent transmit element **3419** (as shown in FIG. **61B**), or from different feeder-link constituent transmit elements **3419** (as shown in FIG. **60B**). As with FIG. **65A**, the combined bandwidths of frequency ranges **6525b** and **6530b** are illustrated to be equal to the bandwidth of frequency range **6520b**. Thus, a first portion **6560a** of return uplink signals **6550** may be frequency converted and transmitted by a first set of return link transponders **3440** as return downlink signals **6555a** while a second portion **6560b** (which may or may not overlap with the first portion **6560a**) may be frequency converted and transmitted by a second set of return link transponders **3440** as return downlink signals **6555b**. Thus, some return user beams may be formed by performing return link beamforming processing on portions of return downlink signals **6555a** and some return user beams

may be formed by performing return link beamforming processing on portions of return downlink signals **6555b**. Additionally or alternatively, some return user beams may be formed by performing return link beamforming processing on portions of return downlink signals **6555a** and return downlink signals **6555b** (e.g., some portions of return downlink signals **6555a** and **6555b** may cooperate to form a single return user beam). In some cases, one or both of frequency ranges **6525b** or **6530b** may have the same bandwidth as frequency range **6520b** (e.g., or the combined bandwidth of frequency ranges **6525b** and **6530b** may exceed the bandwidth of frequency range **6520b**), and thus up to all return user beams may be formed by cooperative superposition of return downlink signals **6555a** and **6555b**.

[0361] FIGS. **66A** and **66B** illustrate example receive/transmit signal paths supporting cooperating AN clusters operating in different frequency ranges in accordance with aspects of the present disclosure. Forward receive/transmit signal path **6600** of FIG. **66A** includes forward-link transponders **34301** coupled between feeder-link constituent receive elements **3416a** and user-link constituent transmit elements **3429** and forward-link transponders **3430m** coupled between feeder-link constituent receive elements **3416b** and user-link constituent transmit elements **3429**. As described above, the forward-link transponder **34301** can include some or all of LNAs **3705a**, frequency converters and associated filters **37101**, channel amplifiers **3715a**, phase shifters **3720a**, power amplifiers **3725a**, and harmonic filters **3730a**. Similarly, forward-link transponder **3430m** can include some or all of LNAs **3705a**, frequency converters and associated filters **3710m**, channel amplifiers **3715a**, phase shifters **3720a**, power amplifiers **3725a**, and harmonic filters **3730a**. In some cases, frequency converter **37101** may be operable to convert signals from a first feeder-link uplink frequency range (e.g., frequency range **6525a** of FIG. **65A**) to a first portion of a user-link downlink frequency range (e.g., frequency range **6521a** of FIG. **65A**) while frequency converter **3710m** is operable to convert signals from a second feeder-link uplink frequency range (e.g., frequency range **6530a** of FIG. **65A**) to a second portion of the same user-link downlink frequency range (e.g., frequency range **6521b** of FIG. **65A**). The forward-link transponders **3430** couple multiple feeder-link constituent receive elements **3416a** and **3416b** to a single user-link constituent transmit element **3429**. Feeder-link constituent receive elements **3416a** and **3416b** may be parts of the same feeder-link antenna element array **3415** or separate feeder-link antenna element arrays **3415a** and **3415b** (as shown). Feeder-link constituent receive element **3416a** may act as input to forward-link transponder **34301** while feeder-link constituent receive element **3416b** may act as input to forward-link transponder **3430m**. The outputs of the forward-link transponders **3430** may be fed to signal combiner **6610** before being transmitted by user-link constituent transmit element **3429** to user terminals **517** in the user coverage areas **3460**. In some cases, components of receive/transmit signal paths **6600** and **6650** may be rearranged (or omitted) e.g., such that signal combiner **6610** may follow harmonic filters **3430b**, splitter **6605** may precede LNAs **3705a**, etc.

[0362] Return receive/transmit signal path **6650** of FIG. **66B** includes return-link transponder **34401** coupled between a user-link constituent receive element **3426** and a corresponding feeder-link constituent transmit element **3419a** and return-link transponder **3440m** coupled between a user-link constituent receive element **3426** and a corresponding feeder-link constituent transmit element **3419b**. As described above, the return-link transponder **34401** can include some or all of LNAs **3705b**, frequency converters and associated filters **3710n**, channel amplifiers **3715b**, phase shifters **3720b**, power amplifiers **3725b**, and harmonic filters **3730b**. Similarly, return-link transponder **3440m** can include some or all of LNAs **3705b** frequency converters and associated filters **3710o**, channel amplifiers **3715b**, phase shifters **3720b**, power amplifiers **3725b**, and harmonic filters **3730b**. In some cases, frequency converter **3710n** may be operable to convert signals from a first portion of a user-link uplink frequency range (e.g., frequency range **6560a** of FIG. **65B**) to a first feeder-link downlink frequency range (e.g., frequency range **6525b** of FIG. **65B**, which may be the same range as the first feeder-link uplink frequency range described with reference to FIG. **66A**) while frequency converter **3710o** is operable to convert signals from a

second portion of the user-link uplink frequency range (e.g., frequency range **6560b** of FIG. **65B**) to a second feeder-link downlink frequency range (e.g., frequency range **6530b** of FIG. **65B**, which may be the same range as the second feeder-link uplink frequency range described with reference to FIG. **66A**). Following receipt by a user-link constituent receive element **3426**, the return uplink signals may be split (e.g., using a splitter **6605**) and the split signals may serve as inputs to return-link transponders **34401** and **3440m**. In some examples, the splitter **6605** splits signals based on frequency ranges (e.g., such that received return uplink signals occupying a first frequency range are fed to forward-link transponder **34301** and received return uplink signals occupying a second frequency range are fed to forward-link transponder **3430m**). In such a scenario, the splitter **6605** may be an example of one or more filters. Accordingly, frequency converters **3710n** and **3710o** may be operable to accept inputs at different frequency ranges or portions of a frequency range and output signals in different frequency ranges in feeder downlink signals **522**.

[0363] As described above, the various feeder-link antenna elements may be part of the same or different feeder-link antenna element arrays **3415**. The feeder-link constituent transmit elements **3419a** and feeder-link constituent transmit elements **3419b** may be interleaved within the same feeder-link antenna element array **3415** as illustrated in FIG. **62**. Where the frequencies supported for the feeder links by the forward-link transponders **34301** and **3430m** and return-link transponders **34401** and **3440m** are substantially different (e.g., one being different by more than $1.5\times$ from the other, etc.), the different subsets of elements **6205a**, **6205b** of the antenna element array **6200** may be sized appropriately for the different supported frequency ranges (e.g., constituent antenna elements **6205b** supporting a higher frequency range than constituent antenna elements **6205a** may have smaller waveguides/horns, etc.).

Access Nodes Supporting Multiple Independent Feeder Link Signals

[0364] In some examples, one or more ANs **515** may support multiple feeder links (e.g., transmission of multiple forward uplink signals and/or reception of multiple return downlink signals). In some cases, ANs **515** supporting multiple feeder links may be used to reduce the number of ANs. For example, instead of having M ANs **515** where each AN **515** supports one feeder link, the system may have $M/2$ ANs **515**, where each AN **515** supports two feeder links. While having $M/2$ ANs **515** reduces spatial diversity of the ANs **515**, signals between the ANs **515** and the end-to-end relay at different frequencies will experience different channels, which also results in channel diversity between the two feeder links. Each AN **515** may receive multiple access node-specific forward signals **516**, where each access node-specific forward signal **516** is weighted according to beamforming coefficients that are determined based on a channel matrix associated with the corresponding transmit frequency range. Thus, where each AN **515** supports two feeder links, each AN **515** may be provided a first access node-specific forward signal determined based in part on a first forward uplink channel matrix for forward uplink channels between the ANs **515** and the end-to-end relay **3403** over a first frequency range and a second access node-specific forward signal determined based in part on a second forward uplink channel matrix for the forward uplink channels between the ANs **515** and the end-to-end relay **3403** over a second frequency range. Similarly, on the return link, each AN **515** may obtain a first composite return signal based on a first return downlink signal in a third frequency range (which may be the same frequency range or in the same band as the first frequency range) and a second composite return signal based on a second return downlink signal in a fourth frequency range (which may be the same frequency range or in the same band as the second frequency range). Each AN **515** may provide the respective first and second composite return signals to the return beamformer **513**, which may apply beamforming coefficients to the first composite return signals determined based in part on a first return downlink channel matrix for the return downlink channels between the end-to-end relay **3403** and the ANs **515** over the third frequency range and apply beamforming coefficients to the second composite return signals determined based in part on a second return downlink channel matrix for the return downlink channels over the fourth frequency range.

[0365] Systems employing M/2 ANs 515 may have reduced system capacity when compared to having M ANs 515, but the system cost reduction (e.g., including set up and maintenance costs) may be substantial while still providing acceptable performance. Additionally, a number of ANs 515 other than M/2 may be used, such as 0.75.Math.M, which may provide similar or greater performance at reduced cost when compared to M ANs 515 each supporting only one feeder link. Generally, where M ANs 515 would be used each supporting a single feeder link (e.g., a single feeder uplink frequency range and a single feeder downlink frequency range), X.Math.M ANs 515 may be used where each AN 515 supports multiple feeder links, where X is in the range of 0.5 to 1.0.

[0366] Returning to FIGS. 45A and 45B, the X.Math.M ANs 515 may be distributed within the access node area 3450 and may service user terminals 517 within user coverage area 3460 via beamformed user beams, where one or more user beams are beamformed using multiple feeder link signals from at least one AN 515. The multiple feeder links may be supported via a single set of feeder-link constituent antenna elements (e.g., a single feeder-link antenna element array 3415), or separate feeder-link constituent antenna elements (separate feeder-link antenna element arrays 3415 for each feeder link).

[0367] A single feeder-link antenna element array 3415 and a single reflector may be used to support multiple feeder links for each AN 515 using either the forward and return receive/transmit signal paths 6000, 6050 of FIGS. 60A and 60B (e.g., separate subsets of feeder-link constituent antenna elements within the same feeder-link antenna element array 3415), or the forward and return receive/transmit signal paths 6100, 6150 of FIGS. 61A and 61B (e.g., splitters and combiners used to multiplex the multiple feeder links using the same set of feeder-link constituent antenna elements). Where the difference in frequency ranges between the multiple feeder links is substantial (which may be desirable to increase channel diversity), the dimensions of the access node area 3450 may depend on the higher frequency feeder link. For example, where a first feeder link is supported in a frequency range around 30 GHz while a second feeder link is supported in a frequency range around 60 GHz, the access node area is limited to the area illuminated by the single feeder-link antenna element array 3415 via the single reflector. Thus, some path diversity for the lower frequency range may be lost. Alternatively, a first feeder-link antenna element array 3415a may be used to support a first frequency range while a second feeder-link antenna element array 3415b is used to support a second frequency range. In this case, separate reflectors may be used, and may be sized appropriately to provide coverage of a same access node area 3450 at the different frequencies. For example, where a first feeder link is supported by a first feeder-link antenna element array 3415a and a first reflector in a frequency range around 30 GHz while a second feeder link is supported by a second feeder-link antenna element array 3415b and a second reflector in a frequency range around 60 GHz, the first reflector may be larger (e.g., having twice the reflector area) than the second reflector to account for the difference in antenna gain at the different frequencies.

[0368] Frequency allocation for the different feeder links may be performed in various ways including that shown in FIG. 64A, 64B, 65A, or 65B. That is, a first feeder link may use carrier frequencies within frequency ranges 6425a and 6430a (e.g., in K/Ka bands) while a second feeder link uses frequency range 6435a (e.g., in V/W bands) as shown in FIG. 64A. Alternatively, the first feeder link may use carrier frequencies within frequency ranges 6430b (e.g., in K/Ka bands) while a second feeder link uses frequency range 6435b (e.g., in V/W bands) as shown in FIG. 64B. In yet another alternative, the first and second feeder links may both use frequencies different from the user links as shown in FIGS. 65A and 65B where a first feeder link uses frequency ranges 6525a and 6525b (e.g., in V/W bands) while a second feeder link uses frequency ranges 6530a and 6530b (e.g., in V/W bands). In some examples, the first feeder link and second feeder link may use frequency ranges that are substantially different (e.g., the lowest frequency in one frequency range may be greater than 1.5 or 2 times the lowest frequency in the other frequency range). As discussed

above, the bandwidth for each feeder link frequency range may be less than the bandwidth for the user link frequency range, or one or more of the feeder link frequency ranges may have the same bandwidth as the user link frequency range. In some cases, the correlation of the signals associated with the first and second feeder links may be inversely proportional to the bandwidth separation between the two signals (e.g., such that two signals whose frequency ranges are adjacent within the Ka-band are more correlated than a Ka-band signal and a V-band signal or two signals with non-adjacent frequency ranges within the Ka-band). This effect is a result of the signals with adjacent frequency ranges experiencing similar atmospheric effects, whereas signals with a greater degree of bandwidth separation will experience different atmospheric effects, which contributes to the induced multipath.

CONCLUSION

[0369] Although the disclosed method and apparatus is described above in terms of various examples, cases and implementations, it will be understood that the particular features, aspects, and functionality described in one or more of the individual examples can be applied to other examples. Thus, the breadth and scope of the claimed invention is not to be limited by any of the examples provided above but is rather defined by the claims.

[0370] Terms and phrases used in this document, and variations thereof, unless otherwise expressly stated, are to be construed as open ended as opposed to limiting. As examples of the foregoing: the term “including” is used to mean “including, without limitation” or the like; the term “example” is used to provide examples of instances of the item in discussion, not an exhaustive or limiting list thereof; the terms “a” or “an” mean “at least one,” “one or more” or the like.

[0371] Throughout the specification, the term “couple” or “coupled” is used to refer broadly to either physical or electrical (including wireless) connection between components. In some cases, a first component may be coupled to a second component through an intermediate third component disposed between the first and second component. For example, components may be coupled through direct connections, impedance matching networks, amplifiers, attenuators, filters, direct current blocks, alternating current blocks, etc.

[0372] A group of items linked with the conjunction “and” means that not each and every one of those items is required to be present in the grouping, but rather includes all or any subset of all unless expressly stated otherwise. Similarly, a group of items linked with the conjunction “or” does not require mutual exclusivity among that group, but rather includes all or any subset of all unless expressly stated otherwise. Furthermore, although items, elements, or components of the disclosed method and apparatus may be described or claimed in the singular, the plural is contemplated to be within the scope thereof unless limitation to the singular is explicitly stated.

[0373] The presence of broadening words and phrases such as “one or more,” “at least,” or other like phrases in some instances does not mean that the narrower case is intended or required in instances where such broadening phrases may be absent. Additionally, the terms “multiple” and “plurality” may be used synonymously herein.

[0374] While reference signs may be included in the claims, these are provided for the sole function of making the claims easier to understand, and the inclusion (or omission) of reference signs is not to be seen as limiting the extent of the matter protected by the claims.

Claims

1-4. (canceled)

5. A satellite configured for operation as an end-to-end relay in a satellite communications system configured for end-to-end beamforming from a plurality of geographically distributed satellite access nodes (SANs) to a plurality of reference locations, each reference location corresponding to a respective one among a plurality of forward user beam coverage areas, the satellite comprising: an array of transmit antenna elements for transmitting a plurality of forward downlink signals, each

forward downlink signal transmitted from a respective one of the transmit antenna elements; and a plurality of transponders, each transponder comprising an electrical signal path establishing one-to-one connectivity between a respective one among an array of receive antenna elements of the satellite and a respective one among the array of transmit antenna elements, each transponder providing power amplification; wherein the receive antenna elements in the array of receive antenna elements are configured with overlapping receive antenna patterns such that each forward downlink signal comprises a composite of two or more weighted forward uplink signals, each weighted forward uplink signal transmitted by a respective one among the plurality of SANs; and wherein the transmit antenna elements in the array of transmit antenna elements are configured with overlapping transmit antenna patterns such that the composites of the weighted forward uplink signals as transmitted by the satellite superpose to form a plurality of forward user beams corresponding to the plurality of forward user beam coverage areas.

6. The satellite according to claim 5, wherein the overlapping receive antenna patterns are configured such that at least one quarter of the receive antenna elements in the array of receive antenna elements have receive antenna patterns that overlap with one another.

7. The satellite according to claim 5, wherein the overlapping receive antenna patterns are configured such that at least one half of the receive antenna elements in the array of receive antenna elements have receive antenna patterns that overlap with one another.

8. The satellite according to claim 5, wherein the overlapping transmit antenna patterns are configured such that at least one quarter of the transmit antenna elements in the array of transmit antenna elements have transmit antenna patterns that overlap with one another.

9. The satellite according to claim 5, wherein the overlapping transmit antenna patterns are configured such that at least one half of the transmit antenna elements in the array of transmit antenna elements have transmit antenna patterns that overlap with one another.

10. The satellite according to claim 5, wherein the overlapping receive antenna patterns are configured such that, with respect to a particular one of the receive antenna elements that exhibits a peak gain with respect to the forward uplink signal from any particular one of the SANs, at least two more receive antenna elements exhibit gains within 3 dB of the peak gain.

11. A satellite communications system comprising the satellite according to claim 5, and further comprising a ground network that includes a central processing system (CPS), wherein the CPS computes forward beamforming weights for generating the weighted forward uplink signals.

12. The satellite communications system according to claim 11, wherein the CPS comprises circuitry configured to compute the forward beamforming weights as an $M \times K$ forward beam weight matrix corresponding to M SANs and K forward beam signals.
